



Mahavir Education Trust's

SHAH & ANCHOR

KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai

Syllabus of Honours Degree Programs in Technology

(With effect from 2025-26)

1. Artificial Intelligence and Machine Learning
2. Augmented Reality and Virtual Reality
3. Blockchain
4. Cyber Security
5. Data Science
6. Internet of Things



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An Autonomous Institute Affiliated to University of Mumbai

Syllabus of Honours Degree Programs in Technology

(with effect from 2025-26)

**Artificial Intelligence and
Machine Learning**



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

CURRICULUM STRUCTURE UG: B.Tech.

Honours/Minor Degree Program

Artificial Intelligence and Machine Learning

REVISION: R1-V0-2025-26

Effective from Academic Year 2025-26

Approved by Board of Studies on 17/06/2025

Approved by Academic Council on 19/06/2025

Dr. T. P. Vinutha

I/C HOD

Chairman BoS
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Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



Dr. Bhavesh Patel

Principal

Dr. Bhavesh Patel
Chairman
Academic Council

Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



PREAMBLE

Shah and Anchor Kutchhi Engineering College was established in 1985 for the purpose of imparting quality technical education. The college is managed by Mahavir Education Trust. The college is approved by AICTE, New Delhi and DTE, Government of Maharashtra and has been conferred autonomous status by University Grants Commission (UGC) and University of Mumbai for a period of 10 Years with effect from the academic year 2024-25. It also has an ISO 9001:2015 Certification. The institute offers eight under graduate courses, three post-graduate courses and PhD research Centre in Information Technology and Computer Engineering Department.

The Honours Program in Artificial Intelligence and Machine Learning (AI-ML), offered by Shah and Anchor Kutchhi Engineering College, is designed to provide students with a robust foundation in the principles and practices of AI and ML. The program integrates rigorous coursework, hands-on laboratory experiences, and real-world applications to equip students with the skills needed to excel in the rapidly evolving field of artificial intelligence and machine learning.

Over the course of two years, students will engage in an interdisciplinary curriculum that covers advanced mathematical techniques, strategic decision-making models, and innovative applications of AI and ML in various domains, including healthcare and natural language processing. The program is structured to ensure a comprehensive understanding of both theoretical concepts and practical implementations, preparing graduates to tackle complex challenges and drive innovation in AI-ML.

Key features of our autonomous syllabus include:

- **Interdisciplinary Learning:** Foster an interdisciplinary approach by integrating mathematical, technical, and application-oriented courses.
- **Practical Expertise:** Provide extensive hands-on laboratory experiences to bridge the gap between theoretical knowledge and real-world application.
- **Innovation and Research:** Encourage innovative thinking and research through project-based learning and exposure to cutting-edge AI and ML technologies.
- **Ethical Considerations:** Instill a deep understanding of the ethical, societal, and privacy implications of AI and ML technologies.

Graduates of the AI-ML Honours Program will be well-prepared for careers in various sectors, including technology, healthcare, finance, and academia. They will possess the skills required to develop innovative AI solutions, conduct advanced research, and contribute to the growing field of artificial intelligence and machine learning.



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Program Structure for Honours Degree in Artificial Intelligence and Machine Learning (with effect from 2025-2026)

			Learning Scheme					Credits	Assessment Scheme												
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total			Theory					Practical				SLA		Total
									Exam	CIA		ESE	Total		CIAP	ESEP	Total				
			Duration in Hrs	MSE	CCE					Max	Max										
				CL	TL					LL	Notional Hours										
1	Mathematics for AI & ML	HNAMCR1501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
2	Game Theory using AI & ML	HNAMCR1601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
3	AI & ML in Healthcare	HNAMCR1701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225
4	Natural Language Processing and Generative AI	HNAMCR1801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
Total			16	4	4	12	540	18		80	80	240	400		40	60	100		100		600

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
SLA: Self Learning Assessment

Note:

- Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
- 1 credit is equivalent to 30 Notional hrs.
- Self learning hours shall not be reflected in the Time Table.
- SLA will have separate head of passing



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Semester V

Program	Honours Degree in Artificial Intelligence and Machine Learning - Semester: V	CL	TL	LL	SL	C
Course Name: Mathematics for AI & ML	Course Code: HNAMECR1501	4	1	–	3	4
Pre-requisite:	Applied Mathematics					

I RATIONALE

This course lays the mathematical and statistical foundation necessary for data-driven problem solving and machine learning. With the growing relevance of AI and data analytics across industries—such as healthcare, finance, marketing, and environmental studies—learners need a strong conceptual and practical understanding of linear algebra, probability distributions, optimization strategies, and dimension reduction techniques. The course emphasizes real-world data handling, model building, and visualization, which are critical skills for any data scientist, machine learning engineer, or applied researcher. It bridges theoretical constructs with computational tools, empowering learners to perform exploratory data analysis, build efficient models, and derive insights that inform decision-making.

II COMPETENCY

- Demonstrate the ability to apply mathematical and statistical principles—including linear algebra, probability, and sampling—to formulate and interpret models relevant to AI and game-theoretic decision-making.
- Exhibit proficiency in exploratory data analysis and optimization techniques, utilizing appropriate tools and methods to extract, visualize, and interpret actionable insights from complex datasets.
- Develop and integrate dimensionality reduction and model simplification techniques to streamline data-driven decision processes and enhance the effectiveness of AI/ML systems.

III COURSE OUTCOMES (COs)

After the completion of course-based learning, learners will be able to:

Course Outcome Number	Course Outcome Statement
1	Apply linear algebra, probability distributions, and sampling distribution concepts to solve real-world problems.
2	Interpret uncertainty in AI/ML applications using probability distributions, sampling techniques, and statistical inference.
3	Select appropriate graph representations and exploratory data analysis techniques to real datasets, interpreting insights through meaningful visualizations.
4	Analyze various optimization techniques through hands-on implementation, while appreciating their relevance in solving practical problems.
5	Evaluate dimensionality reduction algorithms by applying them to real datasets and explaining their impact on data visualization and model performance.
6	Develop frameworks based on mathematical concepts from the course modules for application in real-world AI/ML contexts.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme						Assessment Scheme														
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Credits		Theory					Practical			SLA		Total		
			CL	TL	LL						Exam	CIA		ESE	Total		CIAP	ESEP				Total	
												MSE	CCE		Max	Min						Max	Min
1	Mathematics for AI & ML	HNAMCR1501	4	1	–	3	120	4	2	20	20	60	100	40	–	–	–	–	25	10	125		

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1 & CO2 & CO6	1. Solve systems of linear equations and explore vector spaces 2. Interpret Eigenvalues and Singular Value Decomposition 3. Explain the fundamentals of probability and random variables 4. Demonstrate sampling methods and sample distributions 5. Apply Chi-square, t, and F-distributions for inference 6. Examine the Central Limit Theorem and bivariate distributions	Fundamentals of Probability and Statistics: 1.1 Vectors, Matrices, Linear equations, Fundamental Subspaces 1.2 Eigenvalues, Eigen Vectors, SVD 1.3 Probability, Random Variables, Distributions 1.4 Sampling, Sample Characteristics, Distributions 1.5 Chi-square, t, F Distributions 1.6 Bivariate Normal Sampling, CLT	16

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO3 & CO6	1. Distinguish between quantitative and qualitative data 2. Use appropriate plots to visualize data 3. Perform data cleaning and preparation 4. Apply feature engineering techniques 5. Handle missing values effectively 6. Identify insights and align ML models with EDA	Exploratory Data Analysis: 2.1 Types of Data: Continuous, Discrete, Categorical, Binary 2.2 Data Visualization: Graphs and Plots 2.3 Need for EDA, Cleaning and Preparation 2.4 Feature Engineering 2.5 Handling Missing Values 2.6 Insight Extraction, ML Model Suitability	18
3	CO4 & CO6	1. Differentiate between constrained and unconstrained optimization 2. Apply bracketing methods for optimization 3. Implement numerical optimization techniques 4. Use Bisection and False Position methods 5. Utilize Newton's method 6. Analyze gradient-based approaches like Steepest Descent	Optimization Techniques: 3.1 Types of Optimization 3.2 Numerical Optimization 3.3 Bracketing Methods 3.4 Bisection and False Position Methods 3.5 Newton's Method 3.6 Steepest Descent Method	14

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5 & CO6	1. Explain the need for dimensionality reduction 2. Implement Principal Component Analysis 3. Use Linear Discriminant Analysis 4. Apply non-linear dimensionality reduction 5. Apply Multidimensional Scaling 6. Apply Isometric Feature Mapping	Dimension Reduction Algorithms: 4.1 Introduction to Dimensionality Reduction 4.2 Principal Component Analysis 4.3 Linear Discriminant Analysis (LDA) 4.4 Non-linear Reduction Techniques 4.5 Multidimensional Scaling 4.6 Isometric Feature Mapping	12

VI SUGGESTED SELF LEARNING ACTIVITIES

Self-Learning Activity: Real-World Problem Solving Using Probability, Statistics and Optimization

learners are required to identify a real-world dataset or domain (e.g., healthcare, finance, marketing, agriculture, environment, etc.) and carry out a semester-long data-driven activity structured in four progressive phases. The activity encourages learners to independently explore theoretical concepts through hands-on applications, enabling them to develop analytical thinking and data interpretation skills.

Phase 1: Understanding the Dataset (Weeks 1–3)

Learners will explore the nature of data (qualitative/quantitative), perform initial exploratory data analysis (EDA) using visualizations and descriptive statistics, identify missing values and anomalies, and submit a brief report summarizing data types (discrete, continuous, categorical, etc.), key insights, and possible directions for further analysis.

Phase 2: Statistical Analysis (Weeks 4–6)

Learners will apply core statistical techniques such as mean, median, mode, variance, and standard deviation. They will model the data using appropriate probability distributions and perform hypothesis testing (e.g., t-test, chi-square). Concepts like the Central Limit Theorem and sampling strategies will be implemented. A checkpoint report will be submitted at the end of this phase.

Phase 3: Optimization and Modeling (Weeks 7–9)

Learners will define an objective function or performance metric relevant to their problem statement. They will apply optimization techniques such as gradient descent, Newton's method, or bisection method. Using basic regression or classification models, they will analyze performance and optimize model parameters.

Phase 4: Dimensionality Reduction and Final Interpretation (Weeks 10–13)

Learners will apply dimensionality reduction techniques like PCA or LDA to enhance data interpretability and visualization. They will plot reduced data in 2D/3D space and justify how dimensionality reduction improved computational efficiency or clarity of results.

Deliverables:

- **Initial Checkpoint Report (Week 4)** – Problem Identification Design Planning
- **Mid-Semester Checkpoint Report (Week 8)** – Summary of dataset, EDA, and statistical analysis
- **Final Report and Presentation (Week 14)** – Complete documentation of the problem, methods used, modeling outcomes, key insights, challenges faced, and visual results including dimension-reduced projections

This activity enables learners to bridge theoretical knowledge with practical application and fosters independent inquiry, critical reasoning, and project documentation skills essential for data-centric domains.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Linear Algebra for Everyone	1 st Edition	Gilbert Strang	Wellesley Cambridge Press	2020
2	An Introduction to Probability and Statistics	3 rd Edition	Vijay Rohatgi, A. K. Md. Ehsanes Saleh	Wiley	2015
3	An Introduction to Optimization	4 th Edition	Edwin Chong, Stanislaw Zak	Wiley	2020
4	Mathematics for Machine Learning	1 st Edition	Marc Deisenroth, A. Aldo Faisal, Cheng Soon Ong	Cambridge University Press	2020
5	Exploratory Data Analysis	Reprint Edition	John Tukey	Addison-Wesley (via Princeton/Bell Labs)	1977

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to Linear Algebra	5 th Edition	Gilbert Strang	Wellesley Cambridge Press	2016
2	Advanced Engineering Mathematics	10 th Edition	Erwin Kreyszig	Wiley	2011
3	Foundations of Machine Learning	2 nd Edition	Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar	MIT Press	2018
4	Understanding Machine Learning: From Theory to Algorithms	1 st Edition	Shai Shalev-Shwartz, Shai Ben-David	Cambridge University Press	2014
5	Mathematics and Programming for Machine Learning with R	1 st Edition	William B. Claster	CRC Press	2020

Online References

Sr. No	Website Name
1	https://math.mit.edu/~gs/linearalgebra/ – Gilbert Strang's Linear Algebra Portal
2	https://nptel.ac.in/courses/111/105/111105090/ – NPTEL: Probability and Statistics
3	https://onlinecourses.nptel.ac.in/noc21_ma01/preview – NPTEL: Linear Algebra
4	https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010/video-lectures – MIT OCW: Linear Algebra Video Lectures



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Semester VI



Program	Honours Degree in Artificial Intelligence and Machine Learning - Semester: VI	CL	TL	LL	SL	C
Course Name: Game Theory using AI & ML	Course Code: HNAMECR1601	4	1	–	3	4
Pre-requisite:	Mathematics for AI & ML					

I RATIONALE

This course lays the mathematical and statistical foundation necessary for data-driven problem solving and machine learning. With the growing relevance of AI and data analytics across industries—such as healthcare, finance, marketing, and environmental studies—learners need a strong conceptual and practical understanding of linear algebra, probability distributions, optimization strategies, and dimension reduction techniques. The course emphasizes real-world data handling, model building, and visualization, which are critical skills for any data scientist, machine learning engineer, or applied researcher. It bridges theoretical constructs with computational tools, empowering learners to perform exploratory data analysis, build efficient models, and derive insights that inform decision-making.

II COMPETENCY

- Competence in applying game theory concepts to design strategic solutions in competitive and multi-agent systems relevant to domains such as finance, cybersecurity, and operations research.
- Ability to work with foundational AI models and tools, enabling the development of intelligent systems with awareness of ethical, legal, and societal considerations.
- Skilled in implementing search and optimization techniques to solve real-world computational problems, enhancing performance in areas like logistics, automation, and intelligent interfaces.
- Proficiency in using logical reasoning and structured knowledge representation to support scalable decision-making systems across diverse industry applications.

III COURSE OUTCOMES (COs)

After the completion of course-based learning, learners will be able to:

Course Outcome Number	Course Outcome Statement
1	Apply foundational principles of game theory by solving strategic decision-making problems using example-based models, with an appreciation for rational behavior in competitive scenarios.
2	Apply Bayesian game theory concepts, rationalizability, and equilibrium analysis to model strategic decision-making in uncertain and competitive environments.
3	Analyze the role of search algorithms in problem solving, inference, perception, knowledge representation, and learning.
4	Design knowledge-based systems to solve real-world problems using logical inference and structured planning techniques.
5	Evaluate strategies for repeated and evolutionary games, including stability analysis and maximin approaches, to ensure robust solutions for imperfect information scenarios.
6	Develop a game theory and AI based problem-solving methods to create intelligent decision-making systems under uncertainty.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme					Credits	Assessment Scheme													
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total Notional Hours		Exam	Theory					Practical				SLA		Total	
			Duration in Hrs	CIA					ESE	Total		CIAP	ESEP	Total								
				MSE	CCE					Max	Min	Max	Min	Max	Min	Max	Min	Max	Min			
1	Game Theory using AI & ML	HNAMCR1601	4	1	–	3	120	4	2	20	20	60	100	40	–	–	–	–	25	10	125	

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1 & CO6	1. Apply rational choice theory and strategic decision frameworks 2. Model and solve games with perfect information 3. Apply Nash equilibria using examples from economics and politics 4. Distinguish between cooperative and competitive scenarios 5. Construct best response functions and strategic forms 6. Interpret dominated strategies and randomization in mixed strategies	Fundamentals of Probability and Statistics: 1.1 Rational choice, Games with Perfect Information, Nash Equilibrium 1.2 Game examples: Prisoners Dilemma, BOS, Matching Pennies, etc. 1.3 Cournot and Bertrand Models, Auctions, Electoral Competition 1.4 Mixed Strategy Equilibrium, Randomization 1.5 Extensive Games with Perfect Information 1.6 Dominated Actions and Best Response Functions	14

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2 & CO3 & CO6	1. Evaluate incomplete and imperfect information scenarios 2. Apply Bayesian game frameworks to strategic modeling 3. Investigate rationalizability and maximization 4. Explore evolutionarily stable strategies 5. Model strategies in repeated and polymorphic games 6. Analyze behavior in long-term interaction and strategic learning	Games with Imperfect Information: 2.1 Bayesian Games: Definitions and Examples 2.2 Strictly Competitive Games and Maxmin 2.3 Rationalizability in Uncertain Environments 2.4 Evolutionary Game Theory and Stability 2.5 Repeated Games and Strategy Profiles 2.6 General Results on Equilibria in Repeated Games	18
3	CO4 & CO6	1. Differentiate between classical and modern AI approaches 2. Describe AI evolution and application domains 3. Apply heuristic search algorithms like A* and Hill Climbing 4. Analyze problem-solving through local and nondeterministic search 5. Identify real-world applications for AI search techniques 6. Compare AI and machine learning perspectives in decision-making	Introduction to AI and Problem Solving: 3.1 History and Foundations of AI 3.2 Applications and Classification of AI Systems 3.3 Generate-and-Test, Hill Climbing, Best First Search 3.4 A* and Problem Reduction 3.5 Local Search and Continuous Optimization 3.6 Search under Uncertainty	12

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5 & CO6	1. Explain propositional and first-order logic for knowledge representation 2. Apply theorem proving techniques in structured AI planning 3. Design AI systems using partial order planning 4. Represent uncertainty through probabilistic methods 5. Model temporal inference using dynamic Bayesian techniques 6. Discuss reasoning frameworks for multi-object environments	Knowledge and Reasoning: 4.1 Knowledge Base: Propositional Logic, FOL, Situation Calculus 4.2 Theorem Proving, Planning, Partial Order Planning 4.3 Uncertain Reasoning: Probabilities 4.4 Bayesian Networks and Reasoning Over Time 4.5 Hidden Markov Models and Kalman Filters 4.6 Dynamic Bayesian Networks	8

VI SUGGESTED SELF LEARNING ACTIVITIES

Self-Learning Activity: Strategic Reasoning and Intelligent Systems

Learners are required to select a real-world scenario or industry domain (e.g., digital marketing, e-commerce, public policy, autonomous systems, cybersecurity, etc.) and conduct a semester-long study to analyze strategic behavior and intelligent decision-making using principles of Game Theory and Artificial Intelligence. The activity is divided into structured phases to develop conceptual clarity, critical thinking, and technical skills relevant to real-world AI deployment.

Phase 1: Modeling Strategic Games (Weeks 1–3)

Learners will identify a strategic interaction relevant to their domain (e.g., pricing wars, election campaigns, bidding in online auctions) and formulate it using game theory. They will create payoff matrices, define player strategies, and model interactions using concepts like Nash Equilibrium, best responses, and dominant strategies. They will document a brief report explaining the structure and rationale of the modeled game.

Phase 2: Imperfect Information and Repeated Games (Weeks 4–6)

Extend the modeled interaction to scenarios with incomplete information. Learners will explore Bayesian games, mixed strategies, and repeated game concepts like tit-for-tat. They will simulate how strategy adaptation works in dynamic or uncertain environments. A checkpoint report will be submitted explaining observations and strategic evolution.

Phase 3: Applying AI Problem-Solving Techniques (Weeks 7–9)

Based on the strategic game setup, learners will implement AI-based methods to explore search and decision strategies. Using heuristic or local search algorithms (e.g., hill climbing, A*, best-first search), they will simulate agent-based decision-making. A comparative evaluation of different search strategies in their problem context will be documented.

Phase 4: Reasoning Under Uncertainty (Weeks 10–13)

Learners will build a basic knowledge-based model using propositional or first-order logic to encode domain knowledge. They will extend their system to handle uncertainty using probabilistic reasoning or Bayesian networks. The final implementation should reflect how AI systems reason under dynamic and ambiguous real-world constraints.

Deliverables:

- **Initial Checkpoint Report (Week 4)** - Problem Identification Design Planning
- **Mid-Semester Checkpoint Report (Week 8)** – Strategic game formulation, solution strategies, and imperfect information analysis
- **Final Report and Presentation (Week 13)** – Description of the real-world problem, modeling techniques, AI problem-solving implementation, knowledge representation and reasoning outputs, visual results, and key takeaways

This activity enables learners to bridge theoretical knowledge of game theory and AI with practical applications, enhancing their ability to analyze decision-making, implement intelligent systems, and evaluate strategies in uncertain environments.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	An Introduction to Game Theory	1 st Edition	Martin Osborne	Oxford University Press	2003
2	Artificial Intelligence: A Modern Approach	3 rd Edition	Stuart Russell, Peter Norvig	Prentice Hall	2015
3	Introduction to Machine Learning	2 nd Edition	Ethem Alpaydin	MIT Press	2010

**Reference Books**

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Game Theory	Revised Edition	Thomas Ferguson	World Scientific	2018
2	Introduction to Game Theory	1 st Edition	Stef Tijs	Hindustan Book Agency	2003

Online References

Sr. No	Website Name
1	https://www.javatpoint.com/automata-and-game-theory – Javatpoint: Game Theory and Automata Overview
2	https://www.analyticsvidhya.com/blog/2019/11/game-theory-ai/ – Analytics Vidhya: Game Theory in Artificial Intelligence
3	https://www.kdnuggets.com/2020/03/crash-course-game-theory-machine-learning.html – KDnuggets: Game Theory and Machine Learning Crash Course



Mahavir Education Trust's

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Semester VII

Program	Honours Degree in Artificial Intelligence and Machine Learning - Semester: VII	CL	TL	LL	SL	C
Course Name: AI & ML in Healthcare	Course Code: HNAMECR1701	4	1	4	3	6
Pre-requisite:	1. Mathematics for AI & ML, 2. Game Theory using AI & ML,					

I RATIONALE

The integration of Artificial Intelligence (AI) and Machine Learning (ML) into the healthcare domain has transformed the way medical data is analyzed, diagnoses are predicted, and treatments are optimized. This course is designed to equip learners with the essential theoretical knowledge and practical skills required to solve real-world healthcare challenges using AI/ML techniques. By focusing on data-driven decision-making, ethical considerations, and hands-on applications like Natural Language Processing (NLP) and algorithm design, the course prepares learners for roles that require both domain knowledge and technical expertise. The curriculum is aligned with the current demands of the healthcare industry and emphasizes interdisciplinary collaboration and ethical responsibility in the deployment of AI solutions.

II COMPETENCY

- Apply machine learning models and advanced algorithms to address practical healthcare challenges such as disease prediction, personalized medicine, and medical image analysis.
- Utilize NLP tools and frameworks to extract actionable insights from unstructured medical text such as clinical notes, prescriptions, and EHRs, facilitating informed medical decisions.
- Collaborate effectively in interdisciplinary teams involving data scientists, clinicians, and software developers to build scalable and ethically responsible healthcare solutions using AI and ML.

III COURSE OUTCOMES (COs)

After the completion of course-based learning, learners will be able to:

Course Outcome Number	Course Outcome Statement
1	Evaluate the role of AI and ML in handling healthcare data while organizing and managing datasets with attention to ethical considerations and data privacy.
2	Apply AI techniques for medical imaging analysis and disease detection, including cancer diagnosis, diabetic retinopathy screening, tumor segmentation, and outbreak prediction.
3	Design advanced AI algorithms to healthcare problems using appropriate techniques and tools.
4	Assess various NLP techniques used for healthcare applications using suitable methods on real-world datasets.
5	Develop intelligent telemedicine and EHR-based solutions to support disease information retrieval, recommendations, and continuous patient monitoring.
6	Evaluate emerging AI and ML technologies, including IoMT, digital health, blockchain, AR/VR, and smart hospital solutions, through real-world healthcare case studies.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme					Credits	Assessment Scheme													
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Theory						Practical				SLA		Total	
			CL	TL	LL				Exam	CIA		ESE	Total		CIAP	ESEP	Total					
										Duration in Hrs	MSE		CCE	Max			Min	Max				Min
						Max	Min												Max	Min		
1	AI & ML in Healthcare	HNAMCR1701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225	

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Collect, clean, and preprocess healthcare data for use in AI/ML models.	Collect, Clean, Integrate and Transform Healthcare Data based on specific disease.	CO1
2	Perform exploratory data analysis to uncover patterns and relationships in healthcare datasets.	Perform Exploratory Data Analysis of Healthcare Data.	CO2
3	Apply computer vision-based AI techniques for disease diagnosis using medical imaging data.	AI for Medical Diagnosis based on MRI/X-ray Data.	CO3
4	Implement ML models for medical prognosis to predict treatment outcomes or disease progression.	AI for Medical Prognosis.	CO3
5	Extract medical entities and structured information from unstructured text using NLP.	Natural Language Entity Extraction from Medical Reports.	CO4
6	Train and evaluate ML models to predict disease risks using structured patient data.	Predict Disease Risk from Patient Data.	CO4
7	Analyze patient reviews and sentiments from social media platforms for public health insights.	Medical Reviews Analysis from Social Media Data.	CO5
8	Interpret AI models and visualize decisions for transparency and accountability in healthcare.	Explainable AI in Healthcare for Model Interpretation.	CO6

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Analyze the multifaceted applications of AI and ML in healthcare, including diagnosis, personalized treatment, drug discovery, and follow-up care. 2. Evaluate metrics, model parameters, and hyperparameter tuning techniques including multivariate testing and federated learning. 3. Identify data mining and abstract learning methods applicable to healthcare scenarios.	Overview of AI and ML in Healthcare : 1.1 Overview of AI and ML in Healthcare, A Multifaceted Discipline, Applications of AI in Healthcare - Prediction, Diagnosis, personalized treatment and behavior modification, drug discovery, follow-up care etc., Realizing potential of AI and ML in healthcare, Healthcare Data - Use Cases. 1.2 Evaluating learning for Intelligence Model development and workflow, evaluation metrics, Parameters and Hyperparameters, Hyperparameter tuning algorithms, multivariate testing, Concept of federated learning. 1.3 AI, ML, Deep Learning and Data Mining Methods for Healthcare, Knowledge discovery and Data Mining, Multi classifier Decision Fusion, Ensemble Learning, Meta-Learning and other Abstract Methods. Evolutionary Algorithms, Computational Intelligence Techniques, Unsupervised learning.	14
2	CO2 & CO3	1. Analyze AI applications in medical imaging and diagnosis systems such as diabetic retinopathy and brain tumor segmentation. 2. Interpret AI applications in outbreak prediction and automated screening systems. 3. Evaluate reinforcement learning use in treatment planning and healthcare deployment.	Illustrative Medical Application : 2.1 Introduction to Medical Imaging - X-ray, CT, MRI, and PET scans, radiology, Medical image processing in cancer diagnosis, diabetic Retinopathy detection, Tumor Segmentation in Brain MRI, Multiagent Infectious Disease, Propagation and Outbreak Prediction, Automated Amblyopia Screening System etc. 2.2 Reinforcement Learning in Treatment Planning, Design and deployment of AI solutions in health care.	16

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO4 & CO5	- Understand low-level and high-level NLP components and their role in healthcare tasks. - Apply NLP models using real-world tools and interpret model decisions with Explainable AI. - Demonstrate knowledge of Telemedicine concepts and Electronic Health Records (EHR) integration for guided health recommendations.	Natural Language Processing in Healthcare : 3.1 NLP tasks in Medicine, Low-level NLP components, High level NLP components, NLP Methods. Web-Scraping approaches in Healthcare 3.2 Clinical NLP resources and Tools, NLP Applications in Healthcare. Model Interpretability using Explainable AI for NLP applications 3.3 Definition and Scope of Telemedicine, Telehealth vs. Telemedicine, Telemedicine Technologies, health-insurance claims processing 3.4 Intelligent personal Health Record Introduction, Key Components of EHR (Electronic Health Record), Guided Search for Disease Information, Recommending SCA's. Recommending HHP's, Continuous User Monitoring	16
4	CO6	- Identify and explore emerging trends like connected medicine, AR/VR, blockchain, and smart hospitals. - Analyze case studies of AI in disease risk prediction, robot-assisted surgeries, and junior doctor training. - Understand the role of IoMT and cloud services in next-generation healthcare ecosystems.	Future of Healthcare using AI and ML : 4.1 Evidence based medicine, Personalized Medicine, Connected Medicine, Digital Health and Therapeutics, Conversational AI, Virtual and Augmented Reality, Blockchain for verifying supply chain, patient record access, Robot - Assisted Surgery, Smart Hospitals, Internet of Medical Things (IoMT) 4.2 Case Studies on use of AI and ML for Disease Risk Diagnosis from patient data, Augmented reality applications for Junior doctors. Blockchain for verifying supply chain, patient record access, Robot - Assisted Surgery 4.3 Smart Hospitals, cloud services in healthcare. Case Studies on use of AI and ML for Disease Risk Diagnosis from patient data, Augmented reality applications for Junior doctors	14

VII SUGGESTED SELF LEARNING ACTIVITIES

Self-Learning Activity: AI-Driven Healthcare Innovation – A Domain-Focused Study

Learners are required to independently identify a healthcare problem area (e.g., disease prediction, medical image classification, EHR data analysis, remote monitoring, etc.) and work on a semester-long AI/ML-driven investigation aligned with real-world healthcare use cases. The activity is divided into four progressive phases to build competencies in data preparation, model development, evaluation, and communication.

Phase 1: Domain Understanding and Data Exploration (Weeks 1–3)

Learners will select a problem statement and locate an appropriate dataset (e.g., public health datasets, WHO/CDC data, Kaggle medical datasets). They will explore the dataset through data visualization, outlier detection, missing value handling, and summarize key patterns in the data.

Phase 2: AI/ML Model Development (Weeks 4–6)

Learners will apply suitable ML or DL models such as logistic regression, decision trees, CNN (for images), or NLP-based models (for medical text). Model building will involve data splitting, training, testing, and hyperparameter tuning. At the end of this phase, a checkpoint report will be submitted documenting methodology and initial results.

Phase 3: Explainability and Ethical Review (Weeks 7–9)

Learners will analyze the interpretability of the AI model using techniques like SHAP, LIME, or Explainable AI APIs. A critical review of the ethical and privacy implications (bias, fairness, data confidentiality) of the developed model will be included. Learners will document their understanding of ethical AI in healthcare.

Phase 4: Outcome Communication and Deployment Insights (Weeks 10–13)

Learners will generate outcome dashboards or visual summaries (using tools like Streamlit/Plotly/Power BI) and reflect on how the developed model can assist medical professionals or decision-makers. If applicable, learners may simulate the deployment plan (API endpoints, edge AI, mobile health insights, etc.).

Deliverables:

- **Initial Checkpoint Report (Week 4)** – Problem Identification Design Planning
- **Mid-Semester Checkpoint Report (Week 8)** – Dataset description, domain mapping, model architecture, training insights
- **Final Report and Demonstration (Week 14)** – End-to-end documentation, model performance, ethical analysis, interpretable visuals, and future enhancement suggestions

This self-learning activity enables learners to bridge technical AI/ML skills with domain knowledge in healthcare, nurturing self-directed learning, model-building proficiency, ethical AI awareness, and interdisciplinary problem solving.

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Machine Learning and AI for Healthcare	1 st Edition	Arjun Panesar	A Press	2019
2	Medical Applications of Artificial Systems	1 st Edition	Arvin Agah	CRC Press	2005

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Artificial Intelligence in Medical Imaging: Opportunities, Applications and Risks	1 st Edition	Erik R. Ranschaert, Sergey Morozov, Paul R. Algra	Springer	2019
2	Emerging Technologies for Health and Medicine: Virtual Reality, Augmented Reality, Artificial Intelligence, Internet of Things, Robotics, Industry 4.0	1 st Edition	Dac-Nhuong Le, Chung Van Le, Jolanda G. Tromp, Gia Nhu Nguyen	IGI Global	2018
3	Artificial Intelligence in Healthcare: Review and Prediction Case Studies	1 st Edition	Guoguang Rong, Arnaldo Mendez, Elie Bou Assi, Bo Zhao, Mohamad Sawan	Engineering Journal (Elsevier)	2020

Online References

Sr. No.	Website Name and Link
1	https://ocw.mit.edu/courses/6-s897-machine-learning-for-healthcare-spring-2019/pages/lecture-notes/ – MIT OCW: Machine Learning for Healthcare
2	https://www.coursera.org/learn/introduction-tensorflow?specialization=tensorflow-in-practice – Coursera: Introduction to TensorFlow
3	https://www.coursera.org/learn/convolutional-neural-networks-tensorflow?specialization=tensorflow-in-practice – Coursera: CNN with TensorFlow
4	https://datarade.ai/data-categories/electronic-health-record-ehr-data – Datarade: EHR Datasets
5	https://www.cms.gov/Medicare/E-Health/EHealthRecords – CMS.gov: Medicare E-Health Records



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Semester VIII

Program	Honours Degree in Artificial Intelligence and Machine Learning - Semester: VIII	CL	TL	LL	SL	C
Course Name: Natural Language Processing and Generative AI	Course Code: HNAMECR1801	4	1	–	3	4
Pre-requisite:	1. Mathematics for AI & ML, 2. Game Theory using AI & ML, 3. AI & ML in Healthcare,					

I RATIONALE

This course enables learners to gain a comprehensive understanding of Natural Language Processing (NLP) and its intersection with Generative AI. NLP plays a pivotal role in extracting insights from unstructured textual data. By integrating generative models such as GANs and Transformers, learners will be equipped with the tools to design intelligent systems capable of creating meaningful and coherent text. Applications span domains like healthcare, social media, and customer experience, making this subject essential for solving real-world problems using advanced AI techniques.

II COMPETENCY

Develop the ability to understand, analyze, and implement Natural Language Processing and Generative AI techniques for extracting, interpreting, and generating human language data, fostering intelligent and context-aware AI applications.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Analyze fundamental NLP concepts with core theories, models, and appreciating language diversity in human communication.
2	Apply clustering, classification, and probabilistic modeling techniques to unstructured text datasets, evaluating model performance for domain-specific applications.
3	Explore real-world problems using NLP techniques for text data analysis, coding solutions using NLP libraries, and collaborating responsibly within teams.
4	Design generative models for synthetic data by conceptualizing deep learning architectures, working prototypes, and valuing data ethics and originality.
5	Investigate aspect-based opinion mining by contextual sentiment, model training and evaluation, and showing awareness of biases in data interpretation.
6	Develop solutions integrating NLP, generative AI, and ethical web/social media data practices to address real-world challenges across diverse domains.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme				Credits	Assessment Scheme														
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL		Total	Exam	Theory				Practical				SLA		Total		
			CL	TL	LL					Duration in Hrs	CIA		ESE	Total		CIAP	ESEP				Total	
											MSE	CCE		Max	Min						Max	Min
1	Natural Language Processing and Generative AI	HNAMCR1801	4	1	–	3	120	4	2	20	20	60	100	40	–	–	–	–	25	10	125	

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1 & CO3 & CO6	1. Discuss foundational principles of text mining 2. Recognize key text representation techniques 3. Identify methods in information extraction	Introduction to Text Mining : 1.1 Algorithms for Text Mining, Future Directions 1.2 Information Extraction: NER, POS Tagging, Relation Extraction, Unsupervised IE 1.3 Text Representation: Word2Vec, GloVe, FastText, Tokenization, Stemming, Stop Words, N-gram Modelling	15
2	CO2 & CO3 & CO6	1. Analyze methods for clustering and classification in NLP 2. Compare various classifiers and modeling techniques 3. Apply clustering and classification to text datasets	Text Clustering, Classification and Modeling : 2.1 Text Clustering: Feature Selection, Distance-based and Probabilistic Clustering 2.2 Text Classification: Decision Tree, Rule-based, Probabilistic and Proximity-based Classifiers 2.3 Text Modelling: Bayesian Networks, HMM, MRF, CRF	15
3	CO4 & CO6	1. Explain generative models and their application in NLP 2. Explore architectures like GANs, VAEs, Transformers 3. Apply generative techniques for text generation	Generative Models : 3.1 Introduction to GANs and VAEs 3.2 Text Generation using GPT, Transformer Architecture, Fine-tuning 3.3 Applications of Generative AI	15

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5 & CO6	1. Examine techniques in sentiment and opinion mining 2. Investigate web-based data extraction and analysis 3. Analyze social network data and detect opinion spam	Social Network Analysis : 4.1 Web Scraping, Meta Search, Web Spamming, Legal & Ethical Issues 4.2 Sentiment Analysis and Opinion Mining from Social Media 4.3 Opinion Spam Detection, Basic Search Engine, Social Network Analysis	15

VI SUGGESTED SELF LEARNING ACTIVITIES

Self-Learning Activity: Real-World Problem Solving Using NLP and Generative AI

Learners shall explore and implement real-world problem-solving applications using Natural Language Processing and Generative AI techniques learned throughout the course. They will identify a domain-specific problem—such as automated feedback analysis, resume screening, fake review detection, news classification, chatbot development, or sentiment analysis—and apply appropriate concepts including text mining, word embeddings, clustering/classification, or generative models like GPT to design a working prototype or case study. The activity encourages integration of web scraping, social media analysis, and ethical considerations to build meaningful AI-powered applications that address challenges in domains such as healthcare, business intelligence, e-governance, and digital marketing. Learners are expected to document the problem statement, applied techniques, results, and reflections as part of their final submission.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mining Text Data	1 st Edition	Charu C. Aggarwal, Cheng Xiang Zhai	Springer	2012
2	Speech and Language Processing	3 rd Edition	Dan Jurafsky, James H. Martin	Pearson	2019
3	NLP in Action	1 st Edition	Lane, Howard, Hapke	Manning Publications	2019
4	Foundations of Statistical NLP	1 st Edition	Christopher D. Manning, Hinrich Schütze	MIT Press	1999
5	Deep Learning for NLP	1 st Edition	Palash Goyal, Sumit Pandey, Karan Jain	Packt	2018
6	Generative Deep Learning	1 st Edition	David Foster	O'Reilly Media	2019
7	NLP with Python and NLTK	1 st Edition	Nitin Hardeniya	Apress	2017

**Reference Books**

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Speech and Language Processing	3 rd Edition	Daniel Jurafsky, James H. Martin, James R. Glass	Pearson	2021
2	NLP with Python	1 st Edition	Steven Bird, Ewan Klein, Edward Loper	O'Reilly	2009

VIII Online References

Sr. No	Website Name
1	https://www.nltk.org/ – NLTK Documentation
2	https://spacy.io/ – spaCy Documentation
3	https://platform.openai.com/docs/introduction – OpenAI GPT Docs
4	https://www.tensorflow.org/tutorials/keras/text_classification – TensorFlow Text Guide



Mahavir Education Trust's

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KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai

Syllabus of Honours Degree Programs in Technology

(with effect from 2025-26)

Augmented Reality & Virtual Reality



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CURRICULUM STRUCTURE UG:B.Tech.

Honours/Minors Degree Program

Augmented Reality and Virtual Reality

REVISION: R1-V0-2025-26

Effective from Academic Year 2025-26 onwards

Approved by Board of Studies on 17/06/2025

Approved by Academic Council on 19/06/2025

Dr. Pravin Shinde
I/C HOD



Dr. Bhavesh Patel
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Chairman BoS
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Preamble

In an era where technological disruptions occur at an unprecedented pace, academic autonomy is a significant advantage for the Engineering Institute. It allows Institutions to swiftly adapt and innovate in their curriculum to meet the evolving demands of society and industry. This newly designed syllabus for the Honors Degree in AR-VR Engineering at Shah and Anchor Kutchhi Engineering College, Mumbai exemplifies the benefits of such autonomy. Starting at semester V the students will go through the well designed courses for four semesters transforming them from a curious learner to proficient AR-VR developer. Following are a few features of the honours degree course in AR-VR:

1. Strong Foundations: The curriculum is meticulously crafted to ensure that students build a robust foundation in mathematical and scientific principles. This foundation is crucial as it equips students with the essential knowledge and analytical skills required to understand and innovate in the rapidly advancing field of AR-VR technology.

2. Industry-Relevant Skills: The syllabus is designed with a keen focus on the latest technologies and industry trends. By integrating contemporary topics and hands-on projects, we prepare our students to seamlessly transition from academic learning to professional practice. The theoretical subjects provide deep insights into foundational and advanced concepts, while the practical sessions in the AR-VR Application Development Lab enable students to apply these concepts in real-world scenarios.

3. Skill Development: Practical sessions are integral to our curriculum, ensuring that students do not just learn about AR-VR technologies but also develop the skills to implement and innovate with them. This hands-on experience is invaluable, providing students with the competence and confidence needed to excel in the industry.

4. Clear Objectives and Outcomes: Our syllabus is designed with clearly defined course objectives and outcomes, providing students with a transparent and structured learning path. This clarity helps students understand the specific skills and knowledge they will acquire throughout the course, allowing them to focus their efforts and measure their progress effectively.

In conclusion, the Honors Degree in AR-VR Engineering at our Institute is a comprehensive program that combines strong theoretical foundations with practical skill development. It prepares students to become proficient and innovative professionals in the AR-VR industry, ready to meet the challenges of a rapidly evolving technological landscape.



Program Structure for Honours Degree In Augmented Reality and Virtual Reality (with effect from 2025–2026)

			Learning Scheme					Credits	Assessment Scheme												
Semester	Course Code	Course Title	Actual Contact Hrs/Week			SL	Total Notional Hours			Theory					Practical				SLA		Total
									Paper Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total				
			MSE	CCE	Max					Max	Max										
			SEM V	HNAVCR1501	Foundations of AR-VR					4	1		-	3	120	4	2	20	20	60	
SEM VI	HNAVCR1601	AR & Mixed Reality	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125
SEM VII	HNAVCR1701	AR-VR Application Development	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225
SEM VIII	HNAVCR1801	Game Development with VR	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125
Total			16	4	4	12	540	18		80	80	240	400		40	60	100		100		600

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
 CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
 SLA: Self Learning Assessment

Note:

1. Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
2. 1 credit is equivalent to 30 Notional hrs.
3. Self learning hours shall not be reflected in the Time Table.
4. SLA will have separate head of passing



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Semester V



Program	Honours Degree in Augmented Reality and Virtual Reality(Semester V)	CL	TL	LL	SL	C
Course Name: Foundations of AR-VR	Course Code: HNAVCR1501	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Computer Graphics.					

I RATIONALE

Foundations of AR-VR is a strategic investment in skills that are becoming increasingly vital. It's not just about understanding gadgets; it's about grasping the next major shift in how we compute, interact, and experience the world around us.

II COMPETENCY

Learners will gain the conceptual and foundational technical knowledge necessary to confidently engage with, discuss, design, and even begin to implement immersive and mixed reality experiences. Students will not just know what AR/VR is, but how it fundamentally works.

III COURSE OUTCOMES (COs)

After the completion of course based learning, Learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply AR and VR system architecture and hardware principles to immersive environment development.
2	Analyze the influence of human sensory systems and psychological factors on AR/VR experience design.
3	Evaluate user interfaces in AR/VR systems using HCI principles and standards.
4	Create interactive AR/VR models using 2D and 3D transformations and computer graphics techniques.
5	Organize insights on emerging trends and startup opportunities in AR/VR through collaborative exploration.
6	Characterize ethical and innovative applications of AR/VR technologies in real-world domains.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Theory					Practical				SLA		Total	
				CL	TL	LL				Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total				
											MSE	CCE		Max	Min			Max	Min	Max		Min
1	Foundations of AR-VR	-	HNAVCR1501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO2	1.1 Articulate the fundamental hardware components and their functions associated with typical AR, MR, and VR systems (e.g., displays, sensors, input devices). 1.2 Explain the major driving forces (e.g., computing power, display technology, sensor miniaturization) behind the progression of immersive technologies. 1.3 Explain the core components and their interconnections within a typical Virtual Reality (VR) system architecture (e.g., input processing, rendering pipeline, display management). 1.4 Articulate diverse applications of AR, MR, and VR technologies across various industry sectors (e.g., healthcare, education, manufacturing, entertainment). 1.5 Relate principles of human visual perception to the design and potential challenges (e.g., vergence-accommodation conflict, motion sickness) encountered in immersive VR experiences. 1.6 Justify the importance of spatial audio in creating realistic and immersive AR-VR experiences, by linking it to principles of human auditory processing.	Introduction to AR and VR 1.1 Introduction to AR and VR : AR-VR technology spectrum, comparison of the technologies in this spectrum, hardware associated with them 1.2 Evolution of AR-VR technologies 1.3 VR system architecture 1.4 Applications. 1.5 Sensory organs of human beings and psychology of VR : Physiology of human vision, how human vision works 1.6 Audio, how auditory perception works.	18

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO4	2.1 Explain the fundamental concepts and common applications of computer graphics and 3D modeling in the context of digital content creation. 2.2 Apply basic (translation, rotation, scaling) and composite (sequence of basic) 2D and 3D transformations using homogeneous coordinates to manipulate virtual objects in a defined coordinate space. 2.3 Explain the fundamental concepts of the viewing pipeline in 3D computer graphics, including projection (e.g., orthographic, perspective) and viewport transformations. Also Determine which geometric primitives will be visible within a specified viewing frustum by applying 2D and 3D clipping algorithms.	Computer Graphics and 3D Modelling 2.1 Computer Graphics and 3D Modelling: Introduction 2.2 Homogeneous coordinate system, 2D and 3D basic and composite transformations 2.3 Viewing and clipping.	20
3	CO3	3.1 Identify and apply basic principles of effective user interface (UI) design (e.g., consistency, feedback, learnability, efficiency, error prevention) to critique simple interface examples. 3.2 Select and utilize appropriate user interface evaluation techniques (e.g., usability testing, heuristic evaluation, surveys) for given interface design problems.	Human-computer interaction 3.1 Human-computer interaction : Fundamentals of HCI, such as human factors, user interface design 3.2 User interface software and tools, user interface evaluation, and user interface standards.	12

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5, CO6	4.1 Categorize given real-world AR-VR applications (e.g., specific mobile AR games, business tools, WebAR campaigns) by their dominant trend (e.g., Mobile AR, AI-powered AR-VR) and justify this categorization based on their features. 4.2 Propose a novel AR-VR application idea for a chosen industry (e.g., tourism, healthcare, or retail) by integrating knowledge of relevant AR-VR technologies and market needs. 4.3 Analyze a given scenario for a Metaverse startup and determine how blockchain could be leveraged to enhance its features or address specific trust and security requirements.	Latest Trends, Startup and research Opportunities in AR-VR 4.1 Latest Trends, Startup and research Opportunities in AR-VR : AI powered AR-VR, Metaverse, Mobile AR, - Games, Business Tools, WebAR, Wearable AR, Teleportation. 4.2 Research and Startup Opportunities : Tourism, VR Concerts and festivals, Interactive Theater, Theme parks, Immersive Cinema. VR Social Casinos, AR Game Studios, AR tournaments and events, Virtual Showrooms, Real Estate Visualization, Telemedicine Platforms, Physical therapy Apps, AR Interior Design Tools Architectural Visualisations, AR Educational Apps, Business Opportunities in Metaverse. 4.3 Blockchain : Introduction to blockchain. Using blockchain for Metaverse related startups.	10

VI SUGGESTED SELF LEARNING ACTIVITIES

Sr. No	Activity Title	Description	CO(s)
1	Hardware Identification	Find images of different AR and VR headsets/devices and label their basic components (e.g., display type, camera, sensors, controller).	CO1
2	Use Case Justification	For 3-5 different industries (e.g., healthcare, education, retail), explain why AR or VR is a suitable technology for specific applications within that field (e.g., "VR is good for surgical training because..." or "AR is beneficial for retail because...").	CO2
3	Interaction Prototype (Low-Fidelity)	Choose an AR/VR interaction: (e.g., selecting a virtual object, navigating a virtual space, drawing in AR). Task: Using simple drawings, flowcharts, or even basic presentation slides, design a low-fidelity "prototype" of how a user would perform this interaction. Annotate your design with explanations of how human perception/cognition influenced your choices.	CO3
4	Emerging Trend Impact Assessment	Choose one "Latest Trend" from your course content: (e.g., AI-powered AR/VR, Metaverse, WebAR). Analyze its Foundational Impact: How does this trend fundamentally change or leverage the core foundational concepts of AR/VR (e.g., how does AI impact tracking or content generation? How does WebAR challenge traditional deployment models?) Discuss both opportunities and new foundational challenges.	CO4

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Computer Graphics- C version	2 nd Edition	Hearn and Baker	Pearson	2002
2	Augmented Reality- Principles and Practice	2016 Edition	Dieter Schmalstieg and Tobias Hollerer	TPearson Education, Inc.	2016
3	Computer Graphics with Virtual Reality	3 rd Edition	R. K Maurya	Wiley India	2018
4	Virtual Reality	-	Steven M. LaVelle	Cambridge University press	2019
4	Virtual Reality Systems	1 st Edition	Vince	Pearson Education	2002

Reference Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Foundations of Sensation and Perception	3 rd Edition	George Mather	Psychology Press book	2016
2	Learning Virtual Reality	1 st Edition	Tony Parisi	O'Reilly	2015
3	Understanding virtual reality: Interface, application and design	2 nd Edition	Alan Craig and William Sherman	Morgan Kaufmann Publisher	2019
4	Fundamentals of Computer Graphics	4 th Edition	Peter Shirley, Michael Ashikhmin, and Steve Marschner	A K Peters/CRC Press	2016

Online References

Sr. No	Website Name
1	https://www.coursera.org/learn/what-is-the-metaverse
2	https://www.coursera.org/learn/ar



Mahavir Education Trust's

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Semester VI



Program	Honours Degree in Augmented Reality and Virtual Reality(Semester VI)	CL	TL	LL	SL	C
Course Name: AR & Mixed Reality	Course Code: HNAVCR1601	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Computer Graphics.					

I RATIONALE

To introduce learners to immersive technologies by exploring the principles, tools, and applications of AR and Mixed Reality across various domains.

II COMPETENCY

Learners will be able to design, analyze, and evaluate AR/MR systems using technical and user-centered approaches for real-world problem-solving.

III COURSE OUTCOMES (COs)

After the completion of course based learning, Learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply the architecture and principles of AR and MR systems to real-world applications.
2	Analyze tracking, registration, and calibration techniques used in AR/MR environments.
3	Evaluate software architectures for AR applications based on system and user requirements.
4	Create models and visual elements for AR/MR environments using appropriate techniques.
5	Organize annotation and authoring strategies for effective AR content development.
6	Characterize the evolution and impact of AR/MR technologies across various domains.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Assessment Scheme																			
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Credits	Exam	Theory					Practical			SLA		Total					
				CL	TL	LL						Duration in Hrs	CIA		ESE	Total			CIAP				ESEP	Total			
													MSE	CCE		Max	Max	Min						Max	Max	Max	Min
1	AR & Mixed Reality	-	HNAVCR1601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125					

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO6	1.1 Understand definitions, evolution, architecture, and applications of AR/MR.	Introduction to AR & Mix Reality and Mobile AR 1.1 Introduction to AR & Mix Reality and Mobile AR : Definitions of AR and MR, evolution of AR and MR technologies, architecture of the AR system, applications.	10
2	CO2	2.1 Analyze tracking technologies and techniques used in AR/MR systems. 2.2 Explain registration concepts and techniques for aligning virtual and real-world elements. 2.3 Apply calibration methods to enhance accuracy in AR/MR systems.	Tracking, Registration and Calibration 2.1 Tracking, Registration and Calibration : Concept of tracking, characteristics of tracking technology, navigation, stationary tracking system, tracking techniques. 2.2 Registration : Concept, techniques. 2.3 Calibration : Concept, techniques.	20
3	CO3	3.1 Evaluate software architectures and frameworks for AR application development.	Software Architectures 3.1 Software Architectures : AR application requirements, software engineering requirements, distributed object system, dataflow, scene graph.	10

(Continued on next page)



Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4, CO5	4.1 Create geometric and appearance models for AR/MR environments. 4.2 Organize annotation strategies to enhance user interaction and information delivery. 4.3 Apply authoring tools and techniques for AR content creation.	Modelling, Annotation and Authoring 4.1 Modelling : Concept, need of modelling, specifying Geometry, specifying appearance, semi-automatic reconstruction, free-for modelling. 4.2 Annotation : Concept, need of annotation. 4.3 Authoring : Requirements of AR authoring, its elements, stand alone authoring solution, plug-in approach.	20

VI SUGGESTED SELF LEARNING ACTIVITIES

Sr. No	Activity Title	Description	CO(s)
1	Explore AR/MR Applications	Research and document real-world applications of AR/MR in fields like healthcare, education, and gaming.	CO1, CO6
2	Implement a Simple AR App	Use tools like Unity and Vuforia to build a basic AR application.	CO1
3	Analyze Tracking Techniques	Compare different tracking methods (optical, GPS, inertial) and their use cases.	CO2
4	Evaluate AR Software Frameworks	Study and critique AR SDKs like ARCore, ARKit, and their architectural models.	CO3
5	Create a 3D Model for AR	Design and render a 3D object using Blender or similar tools for AR integration.	CO4
6	Develop AR Content with Authoring Tools	Use authoring platforms to create annotated AR experiences.	CO5
7	Reflect on AR/MR Evolution	Write a short essay on the historical development and future potential of AR/MR	CO6

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Computer Graphics- C version	2 nd Edition	Hearn and Baker	Pearson	2002
2	Augmented Reality- Principles and Practice	2016 Edition	Dieter Schmalstieg and Tobias Hollerer	TPearson Education, Inc.	2016
3	Computer Graphics with Virtual Reality	3 rd Edition	R. K Maurya	Wiley India	2018
4	Virtual Reality	-	Steven M. LaVelle	Cambridge University press	2019
4	Virtual Reality Systems	1 st Edition	Vince	Pearson Education	2002

**Reference Books**

Sr. No	Title	Edition	Authors	Publisher	Year
1	Foundations of Sensation and Perception	3 rd Edition	George Mather	Psychology Press book	2016
2	Learning Virtual Reality	1 st Edition	Tony Parisi	O'Reilly	2015
3	Understanding virtual reality: Interface, application and design	2 nd Edition	Alan Craig and William Sherman	Morgan Kaufmann Publisher	2019
4	Fundamentals of Computer Graphics	4 th Edition	Peter Shirley, Michael Ashikhmin, and Steve Marschner	A K Peters/CRC Press	2016

Online References

Sr. No	Website Name
1	https://www.coursera.org/learn/augmented-reality
2	https://www.freecodecamp.org/news/augmented-reality-full-course/
3	https://alison.com/tag/augmented-reality



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Semester VII



Program	Honors Degree in Augmented Reality and Virtual Reality (Semester VII)	CL	TL	LL	SL	C
Course Name: AR-VR Application Development	Course Code: HNAVCR1701	4	1	4	3	6
Course Type:	-					
Pre-requisite:	Foundations of AR-VR					

I RATIONALE

This course bridges the gap between conceptual design and functional AR-VR systems by focusing on application development using industry-standard tools. Positioned after foundational and design-focused courses, it enables students to prototype, develop, and test immersive AR-VR applications. The integration of practical sessions equips learners with hands-on experience in interaction design, UI development, and hardware interfacing. This subject plays a pivotal role in the preparation of students for professional roles in the immersive technology industry.

II COMPETENCY

After completing this course, learners will be competent in designing and developing functional AR-VR applications by applying software engineering principles, prototyping techniques, and interaction design. They will gain the ability to integrate hardware and software components, build intuitive user interfaces, and address real-world challenges in immersive environments. This competency will enable them to contribute effectively to the AR-VR industry and innovation-driven projects.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Design software architecture for AR-VR applications incorporating systematic requirement analysis and structured development principles.
2	Construct interactive AR-VR prototypes using industry-standard tools and development frameworks.
3	Evaluate user interaction mechanisms and interfaces for AR-VR applications through systematic usability testing.
4	Analyze AR-VR application challenges considering technical feasibility, ethical implications, and societal impact.
5	Integrate hardware and software components to develop functional AR-VR solutions aligning with specified design requirements.
6	Demonstrate proficiency in deploying AR-VR application.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory					Practical				SLA	Total	
				CL	TL	LL					CIA	CCE	ESE	Total	CIAP	ESEP	Total					
																		Max	Max			Max
1	AR-VR Application Development	-	HNAVCR1701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS
 (Minimum 8)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Describe components, setup, and usage of Meta Quest 2 VR headset.	Introduction to VR Hardware (Meta Quest 2):a. Understanding the Meta Quest 2 headset. b. Explain its components, controls, and setup process. c. Demonstrate how to wear and adjust the headset.	CO1
2	Install and configure Unity for AR-VR development.	Unity Installation and Initial Setup	CO1
3	Create and manipulate 3D objects in a Unity scene.	Creating a Simple 3D Scene:a. Add basic 3D objects (e.g., cubes, spheres) to the scene. b. Manipulate their position, rotation, and scale. c. Apply materials and textures	CO2
4	Navigate Unity toolbars and menus effectively for project development.	Understanding Unity Toolbars and Menus:a. Explore the main Unity toolbar (File, Edit, Assets, GameObject, etc.). b .Explain common menu options (e.g., File-Save,Edit-Undo). c. Introduce keyboard shortcuts	CO2
5	Build a basic VR environment with terrain, lighting, and skybox.	Building a VR Environment:a. Set up a basic VR environment (e.g., room, landscape). b. Add terrain, lighting, and skyboxes. c. Test the scene in VR mode	CO2
6	Implement object interactions in VR using XR Interaction Toolkit.	User Interaction with VR Objects:a. Implement basic interactions (grabbing, throwing) using Unity's XR Interaction Toolkit. b. Attach colliders and rigid bodies to objects. c. Create interactive buttons or switches.	CO2
7	Simulate physics in VR by applying Unity physics components.	Physics in VR a. Explore physics simulations (gravity, collisions) in VR. b. Create a simple physics-based game (e.g., rolling ball, falling blocks).	CO2, CO5
8	Design and implement basic UI elements suitable for VR interfaces.	UI Elements in VR:a. Design UI elements (buttons, sliders) for VR.b. Use Unity's Canvas system to create VR menus.c. Implement gaze-based interaction.	CO3
9	Integrate spatial audio elements to enhance immersive experiences.	Spatial Audio and Ambisonics: a. Introduce spatial audio principles. b. Add 3D sound sources to the scene. c. Discuss ambisonics for immersive audio.	CO4, CO 5
10	Develop basic scripts to control object behavior in Unity using C#.	Introduction to Scripting in Unity a. Create basic C scripts b. Attach scripts to objects for simple behaviors (e.g., rotation, movement). c. Explain the MonoBehaviour lifecycle.	CO3, CO 5

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Explain the architectural requirements of AR-VR applications. 2. Apply software engineering principles to identify and define system requirements for AR-VR applications. 3. Describe the role of distributed object systems in AR application development. 4. Analyze the flow of data in AR systems using data flow models. 5. Explain the use of scene graphs in structuring AR content.	Software Architectures in AR 1.1 AR-VR Application Requirement 1.2 Software Engineering Requirements 1.3 Distributed Objects System 1.4 Data Flow 1.5 Scene Graph	16
2	CO2, CO5	1. Explain the principles of AR and VR design. 2. Compare and contrast physical and digital prototyping in AR-VR application development. 3. Apply XR development approaches to prototype AR-VR experiences. 4. Describe the XR design process for developing immersive applications.	XR Application Design 2.1 UX design methodology 2.2 User and business requirements gathering 2.3 Mental models, wireframes, prototyping 2.4 Usability testing basics	16

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3	CO3, CO5, CO6	1. Describe different types of 3D interaction in VR environments. 2. Demonstrate navigation techniques in virtual spaces. 3. Implement object interaction techniques within VR environments. 4. Design graphical user interfaces (GUI) using VR tools. 5. Identify challenges and limitations in VR interaction design.	3D Interaction in VR 3.1 Types and processes of usability testing 3.2 Prototyping tools and usability test execution 3.3 Iterative feedback application 3.4 UX deliverables and implementation communication	16
4	CO4	1. Identify real-world applications of AR and MR technologies. 2. Discuss current issues faced in AR-VR application development. 3. Analyze major challenges in the deployment of AR-VR systems. 4. Evaluate AR-VR applications and assess their physical, ethical, and social implications.	Applications issues and Challenges 4.1 Applications of AR-MR 4.2 Issues in AR-VR development 4.3 Representation techniques for virtual environments 4.4 Challenges in AR-VR deployment 4.5 Physical, ethical, and social implications of AR-VR	12

VII SUGGESTED SELF LEARNING ACTIVITIES

Sr. No	Activity Title	Description	CO(s)
1	Mini/ Micro Project:e.g. AR Navigation System for Campus	Design and implement a simple AR app to help new students navigate campus buildings using marker-based AR. Document the process and challenges.	CO3, CO4
2	YouTube Video:e.g. Explain Marker-based vs Marker-less AR	Create an engaging explainer video comparing marker-based and marker-less AR, including visuals and real-world examples.	CO2
3	Comparative Study of AR-VR tools or technologies: e.g. Unity vs Unreal Engine for AR-VR	Conduct a detailed comparison between Unity and Unreal Engine based on features, usability, learning curve, and AR/VR capabilities.	CO4
4	Blog Post: e.g. Ethical Implications of AR in Healthcare or advanced applications of AR-VR like Metaverse.	Write a blog analyzing ethical concerns like privacy, addiction, and misinformation in AR applications used in the healthcare sector. Blog/Vlog post on Metaverse etc.	CO1,CO5
5	GitHub Repository: e.g. Interactive VR Museum Tour	Develop a VR app that simulates a museum tour experience and upload code, assets, and documentation to a GitHub repository.	CO3,CO4

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Computer Graphics- C version	2 nd Edition	Jesse James Garrett	Pearson	2002
2	Computer Graphics with Virtual Reality	3 rd Edition	R. K. Maurya	Wiley India	2019
3	Virtual Reality	-	Steven M. LaValle	Cambridge University Press	2012

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Foundations of Sensation and Perception	3 rd Edition	George Mather	Psychology Press	2016
2	Learning Virtual Reality	1 st Edition	Tony Parisi	O'Reilly	2015
3	Understanding Virtual Reality: Interface, Application and Design	-	Alan Craig, William Sherman	Morgan Kaufmann Publisher	2019
4	Fundamentals of Computer Graphics	4 th Edition	Peter Shirley, Michael Ashikhmin, Steve Marschner	A K Peters/CRC Press	2016

Online References

Sr. No	Website Name
A	A-Frame(aframe.io) , A web framework for building VR experiences using HTML. Great for beginners and web developers exploring WebVR/WebXR.)
B	https://www.youtube.com/playlist?list=PL_ezWOhnpakMojjJGm-YiCz5zr4GpuLG_ , Virtual Reality – Full Course Playlist by Steven Lavelle
C	https://www.freecodecamp.org/news/augmented-reality-full-course/ , A 12-hour YouTube course covering AR fundamentals, Unity, C, Vuforia, and even Spark AR for filters. Perfect for structured, hands-on learning.
D	BanubasARWebsiteShowcase , A curated list of interactive AR websites—great for inspiration and understanding real-world applications.
E	AppMasterBlogARinWebDevelopment , Offers insights into integrating AR into web apps, including use cases and tools like AR.js and WebXR



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Semester VIII



Program	Honors Degree in Augmented Reality and Virtual Reality (Semester VIII)	CL	TL	LL	SL	C
Course Name: Game Development with VR	Course Code: HNAVCR1801	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Foundations of AR-VR, AR-VR Application Development					

I RATIONALE

Game Development with VR serves as the capstone theory course in the Honors Degree in Augmented Reality and Virtual Reality, integrating and advancing the knowledge gained from the preceding subjects — Foundations of AR-VR, AR & Mixed Reality, and AR-VR Application Development. This subject equips learners with the theoretical understanding and design strategies required to conceptualize, architect, and optimize immersive VR game experiences using industry-standard tools and frameworks. Emphasizing software engineering principles, interaction design, and performance optimization, it prepares students to address the creative, technical, and user-experience challenges unique to VR game development. The course also fosters problem-solving skills, innovation, and readiness for roles in the gaming and interactive media industries, as well as applications of VR in training, education, and simulation.

II COMPETENCY

Learners will be able to design and develop comprehensive concepts, architectures, and optimization strategies for immersive VR games by applying software engineering principles, game design methodologies, and interaction techniques, enabling them to contribute effectively to the gaming, simulation, and interactive media industries.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply software engineering principles to plan and structure VR game development projects.
2	Design system architecture and interaction models for immersive VR games.
3	Implement core VR game components and mechanics using Unity features.
4	Develop scripts in C# to control game play functionality and user interaction.
5	Optimize VR game performance through effective resource management and technical enhancements.
6	Evaluate VR game designs for user experience, playability, and compliance with industry standards.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory				Practical				SLA		Total	
				CL	TL	LL					CIA	MSE	CCE	ESE	Total	CIAP	ESEP	Total				
																		Max				Min
1	Game Development with VR	-	HNAVCR1801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO2, CO6	1. Apply software engineering models and pipelines to structure VR game development (CO 1). 2. Identify functional, non-functional, and experiential requirements for immersive VR games. (CO 1). 3. Design a high-level architecture integrating game mechanics, interaction models, and asset workflows (CO2). 4. Document VR game concepts using Game Design Documents and technical specifications (CO1, CO2). 5. Analyze ethical, accessibility, and inclusivity factors in VR game design (CO6).	Software Engineering for VR Game Development 1.1 VR Game Development Process • Overview of VR game production pipeline • Iterative and agile models for game dev • Unity project setup for VR games 1.2 Requirement Gathering for VR Games • Identifying game goals, target audience, and genres • Player experience design considerations • Functional vs. non-functional requirements in VR 1.3 System and Asset Architecture • Core subsystems: input, physics, rendering, audio, AI, networking • Asset workflow: 3D models, animations, textures, audio • Version control in VR game projects (Git, Unity Collaborate) 1.4 Game Design Documentation (GDD) • Components storyline, mechanics, gameplay loop, UI flow • Technical specs for VR hardware & performance targets 1.5 Ethics & Accessibility in VR Games • Safety considerations (motion sickness, play area) • Inclusive design for diverse players • Regulatory and content rating systems	12

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2	CO2, CO3, CO6	1. Design gameplay loops, progression systems, and interaction models for VR games (CO2). 2. Implement interaction design principles for VR input devices and feedback systems (CO3). 3. Develop UI/UX concepts optimized for immersive VR environments (CO2, CO3). 4. Apply level design techniques to create engaging and navigable VR worlds (CO2). 5. Integrate narrative elements with game mechanics to enhance player engagement (CO2, CO6).	Design and Interaction Architecture for VR Games 2.1 Core Gameplay and Progression Systems • Understanding gameplay loops • Player progression: skills, challenges, rewards • Balancing difficulty in VR 2.2 VR Interaction Models • Motion controllers, gesture recognition, gaze tracking • Haptic feedback and spatial audio cues for immersion 2.3 UI/UX for VR • Diegetic UI (in-world interface) vs non-diegetic UI (HUD) • Spatial menus, VR tooltips, VR-based accessibility features 2.4 Level Design in VR • Spatial planning for player movement • Scene composition and environmental storytelling • Avoiding player discomfort (motion sickness mitigation) 2.5 Narrative Integration • Storytelling in VR environments • Branching narratives and player agency • Aligning narrative pacing with gameplay mechanics	16
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3	CO3, CO4, CO5, CO6	1. Implement VR-specific features and mechanics in Unity using built-in and XR toolkits (CO3). 2. Develop interactive VR gameplay elements integrating physics, AI, and player controls (CO3, CO4). 3. Apply Unity-based networking for multiplayer VR game scenarios (CO3). 4. Optimize asset integration and performance during the development phase (CO5). 5. Evaluate gameplay prototypes for functionality and player experience (CO6).	VR Game Development and Mechanics Implementation in Unity 3.1 Unity VR Development Setup • Project configuration for VR platforms • Unity XR Interaction Toolkit basics • Scene setup for VR gameplay 3.2 Core VR Mechanics Implementation • Player movement and teleportation systems • Interaction with objects (grab, throw, push) • Physics-based puzzles and environment manipulation 3.3 AI in VR • NPC behaviors and pathfinding in immersive spaces • Enemy AI and difficulty balancing 3.4 Multiplayer VR Basics • Introduction to Unity Netcode / Photon • Synchronizing VR interactions in multiplayer 3.5 Asset Integration and Optimization • Importing and managing 3D models, animations, audio • Draw call reduction, LOD (Level of Detail) usage • Scene baking and performance profiling	16
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4	CO4, CO5, CO6	1. Develop gameplay scripts in C# to implement interactivity, AI, and UI logic (CO4). 2. Optimize VR game performance through memory, rendering, and physics improvements (CO5). 3. Apply debugging and profiling tools to identify and fix gameplay issues (CO5). 4. Evaluate game builds for quality, usability, and platform compliance (CO6). 5. Document and prepare VR games for publishing on relevant platforms (CO6). 6. Analyze issues, deployment challenges, and ethical considerations in VR game distribution (CO6).	VR Game Scripting, Optimization, and Deployment Challenges with Ethical Considerations 4.1 C# Scripting for VR Games • Game event handling and triggers • Player input mapping for VR controllers • AI behavior scripting and state machines • UI interaction scripting in VR 4.2 Performance Optimization • Frame rate targets and VR hardware limits • Object pooling, memory management, physics tuning • Occlusion culling and lightmap baking 4.3 Testing and Debugging • Unity Profiler and Frame Debugger • VR-specific bug detection (tracking glitches, motion sickness issues) 4.4 Publishing and Deployment • Preparing builds for VR headsets (Meta Quest, HTC Vive, etc.) • Store submission guidelines (Oculus, SteamVR) enditemize • Challenges and Ethical Considerations in VR Game Deployment – Issues in VR game development (hardware compatibility, performance bottlenecks, user on-boarding) – Deployment challenges (platform certification, cross-platform optimization, distribution restrictions) – Physical considerations (motion sickness, ergonomics, safety zones) – Ethical and social implications (content ratings, inclusivity, psychological effects, privacy in online VR)	16
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VI SUGGESTED SELF LEARNING ACTIVITIES

Sr. No	Activity Title	Description	CO(s)
1	Mini/ Micro Project:e.g. AR Navigation System for Campus	Design and implement a simple AR app to help new students navigate campus buildings using marker-based AR. Document the process and challenges.	CO3, CO4
2	YouTube Video:e.g. Explain Marker-based vs Marker-less AR	Create an engaging explainer video comparing marker-based and marker-less AR, including visuals and real-world examples.	CO2
3	Comparative Study of AR-VR tools or technologies: e.g. Unity vs Unreal Engine for AR-VR	Conduct a detailed comparison between Unity and Unreal Engine based on features, usability, learning curve, and AR/VR capabilities.	CO4
4	Blog Post: e.g. Ethical Implications of AR in Healthcare or advanced applications of AR-VR like Metaverse.	Write a blog analyzing ethical concerns like privacy, addiction, and misinformation in AR applications used in the healthcare sector. Blog/Vlog post on Metaverse etc.	CO1,CO5
4	GitHub Repository: e.g. Interactive VR Museum Tour	Develop a VR app that simulates a museum tour experience and upload code, assets, and documentation to a GitHub repository.	CO3,CO4

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

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Sr. No.	Title	Edition	Authors	Publisher	Year
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4	Fundamentals of Computer Graphics	4 th Edition	Peter Shirley, Michael Ashikhmin, Steve Marschner	A K Peters/CRC Press	2016

Online References

Sr. No	Website Name
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C	https://www.freecodecamp.org/news/augmented-reality-full-course/ , A 12-hour YouTube course covering AR fundamentals, Unity, C, Vuforia, and even Spark AR for filters. Perfect for structured, hands-on learning.
D	BanubasARWebsiteShowcase , A curated list of interactive AR websites—great for inspiration and understanding real-world applications.
E	AppMasterBlogARinWebDevelopment , Offers insights into integrating AR into web apps, including use cases and tools like AR.js and WebXR



Mahavir Education Trust's

SHAH & ANCHOR

KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai

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Mahavir Education Trust's

SHAH & ANCHOR

KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai

Syllabus of Honours Degree Programs in Technology

(With effect from 2025-26)

Blockchain



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

CURRICULUM STRUCTURE UG: B.Tech.

Honours/Minor Degree Program

Blockchain

REVISION: R1-V0-2025-26

Effective from Academic Year 2025-26

Approved by Board of Studies on 17/06/2025

Approved by Academic Council on 19/06/2025

Dr. Vidyullata Devmane

I/C HOD

Chairman BoS

Department of Computer Engineering
Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



Dr. Bhavesh Patel

Principal

Dr. Bhavesh Patel

Chairman

Academic Council

Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

PREAMBLE

Shah and Anchor Kutchli Engineering College was established in 1985 for the purpose of imparting quality technical education. The college is managed by Mahavir Education Trust. The college is approved by AICTE, New Delhi and DTE, Government of Maharashtra and has been conferred autonomous status by University Grants Commission (UGC) and University of Mumbai for a period of 10 Years with effect from the academic year 2024-25. It also has an ISO 9001:2015 Certification. The institute offers eight undergraduate courses, three post-graduate courses and one PhD research Center in IT

The Honours Program in Blockchain, offered by Shah and Anchor Kutchhi Engineering College, is designed to equip students with comprehensive knowledge and practical skills in blockchain technology. The program integrates rigorous coursework, hands-on laboratory experiences, and real-world applications to prepare them for successful careers in this rapidly evolving field in Blockchain.

Over the course of two years, students will engage in an interdisciplinary curriculum that covers exploring Blockchain foundational principles, the underlying cryptographic mechanisms, deployment of different Blockchain platforms and its diverse applications. By examining both theoretical frameworks and practical implementations, this program seeks to provide a comprehensive understanding of how blockchain technology can enhance transparency, security, and efficiency in various sectors. This program aspires to contribute to the ongoing discourse in the field of blockchain technology, underscoring its significance and potential impact on the digital landscape.

Key features of our autonomous syllabus include:

- 1 Interdisciplinary Learning: Foster an interdisciplinary approach by integrating technical, and application-oriented courses.
2. Practical Expertise: Provide extensive hands-on laboratory experiences to bridge the gap between theoretical knowledge and real world application.
- 3 Innovation and Research: Encourage innovative thinking and research through project-based learning and exposure to cutting-edge Blockchain technologies.
- 4 Ethical Considerations: Ensuring ethical awareness and responsibility in blockchain development, deployment, and use to mitigate risks and promote fair and transparent practices.

Graduates of the Blockchain Honours Program will be well-prepared for careers in various sectors, including technology, healthcare, finance, and academia. They will possess the skills required to develop innovative Blockchain solutions, conduct advanced research, and contribute to the growing field of Blockchain Technology.



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Program Structure for Honours Degree in Blockchain (with effect from 2025–2026)

Semester V To Semester VIII

				Learning Scheme						Assessment Scheme												
Semester	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total		Exam	Theory					Practical				SLA		Total
				CL	TL	LL					CIA		ESE	Total		CIAP	ESEP	Total				
							Duration in Hrs	MSE			CCE	Max								Max	Max	
SEM V	Fundamentals of Blockchain Technology	PC	HNBKCR1501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
SEM VI	Blockchain Platform I	PC	HNBKCR1601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
SEM VII	Blockchain Platform II	PC	HNBKCR1701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225
SEM VIII	Blockchain Applications	PC	HNBKCR1801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
Total				16	4	4	12	540	18		80	80	240	400		40	60	100		100		600

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
SLA: Self Learning Assessment

Note:

1. Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
2. 1 credit is equivalent to 30 Notional hrs.
3. Self learning hours shall not be reflected in the Time Table.
4. SLA will have separate head of passing



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Semester V



Program	Honours Degree in Blockchain	CL	TL	LL	SL	C
Course Name: Fundamentals of Blockchain Technology	Course Code: HNBKCR0501	4	1	-	3	4
Pre-requisite:	Basic knowledge of cryptography and computer networks					

I RATIONALE

The subject Fundamentals of Blockchain Technology offers a foundational understanding of blockchain as a secure, transparent, and decentralized system. With growing relevance across industries like finance, healthcare, and supply chain, the course covers core concepts such as architecture, cryptography, consensus mechanisms, and practical applications. It prepares students for roles in blockchain development, fintech, and cybersecurity, making it essential for modern technical and business education.

II COMPETENCY

Learners will be able to analyze, develop, and implement blockchain-based solutions using public and private blockchain platforms, smart contracts, and decentralized technologies to solve real-world problems in various domains.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learner will be able to

Course Outcome Number	Course Outcome Statement
1	Apply blockchain technology's underlying principles, architecture, and operational mechanisms.
2	Evaluate the impact of blockchain adoption on existing systems and processes in industries.
3	Deploy smart contracts using solidity by applying key programming constructs.
4	Evaluate public blockchain concepts and Ethereum architecture.
5	Analyze consensus algorithms and fault tolerance mechanisms in a private blockchain environment.
6	Compare public and private blockchain platforms by analyzing their architecture, features, and real-world applications.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme						Assessment Scheme														
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Credits		Theory						Practical				SLA		Total
			CL	TL	LL						Exam	CIA		ESE	Total		CIAP	ESEP	Total				
												MSE	CCE		Max	Min			Max	Min			
1	Fundamentals of Blockchain Technology	HNBKCR0501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125		

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1,CO2	1. Apply the Importance cryptographic principles that form the foundation of blockchain technology. 2. Interpret the core concepts, architecture, and internal structure of a blockchain, including how blocks are linked and verified. 3. Compare various types of blockchain system and consensus mechanisms and evaluate their advantages, limitations, and use cases. 4. Describe the fundamental concepts of crypto currencies and their ecosystem, including types, usage, and the role of wallets and ICOs. 5. Identify and evaluate the potential and real-world applications of blockchain technology in different sectors.	Introduction to Blockchain 1.1 Prerequisite: Hash functions, Public key cryptography, Digital Signature (ECDSA) 1.2 Introduction to Blockchain: Evolution, Working, Components, Block structure, Block header, Block identifiers, Genesis block, Merkle trees 1.3 Distributed Ledger Technology (DLT): Benefits, Types (Public, Private, Consortium, Hybrid), Consensus Mechanisms (PoW, PoS, PoET) 1.4 Cryptocurrency Basics: Characteristics, Wallets, Types (Bitcoin, Altcoin, Token), Cryptocurrency Usage, ICOs 1.5 Blockchain Applications: FinTech, IoT, Industry, Energy, Supply Chain, Records, Healthcare	14

(Continued on next page)



Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO3	1. Make use of the concept of smart contracts, their characteristics, various types, the role of oracles, implementation approaches, and limitations in blockchain applications. 2. Develop the ability to write, deploy, and debug smart contracts using Solidity by applying key programming constructs and understanding the Solidity development workflow, including ABI integration.	Smart Contracts and Solidity Programming 2.1 Introduction to Smart Contracts, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contract Approaches, and Limitations of Smart Contracts 2.2 Introduction to Programming: Solidity Programming – Basics, functions, Visibility and Activity Qualifiers, Address and Address Payable, Bytes and Enums, Arrays-Fixed and Dynamic Arrays, Special Arrays-Bytes and strings, Struct, Mapping, Inheritance, Error handling, Application Binary Interface (ABI).	14
3	CO4	1. Apply knowledge of public blockchain architecture to explain the functioning of Bitcoin and Ethereum, including components like EVM, Ether, Gas, and mining. 2. Demonstrate the use of Ethereum tools such as test networks, wallets, frameworks, and simulate contract deployment using Ganache.	Public Blockchain 3.1 Introduction to Public Blockchain, Bitcoin Blockchain, Common Terminologies, Ethereum and its Components, Mining in Ethereum, Ethereum Virtual Machine (EVM), Ether, Gas, Accounts, Architecture and Workflow, Comparison between Bitcoin and Ethereum. 3.2 Types of test-networks used in Ethereum, Transferring Ethers using Metamask, Mist Wallet, Ethereum frameworks, Case study of Ganache for Ethereum blockchain.	14

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5,CO6	1. Analyze the role and applications of private blockchain platforms in e-commerce by examining their features, use cases, and implementation using blockchain commands. 2. Explain smart contracts, consensus mechanisms (PAXOS, RAFT), and fault tolerance (BFT, PBFT) in a private blockchain environment.	Private Blockchain 4.1 Introduction to Private Blockchain, Key characteristics, Need of Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Commands in E-commerce Blockchain 4.2 Smart Contract in a Private Environment, State Machine Replication, Consensus Algorithms for Private Blockchain - PAXOS and RAFT, Byzantine Faults: Byzantine Fault Tolerant (BFT) and Practical BFT	10

VI SUGGESTED SELF LEARNING ACTIVITIES

An application development using Fundamentals of Blockchain Technology (Micro-Project) / Case Study of an application where Fundamentals of Blockchain Technology are applied for effective implementation / MOOCs, etc.

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VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mastering Ethereum: Building Smart Contracts and DApps	1 st Edition	Andreas M. Antonopoulos, Gavin Wood	O'Reilly Media	2018
2	Mastering Bitcoin: Programming the Open Blockchain	2 nd Edition	Andreas M. Antonopoulos	O'Reilly Media	2017
3	Blockchain Technology	1 st Edition	Chandramouli Subramanian, Asha A. George, Abhilash K. A, Meena Karthikeyan	Universities Press	2021
4	Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction	1 st Edition	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder	Princeton University Press	2016

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Ethereum Smart Contract Development: Build blockchainbased decentralized applications using Solidity	1 st Edition	Mayukh Mukhopadhyay	Packt Publishing	2018
1	Mastering Blockchain	1 st Edition	Imran Bashir	Packt Publishing	2020

Online References

Sr. No	Website Name
A	https://onlinecourses.nptel.ac.in/noc22_cs44/preview , NPTEL Course: Blockchain and its Applications By Prof. Sandip Chakraborty, Prof. Shamik Sural (IIT Kharagpur)
B	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0133014807221780482223_shared/overview , infys pringboard: Block chain - A technical introduction By Prof. Krishnendu Chakraborty
C	BlockchainandCryptographyOverview CourseraIntroductiontoEthereumBlockchain Coursera Coursera



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Semester VI



Program	Honours Degree in Blockchain(Semester VI)	CL	TL	LL	SL	C
Course Name: Blockchain Platform I	Course Code: HNBKCR1601	4	1	-	3	4
Pre-requisite:	Cryptography, Blockchain Fundamentals					

I RATIONALE

The increasing adoption of decentralized technologies is revolutionizing various industries by offering transparency, security, and trustless execution of digital transactions. Among these technologies, blockchain platforms particularly Ethereum stand out for their programmable infrastructure, which enables the creation and deployment of decentralized applications (DApps). This subject, Blockchain Platform, is designed to provide learner with a comprehensive understanding of Ethereum-based blockchain systems and their practical applications. The course is structured to build foundational knowledge and gradually introduce more complex concepts, tools, and real-world use cases.

II COMPETENCY

Demonstrate the ability to design, develop, and deploy secure and efficient decentralized applications (DApps) on blockchain platforms by applying knowledge of Ethereum architecture, smart contracts, token standards, transaction mechanics, Ethereum Virtual Machine (EVM) operations, and integration of oracles, using appropriate tools and clients.

III COURSE OUTCOMES (COs)

After the completion of course based learning, the learner will be able to

Course Outcome Number	Course Outcome Statement
1	Analyze Ethereum's architecture, functionalities, and its applications within decentralized ecosystems.
2	Analyze different Ethereum client implementations and its essential components for ethereum transactions.
3	Simplify the significance of Turing Completeness in the EVM and its implications on smart contract functionality.
4	Compare PoW and PoS consensus mechanisms.
5	Design comprehensive token models with appropriate Ethereum token standards.
6	Build the decentralized applications (DApps) on the Ethereum platform, including smart contract development.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme					Credits	Assessment Scheme												
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Exam	Theory					Practical				SLA		Total
										Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total			
			CL	TL	LL	Notional Hours	Max				Max	Max		Max	Min			Max	Max	Max	
1	Blockchain Platform-I	HNBKCR1601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Explain the evolution of Ethereum through its four developmental stages (Frontier, Homestead, Metropolis, Serenity), highlighting the key features and improvements at each stage. 2. Identify and describe the core components of the Ethereum ecosystem, including the Ethereum Virtual Machine (EVM), smart contracts, accounts, and gas. 3. Demonstrate an understanding of Ether currency units (Wei, Gwei, Ether) and their relevance in Ethereum transactions. 4. Make Use of MetaMask wallet to manage accounts, switch between Ethereum networks (Mainnet, testnets), and perform basic operations such as sending and receiving ethers. 5. Utilize test Ether from faucets and perform test transactions to understand the Ethereum transaction lifecycle. 6. Analyze the structure and roles of EOAs and contract accounts in Ethereum, including their interaction with smart contracts. 7. Create a basic smart contract (Faucet Contract) using a development environment (e.g., Remix or Hardhat), and interact with it via MetaMask. 8. Identify transaction history using blockchain explorers (e.g., Etherscan), identifying key elements like sender, receiver, gas usage, and status.	Ethereum Basics 1.1 Ethereum's Four Stages of Development, Ethereum's Components, Decentralized Applications (DApps), Web 3.0. 1.2 Ether Currency Units, Ethereum Wallet, MetaMask, Switching Networks, Getting Test Ethers, Sending Ether from MetaMask, Exploring the Transaction History of an Address, Externally Owned Accounts (EOAs) and Contracts, Compiling the Faucet Contract, Creating the Contract on the Blockchain, Interacting with the Contract.	12

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2	1. Describe the function and role of full nodes in the Ethereum network, including their responsibilities in transaction validation, block propagation, and chain synchronization. 2. Demonstrate the ability to run an Ethereum client, connect to testnets or mainnet, and manage interactions through command-line interfaces or GUI tools. 3. Compare remote Ethereum clients, such as mobile wallets and browser wallets, and explain their advantages, limitations, and security considerations. 4. Explain the fundamentals of public key cryptography(ECC,Cryptographic Hash Functions) and its role in identity, transactions, and security within Ethereum. 5. Analyze the structure of an Ethereum transaction, identifying and explaining its key components: nonce, gas price, gas limit, recipient address, value, data. 6. Construct and send transactions that transmit value and/or data, including transferring Ether and invoking smart contract functions with encoded payloads. 7. Identify the propagation of a transaction across the Ethereum network, from signing and broadcasting to validation and block inclusion.	Ethereum Clients and Transactions 2.1 Full Node , Running an Ethereum Client, Parity, Go-Ethereum (Geth), Remote Ethereum Clients - Mobile (Smartphone) Wallets, Browser Wallets , Keys and Addresses, Public Key Cryptography, Private Keys, Public Keys, Elliptic Curve Cryptography, Cryptographic Hash Functions, Ethereum's Cryptographic Hash Function: Keccak-256, Ethereum Addresses, and Inter Exchange Client Address Protocol 2.2 Transactions - Structure of a Transaction, Transaction Nonce , Transaction Gas, Transaction Recipient, Transaction Value and Data, Transmitting Value to EOAs and Contracts, Transmitting a Data Payload to an EOA or Contract, Contract Creation, Elliptic Curve Digital Signature Algorithm, ECDSA Math, Separating Signing and Transmission (Offline Signing), Transaction Propagation, and Multiple-Signature (Multisig) Transactions	18

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3	1. Identify the role of the EVM in executing smart contracts. 2. Recognize how smart contract code is compiled into EVM bytecode. 3. Compare between externally owned accounts (EOA) and contract accounts. 4. Understand how the initialization code results in the final contract code stored on-chain 5. Illustrate how gas is calculated and accounted for during transaction execution, how gas limits and gas usage affect transaction execution. 6. Explain the fundamentals of the Proof of Work and Proof of Stake consensus mechanism. 7. Illustrate the concept of memory-hard functions and how Ethash resists ASIC mining. 8. Examine the design goals of Casper as Ethereum's PoS consensus protocol and consensus algorithms achieve agreement in a decentralized environment. 9. Explain the benefits and challenges of DAOs in decentralized decision-making.	Ethereum Virtual Machine and Consensus 3.1 EVM architecture and execution context, EVM Instruction Set, Ethereum State, Contract Deployment Code, Turing Completeness and Gas, Gas Accounting, Gas Cost Versus Gas Price, Block Gas Limit 3.2 Consensus via Proof of Work, Consensus via Proof of Stake (PoS), Ethash, Casper, Principles of Consensus, Ethereum Classic (ETC), Decentralized Autonomous Organization (DAO), DAO Soft fork and Hard Fork	18

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	- Outline the concept of tokens as digital assets built on Ethereum, ERC20, ERC223, ERC777 - Compare different classes of token standards (fungible, non-fungible, multi-token). - Plan about optional extensions to basic token standards that add functionality (e.g., metadata, enumerable tokens) which helps to improve token usability, tracking, and integration. - Illustrate the role of tokens in Initial Coin Offerings (ICOs) and fundraising. - Understand the role of oracles as bridges between blockchain smart contracts and external data sources. - Identify the role of blockchain technology in enabling DApps.	Tokens, Oracles and DApps 4.1 Tokens on Ethereum, ERC20 Token Standard, ERC223, ERC777, ERC721: Non-fungible Token, Token Standards, Extensions to Token Interface Standards, Tokens and ICOs 4.2 Oracles, Oracle Use Cases, Oracle Design Patterns, request-response, publish-, subscribe, and immediate-read, Data Authentication 4.3 Decentralized Applications (DApps), Backend (Smart Contract), Frontend (Web User Interface), Data Storage-IPFS, Swarm, Decentralized Message Communications Protocols, Auction DApp, Ethereum Name Service	12

VI SUGGESTED SELF LEARNING ACTIVITIES

Research and write a short summary (1-2 pages) on how blockchain works, focusing on decentralized ledgers, consensus mechanisms, and smart contracts/An application development using Blockchain Technology (Micro-Project) / Case Study of real time application where Blockchain are applied for effective implementation / MOOCs, etc.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mastering Ethereum: Building Smart Contracts and DApps	1 st Edition	Andreas M. Antonopoulos, Dr. Gavin Wood	O'Reilly Media	2018
2	Learn Ethereum: A Practical Guide to Help Developers Set Up and Run Decentralized Applications with Ethereum 2.0	2 nd Edition	Xun (Brian) Wu, Zhihong Zou, Dongying Song	Packt Publishing	2023
3	Blockchain Technology	1 st Edition	Chandramouli Subramanian, Asha A. George, Abhilash K. A., Meena Karthikeyan	Universities Press (India) Pvt. Ltd.	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mastering Blockchain: A Deep Dive into Distributed Ledgers, Consensus Protocols, Smart Contracts, DApps, Cryptocurrencies, Ethereum, and More	3 rd Edition	Imran Bashir	Packt Publishing	2020
2	Blockchain for Beginners	1 st Edition	Yathish R and Tejaswini N	Shroff Publishers Distributors (SPD)	2019
3	Blockchain Basics: A Non-Technical Introduction in 25 Steps	1 st Edition	Daniel Drescher	Apress	2017

Online References

Sr. No	Website Name
A	https://www.globalxetfs.com/articles/ethereum-the-basics , Basics of Ethereum
B	https://101blockchains.com/free-blockchain-course , Blockchains
C	https://masterthecrypto.com/guide-to-ethereum-blockchain-protocol/ , Tokens, Gas
D	https://www.coursera.org/specializations/building-dapps-ethereum-blockchain , Building DApps



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Semester VII



Program	Honours Degree in Blockchain (Semester VII)	CL	TL	LL	SL	C
Course Name: Blockchain Platform II	Course Code: HNBKCR1701	4	1	4	3	6
Pre-requisite:	Fundamentals of Blockchain Technology, Blockchain Platform I					

I RATIONALE

Blockchain technology has revolutionized the digital landscape by introducing decentralized, transparent, and secure systems for various industries. As enterprises increasingly adopt blockchain solutions, particularly permissioned frameworks like Hyperledger Fabric, it is essential for learners to gain both conceptual understanding and hands-on experience. This course bridges the gap between theoretical blockchain foundations and real-world application development using platforms such as Ethereum and Hyperledger Fabric. Learners will explore architectural components, develop and test smart contracts, and analyze diverse use cases of both public and private blockchains. The course equips learners with the knowledge and skills required to contribute effectively to enterprise-grade blockchain projects across domains like finance, supply chain, healthcare, and governance.

II COMPETENCY

Apply foundational and advanced concepts of public and private blockchain technologies to design, develop, and evaluate decentralized applications using platforms like Ethereum and Hyperledger Fabric.

III COURSE OUTCOMES (COs)

After the completion of course based learning, the learners will be able to

Course Outcome Number	Course Outcome Statement
1	Analyze the architecture and components of Hyperledger Fabric by setting up the test network.
2	Deploy a Hyperledger Fabric network with channels and organizations.
3	Construct smart contracts (chaincode) on a Fabric network.
4	Implement identity management and privacy features in fabric.
5	Design blockchain-based applications for real-world use cases.
6	Evaluate the impact of blockchain adoption on existing systems and processes in industries.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme				Credits	Assessment Scheme													
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL		Total	Exam	Theory					Practical				SLA		Total
										Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total			
			MSE	CCE	Max						Min	Max		Min	Max			Min	Max	Min	
1	Blockchain Platform II	HNBKCR1701	4	1	4	3	180	6	2	20	20	60	100	40	40	60	100	40	25	10	225

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS
 (Minimum 10)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Analyze Hyperledger Fabric Components and Diagram the Architecture.	Introduction to Hyperledger Fabric and Architecture.	CO1
2	Install Fabric prerequisites, launch the test network, explore logs and CLI.	Set Up Hyperledger Fabric Test Network	CO1
3	To write a basic chaincode (smart contract) for asset transfer in Hyperledger Fabric, deploy it on a Fabric network, and interact with it using the CLI.	Write and Deploy Basic Chaincode in Fabric.	CO1, CO3
4	Apply the process of creating and configuring a channel in Hyperledger Fabric, and analyze how channels enable data isolation and controlled communication between organizations within a blockchain network.	Create and Use Fabric Channels	CO2
5	Use Fabric CA to issue identities, enroll users, and connect them.	Set Up Hyperledger CA and Enroll Users.	CO4
6	Build a basic app using the Fabric SDK to submit and evaluate transactions from your chaincode.	Fabric SDK Integration (Node.js or Java)	CO2, CO4
7	Implement different Fabric chaincode policies, including endorsement, modification, and signature policies, and see how they impact transaction validation.	Enforce Endorsement and Signature Policies in Fabric.	CO4

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
8	To set up Fabric CA, issue identities with specific roles (peer, client, admin), and manage user credentials securely.	Fabric Certificate Authority (CA) with Custom Identity Roles.	CO4
9	To dynamically add a new organization to an existing Fabric channel and verify its participation in transaction processing.	Organization in an Existing Channel.	CO2
10	Apply Fabric's data privacy features using Private Data Collections (PDC) between selected organizations.	Chaincode with Private Data Collections for Confidential Transactions	CO4
11	To apply Fabric concepts to a real-world application, simulating an end-to-end use case like KYC verification, food traceability, or digital asset transfer.	Mini Use Case Simulation (e.g., KYC or Food Supply Chain) Using Fabric.	CO1-CO5

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Differentiate between permissioned and permissionless blockchain platforms concerning access control, trust models, and governance structures. 2. Explain the modular architecture of Hyperledger Fabric, highlighting the roles of pluggable components such as consensus, membership, and data stores. 3. Describe the concept of smart contracts (chaincode) in Hyperledger Fabric and their role in implementing business logic. 4. Illustrate the execute-order-validate transaction flow model in Hyperledger Fabric and contrast it with traditional order-execute models. 5. Analyze how Hyperledger Fabric ensures privacy and confidentiality through channels, private data collections, and membership services. 6. Apply knowledge of Fabric's ledger architecture, transaction lifecycle, and consensus mechanism to evaluate its suitability for enterprise applications.	Introduction to Hyperledger Fabric 1.1 Hyperledger Fabric, Modularity, Permissioned vs Permissionless Blockchains, Smart Contracts- Chain code, execute-order-validate model, Privacy and Confidentiality, Pluggable Consensus, Performance and Scalability 1.2 Hyperledger Fabric Model- Assets, Chaincode, Ledger Features, Privacy, Security and Membership Services, Consensus, Transaction Flow.	10

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2	1. Demonstrate the steps to create and configure a sample Hyperledger Fabric network, including channel creation and component joining. 2. Apply the process of installing, approving, and committing chaincode to a channel using the Fabric CLI or scripts. 3. Illustrate how to add components and organizations to existing and new channels in a Hyperledger Fabric network. 4. Explain the role of Certificate Authorities (Root CAs, Intermediate CAs) and their importance in establishing trust in a Fabric network. 5. Describe the concept of identity and certificate revocation in Fabric using Fabric CA and Certificate Revocation Lists (CRLs). 6. Analyze the structure and function of the Membership Service Provider (MSP), including organizational units (OUs), node roles, and domain configurations.	Hyperledger Clients and Transactions 2.1 Creating the sample network, Certificate Authorities, Join nodes to the channel, Install, approve, and commit a chaincode, Using an application on the channel, Joining components to multiple channels, Creating the new channel configuration, Adding components, Adding an organization to an existing channel, Adding existing components to the newly joined channel. 2.2 Identity, Root CAs, Intermediate CAs and Chains of Trust, Fabric CA, Certificate Revocation Lists. 2.3 Membership Service Provider(MSP), MSP domains, Organizational Units (OUs), Node OU Roles - Client, peer, admin, orderer, MSP Structure	20

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3,CO4	1. Explain various policy types in Hyperledger Fabric, including endorsement policies, signature policies, and implicit meta policies, and their role in securing blockchain operations. 2. Describe the relationships among peers, chaincode, ledgers, channels, and organizations in a Hyperledger Fabric network. 3. Illustrate the structure and function of the ledger, including the world state, transaction log, and database options used for asset tracking and query. 4. Analyze the interaction between orderer nodes, channel configuration, and the transaction flow from proposal to commitment. 5. Develop and deploy smart contracts using Fabric chaincode lifecycle processes including installation, approval, and commitment phases. 6. Evaluate different chaincode deployment scenarios in Hyperledger Fabric for scalability, upgradability, and modular design.	Policies, Peers and Ledger 3.1 Policies, Smart contract endorsement policies, Modification policies, Signature policies, Implicit Meta policies. 3.2 Peers, Ledgers and Chaincode, Multiple Ledgers, Multiple Chaincodes, Applications and Peers, Peers and Channels, Peers and Organizations, Peers and Identity, Peers and Orderers. 3.3 Ledger, World State, World State database, Ledger: Basic Asset Transfer, ordering, Orderer nodes and channel configuration, Orderer nodes and identity, Orderers and the transaction flow, Orderers and the transaction flow. 3.4 Smart Contracts and Chaincode- Ledger, Development, Endorsement, Valid transactions, Channels. 3.5 Fabric chaincode lifecycle- Deploying a chaincode, Install and define a chaincode, Upgrade a chaincode, Deployment scenarios.	20

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5,CO6	1. Identify real-world blockchain applications across diverse sectors such as finance, healthcare, supply chain, and public governance. 2. Explain the role of blockchain in enhancing transparency, security, and efficiency in applications like digital identity, voting, and cross-border payments. 3. Analyze the implementation of blockchain in selected domains such as education, energy, and IoT by evaluating platforms, use cases, and challenges. 4. Evaluate the impact of blockchain adoption on existing systems and processes in industries such as real estate, entertainment, and digital asset management.	HyperLedger real world applications and Use cases 4.1 Financial Services , Land and Property Management, Cross Border Payments, Government /Public Sector, Letters Of Credit, Food /Food Delivery, Green Assets Management, Real Estate Transactions, Digital Assets, Healthcare, Smart Contracts, Education, Internet Of Things, Supply Chain Management, Energy, Information Technology, Ticketing /Entertainment, Identity Management, KYC, Voting	10

VII SUGGESTED SELF LEARNING ACTIVITIES

Deploy a real-world decentralized application (DApp) using blockchain principles such as immutability, decentralization, and trustless transactions, using either public platforms (like Ethereum) or permissioned platforms (like Hyperledger Fabric) (Micro- Project)/ Case study on Emerging Blockchain Platforms / MOOCs, etc.

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Blockchain with Hyperledger Fabric	2 nd Edition	Nitin Gaur, Anthony O'Dowd, Petr Novotny, Luc Desrosiers, Venkatraman Ramakrishna, Salman A. Baset	Packt Publishing	2020
2	Hands-On Blockchain with Hyperledger: Building decentralized applications with Hyperledger Fabric and Composer	1 st Edition	Petr Novotny, Luc Desrosiers, Nitin Gaur	Packt Publishing	2018
3	Blockchain Technology	1 st Edition	Chandramouli Subramanian, Asha A. George, Abhilash K. A and Meena Karthikeyan	Universities Press	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mastering Blockchain: A deep dive into distributed ledgers, consensus protocols, smart contracts, DApps, cryptocurrencies, Ethereum, and more	3 rd Edition	Imran Bashir	Packt Publishing	2020
2	Blockchain Basics, A non Technical Introduction in 25 Steps	1 st Edition	Daniel Drescher	Apress Berkeley, CA	2017

Online References

Sr. No	Website Name
A	https://hyperledger-fabric.readthedocs.io/en/latest/key_concepts.html , Hyperledger Fabric Docs
B	https://hyperledger-fabric.readthedocs.io/en/latest/tutorials.html , Hyperledger Fabric Tutorials



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Semester VIII



Program	Honours Degree in Blockchain (Semester VIII)	CL	TL	LL	SL	C
Course Name: Blockchain Applications	Course Code: HNBKCR1801	4	1	-	3	4
Pre-requisite:	Cryptography, Blockchain Fundamentals					

I RATIONALE

Blockchain is considered a foundational technology, much like the internet. It enables decentralized, transparent, and secure systems, disrupting traditional centralized models in Finance, Supply chains, Healthcare, Voting systems, etc. Understanding how blockchain applications work prepares the learners to participate in and lead digital transformation. The study of blockchain applications equips the learners with the knowledge, skills, and critical perspective needed to engage with one of the most transformative technologies.

II COMPETENCY

The learners will be equipped to Present blockchain solutions and justify design decisions to both technical and non-technical stakeholders, Evaluate the suitability of blockchain for various real-world applications, Analyze blockchain-based solutions for efficiency, scalability, and security. Assess risks, limitations, and challenges in deploying blockchain systems, Build decentralized applications (dApps) using blockchain platforms, and work effectively in teams to develop and critique blockchain projects.

III COURSE OUTCOMES (COs)

After the completion of course based learning, the learners will be able to

Course Outcome Number	Course Outcome Statement
1	Analyze Bitcoin blockchain data for transaction immutability and transparency.
2	Evaluate Bitcoin mining process and consensus mechanism.
3	Analyze Node Roles and Transaction Verification in a Bitcoin Network.
4	Compare the Decentralized finance as an alternative to traditional financial infrastructure by deconstructing its key components.
5	Analyze the core primitives of the DeFi ecosystem to explain the roles of tokens, incentives, and protocols.
6	Assess the risks of DeFi protocols for evaluation of their impact on the efficiency of decentralized financial platforms.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme						Assessment Scheme												
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Credits	Exam	Theory				Practical				SLA		Total
											CIA		ESE	Total		CIAP	ESEP	Total			
			Duration in Hrs	MSE	CCE	Max	Min			Max	Min	Max		Min	Max			Min			
				Max	Max								Max			Max	Max		Max	Max	
1	Blockchain Applications	HNBKCR1801	4	1	-	2	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Demonstrate the working of bitcoin cryptocurrency. 2. Describe the working and types of a bitcoin wallet. 3. Describe the bitcoin cryptocurrency transactions basics. 4. Describe about fees and scripting of transactions in bitcoin. 5. Discuss what are different types of transactions and scripts in bitcoin.	Introduction to Bitcoin, wallet and transaction 1.1 What is Bitcoin, Digital Currencies before Bitcoin, History of Bitcoin, Bitcoin Wallet, Sending and receiving Bitcoin 1.2 Nondeterministic (Random) Wallets, Deterministic (Seeded) Wallets, HD Wallets (BIP-32/BIP-44), Wallet Best Practices, Using Bitcoin Wallets. 1.3 Bitcoin Transaction - Transaction Lifecycle, Transaction Structure, Transaction Outputs and Inputs, Transaction Fees, Transaction Chaining and Orphan Transactions. 1.4 Transaction Scripts and Script Language-Script Construction (Lock + Unlock), Scripting Language, Standard Transactions-Pay-to-Public-Key-Hash (P2PKH), Pay-to-Public-Key, Pay-to-Script-Hash (P2SH), Benefits of pay-to-script-hash.	10

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2, CO3	1. Understand details of bitcoin network and types of nodes in bitcoin. 2. compare working of SPV node with other types of nodes. 3. Describe the concept and importance of decentralized consensus in blockchain systems. 4. Demonstrate how mining nodes compete to solve cryptographic puzzles to validate transactions. 5. Analyze the components of the Coinbase reward and how Coinbase data is utilized in the mining process, particularly in nonce variation and block hashing. 6. Describe each field within the block header (version, previous block hash, Merkle root, timestamp, difficulty target, nonce) and its role.	Bitcoin Network, Mining and Consensus 2.1 Peer-to-Peer Network Architecture, Node Types and Roles, Extended Bitcoin Network, Bitcoin Relay Networks, Network Discovery, Full Nodes, Exchanging "Inventory". 2.2 Simplified Payment Verification (SPV) Nodes, Merkle Trees and Simplified Payment Verification (SPV), Bloom Filters, SPV Nodes and Privacy, Encrypted and Authenticated Connections, Transaction Pools. 2.3 Decentralized Consensus, Independent Verification of Transactions, Mining Nodes, Aggregating Transactions into Blocks. 2.4 Coinbase Transaction, Coinbase Reward and Fees, Structure of the Coinbase Transaction, Coinbase Data, Constructing the Block header.	18

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO4	1. Analyze the efficiency, latency, and security challenges associated with traditional payment systems. 2. Describe the factors affecting accessibility of banking services such as geographic location, KYC/AML regulations, and digital divides. 3. Compare and Contrast decentralized finance and traditional finance with respect to structure, accessibility, trust models, and operational transparency. 4. Explain the concept of stablecoins and the mechanisms used to peg their value to fiat currencies or other assets. 5. Explain the functioning of decentralized exchanges (DEXs) and their difference from centralized exchanges. 6. Describe how DeFi platforms create and trade derivative products like futures and options. 7. Describe how encrypted and authenticated connections ensure security and privacy 8. Explain how DeFi enables peer-to-peer payment systems without intermediaries.	Introduction to Decentralized Finance 3.1 Traditional Financial Institution-Banks: 1. Payment and Clearance systems, 2. Accessibility, 3. Centralization and Transparency. 3.2 Decentralized Finance Vs Traditional Finance. 3.3 Defi key Categories:-Stablecoins, Stablecoin and pegging, Lending and Borrowing, Exchanges, Derivations, Fund Management, Lottery, Payments, Insurance. 3.4 DeFi Architecture: Consumer Layer: Blockchains, Cross-Blockchain networks, Oracles, Digital Asset Layer: Cryptocurrencies, Infrastructure Layer: Wallets and Asset Management 3.5 DEXes and Liquidity, Lending and Borrowing, Prediction Markets, Synthetic Assets, Insurance	16

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5, CO6	- Describe the characteristics, models of asset custody and purposes of different fungible token types within DeFi. - Analyze the role of incentives in promoting network participation and penalizing malicious behavior. - Compare the efficiency, liquidity, and user experience between order book and AMM models. - Describe how loanable fund markets operate within DeFi ecosystems using on-chain assets as collateral. - Explain the role of stablecoins in maintaining value stability and enabling transactions in DeFi. - Explain common technical risks, usability challenges and risks arising from central points of control.	DeFi Primitives and Ecosystem 4.1 DeFi Primitives: Transactions Fungible Token: Equity Tokens, Utility Tokens and Governance Tokens NFT: NFT Standard, Multi-token standard Custody Supply Adjustment: Burn-Reduce Supply, Mint-Increase Supply, Bonding Curve-Pricing Supply. 4.2 Incentives: Staking Rewards, Slashing, Direct Rewards and Keepers, Fees. 4.3 Swap: Order Book Matching, Automated Market Makers, Collateralized Loans Flash Loans (Uncollateralized Loans). 4.4 DeFi Key Metrics: Total Value Locked, Daily Active Users, Market Cap. 4.5 DeFi EcoSystem and Protocols: On-chain Asset Exchange, Loanable Fund Markets on-chain assets, Stablecoins, Portfolio Management, Derivatives, Privacy-preserving mixers. 4.6 DeFi Risk and Challenges: Technical Risks, Usability Risks, Centralization Risks, Liquidity Risks, Regulation Risk.	16

VI SUGGESTED SELF LEARNING ACTIVITIES

Case Study of an application where blockchain is applied for effective implementation / MOOCs, etc.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mastering Bitcoin	2 nd Edition	Andreas M. Antonopoulos	O'Reilly Media	2017
2	How to DeFi	1 st Edition	Darren Lau, Daryl Lau, Teh Sze Jin, Kristian Kho, Erina Azmi, TM Lee	Bobby Ong	2020
3	DeFi and the Future of Finance	1 st Edition	Campbell R. Harvey	John Wiley & Sons Inc	2021
4	Defi Adoption 2020 A Definitive Guide to Entering the Industry	1 st Edition	Demelza Hays, Ben Yorke, Helen Rosenberg	Cointelegraph	2020
5	SoK: Decentralized Finance (DeFi)	1 st Edition	Sam M. Werner, Daniel Perez, Lewis Gudgeon, Arian Klages-Mundt, Dominik Harz, William J. Knottenbelt	ResearchGate	2021

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction	1 st Edition	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder	Princeton University Press	2016

Online References

Sr. No	Website Name
A	https://berkeley-defi.github.io/assets/material/defi-sok-ariah-2101.08778.pdf
B	https://www.coursera.org/learn/cryptocurrency . MOOC
C	https://www.coursera.org/specializations/decentralized-finance-duke . MOOC



Mahavir Education Trust's

SHAH & ANCHOR

KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai

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Syllabus of Honours Degree Programs in Technology

(with effect from 2025-26)

Cyber Security



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

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CURRICULUM STRUCTURE UG: B.Tech.

Honours/Minor Degree Program

Cyber Security

REVISION: R1-V0-2025-26

Effective from Academic Year 2025-26 Onwards

Approved by Board of Studies on 17/06/2025

Approved by Academic Council on 19/06/2025

Dr. Nilakshi Jain

I/C HOD

Chairman BoS

Department of Cyber Security
Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



Dr. Bhavesh Patel

Principal

Dr. Bhavesh Patel

Chairman

Academic Council

Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



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UG Program in Cyber Security

PREAMBLE

Cyber Security Department: Guardians of the Digital Realm: SAKEC Cyber Security Engineering –Protecting Tomorrow in Shades of Purple !

Shah and Anchor Kutchhi Engineering College was established in 1985 for the purpose of imparting quality technical education. The college is managed by Mahavir Education Trust. The college is approved by AICTE, New Delhi and DTE, Government of Maharashtra and has been conferred autonomous status by University Grants Commission (UGC) and University of Mumbai for a period of 10 Years with effect from the academic year 2025-26. It also has an ISO 9001:2015 Certification. The institute offers eight undergraduate courses, three post-graduate courses and one PhD research Center in IT.

The Honours Degree program is designed for students who are passionate about understanding and mastering the technologies and techniques vital to the field of Cyber Security. This curriculum covers foundational to advanced concepts, equipping students to protect computers, operating systems, networks, and data from cyber-attacks and unauthorized access. Total credits of the entire program will be of 18, wherein focus is not only on providing knowledge but also on building skills, attitude and self learning.

The program emphasizes the identification and mitigation of vulnerabilities across computing systems, preparing students to effectively safeguard sensitive information. Through hands-on projects, simulated cyber-attacks, and real-world scenarios, students gain practical experience and confidence in combating cyber threats.

This program is the result of invaluable contributions from both academia and industry, with special recognition to the Members of the Board of Studies and Academic Council for their critical input in shaping the curriculum. We believe this curriculum not only meets but exceeds the expectations of all stakeholders.

Embark on a journey where your passion for cybersecurity transforms into expertise, and together, let's safeguard the integrity of tomorrow's digital frontiers.



Program Structure for Honours Degree in Cyber Security (with effect from 2025–2026)

			Learning Scheme						Assessment Scheme													
Semester	Course Code	Course Title	Actual Contact Hrs/Week			SL	Total Notional Hours		Credits		Theory					Practical				SLA		Total
				CIA							ESE	Total		CIAP	ESEP	Total						
				CL	TL							LL	Paper Duration in Hrs			MSE Max	CCE Max	Max	Max	Min	Max	
SEM V	HNCSCR1501	Computer Network	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125	
SEM VI	HNCSCR1601	Ethical Hacking	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125	
SEM VII	HNCSCR1701	Advanced Digital Forensics	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225	
SEM VIII	HNCSCR1801	Penetration Testing (PENT)	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125	
Total			16	4	4	12	540	18		80	80	240	400		40	60	100		100		600	

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
 CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
 SLA: Self Learning Assessment

Note:

1. Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
2. 1 credit is equivalent to 30 Notional hrs.
3. Self learning hours shall not be reflected in the Time Table.
4. SLA will have separate head of passing



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Semester V



Program	Honours Degree in Cyber Security (Semester V)	CL	TL	LL	SL	C
Course Name: Computer Network	Course Code: HNCSCR0501	4	1	-	3	4
Course Type:	-	-				
Pre-requisite:	-					

I RATIONALE

Explore the subject Computer Networks to gain insights into the structure, functioning, and protocols of modern computer networks, forming a core foundation for understanding cybersecurity threats, defenses, and secure communication.

II COMPETENCY

Enable learners to explain how computer networks operate and how protocols enable communication, providing the knowledge required to assess and secure networked systems in cybersecurity domains.

III COURSE OUTCOMES (COs)

After the completion of course-based learning, learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply networking concepts including topologies, protocols, media, and OSI layers.
2	Apply data link layer design principles, error detection/correction methods, and data link layer protocols to solve network communication problems.
3	Apply IP addressing, subnetting, NAT, and routing algorithms to design and troubleshoot efficient network routing and addressing schemes.
4	Apply transport and application layer protocols to design and troubleshoot network communication and services.
5	Analyze Standard Operating Procedures, disaster recovery strategies, and network documentation for optimizing infrastructure management.
6	Troubleshoot connectivity, service issues, and performance problems using standard tools.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme																
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory					Practical				SLA		Total		
				CL	TL	LL					Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total					
												Max	Max		Max	Max			Min				Max	Max
												1	Computer Network		-	HNCSCR0501			4	1	-		3	120

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module No.	Mapped CO	Major Learning Outcomes	Theory Content	Hours
1	CO1	1. Explain basic networking topologies, devices, and their roles in OSI layers. 2. Classify various types of cables, connectors, and standards used in media transmission. 3. Identify purposes and functioning of ports, protocols, and IP addressing models. 4. Illustrate virtual network components and Ethernet deployment techniques. 5. Compare OSI and TCP/IP models with real-world implementations.	Basics of Networking & Infrastructure 1.1 Network topologies, types and technologies. Devices and services at their appropriate OSI layers. 1.2 Purposes and uses of ports and protocols, TCP/IP models. 1.3 Media types: Copper, UTP, STP, Coaxial, Fiber, Single-mode, Multimode, Plenum vs. PVC, Connector types, Copper cable standards, Copper termination standards. 1.4 Ethernet deployment standards, Virtual networking components.	15
2	CO2	1. Describe data link layer functions and common design issues. 2. Apply error detection/correction mechanisms and reliable transmission techniques. 3. Solve subnetting and supernetting problems in IPv4 address space. 4. Analyze routing protocols and algorithms for efficient network design. 5. Implement NAT and evaluate IPv6 addressing mechanisms.	Switching & Routing Concepts 2.1 DLL Design Issues (Services, Framing, Error Control, Flow Control). 2.2 Error Detection and Correction (Hamming Code, CRC, Checksum), Elementary Data Link protocols Stop and Wait, Sliding Window (Go Back N, Selective Repeat), CSMA/CD. 2.3 IPv4 Addressing (classful and classless), Subnetting, Supernetting design problems, IPv4 Protocol, 2.4 Network Address Translation (NAT), IPv6, Routing algorithms : Shortest Path (Dijkstra's), Link state routing, Distance Vector Routing, Protocols - ARP, RARP, ICMP, IGMP.	15

3	CO3 ,CO4	1. Explain transport layer services, sockets, and TCP/UDP communication. 2. Demonstrate TCP connection setup, flow control, and congestion handling. 3. Distinguish between HTTP, DNS, SMTP and other application layer protocols. 4. Configure secure remote access using SSH, VNC, and FTP protocols.	Higher Layer Protocols 3.1 The Transport Service: Transport service primitives, Berkeley Sockets, Connection management (Handshake), UDP, TCP, TCP state transition, TCP timers TCP Flow control (sliding Window), TCP Congestion Control. 3.2 Application Layer: DNS: Name Space, Resource Record and Types of Name Server.HTTP,SMTP, 3.3 DHCP, remote access methods: SSH, VNC, Telnet, FTP/FTPS.	15
4	CO5,CO6	1. Interpret standard operating procedures and disaster recovery concepts. 2. Apply troubleshooting methodologies using hardware/software tools. 3. Diagnose wired and wireless connectivity and performance issues. 4. Evaluate network diagrams and patching techniques.	Network Operations, Troubleshooting and Tools 4.1 Diagram symbols, Standard operating procedures/ work instructions, Logical vs. physical diagrams, Rack diagrams, disaster recovery concepts, scanning, monitoring and patching processes. 4.2 Network troubleshooting methodology, Hardware tools, Software tools, wired/wireless connectivity and performance issues. 4.3 Common network service issues	15



VI SUGGESTED SELF LEARNING ACTIVITIES

- Design and simulate a network topology for a medium-sized organization using tools like Cisco Packet Tracer or GNS3. Include addressing schemes, routing protocols, VLANs, and security controls.

OR

- Create a detailed video or presentation comparing IPv4 and IPv6 addressing and routing with real-time simulations and packet captures using Wireshark.

OR

- Participate in a network troubleshooting lab challenge (e.g., on TryHackMe, INE, or Cisco NetAcad). Document issues identified and resolved using a structured troubleshooting methodology.

OR

- Appear for 10 GATE/Networking mock tests (via SAKEC software: <https://115.113.39.68:4433/remote/login?lang=en>)

OR

- Obtain a certification in networking or network security (e.g., CCNA, CompTIA Network+, or NPTEL/Infosys Springboard/Coursera) and submit the certificate along with a short report summarizing key learnings.

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**VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS****Text Books**

Sr. No	Title (with Link)	Edition	Authors	Publisher	Year
1	Computer Networks	4 th Edition	A.S. Tanenbaum	Pearson Education	2003
2	Data Communications and Networking	5 th Edition	B.A. Forouzan	TMH	2013
3	Computer Networking: A Top-Down Approach Featuring the Internet	6 th Edition	James F. Kurose, Keith W. Ross	Addison Wesley	2012

Reference Books

Sr. No	Title (with Link)	Edition	Authors	Publisher	Year
1	An Engineering Approach to Computer Networking	1 st Edition	S. Keshav	Pearson Education	1997
2	Computer Networks: Principles, Technologies & Protocols for Network Design	1 st Edition	Natalia Olifer, Victor Olifer	Wiley India	2006
3	Computer Networks: A Systems Approach	2 nd Edition	Larry L. Peterson, Bruce S. Davie	Morgan Kaufmann	2000

Online References

Sr. No	Website Name
A	NPTEL Course: Computer Networks
B	GeeksforGeeks: Computer Networks Tutorials
C	Coursera: Computer Networking by Stanford University



Mahavir Education Trust's

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Semester VI



Program	Honours Degree in Cyber Security (Semester VI)	CL	TL	LL	SL	C
Course Name: Ethical Hacking	Course Code: HNCSCR0601	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Computer Network, Operating Systems					

I RATIONALE

Develop core cybersecurity skills with Ethical Hacking by learning how hackers think and operate. This subject equips learners to test vulnerabilities, apply penetration techniques, and design secure systems to defend against real-world cyber attacks.

II COMPETENCY

Identify and ethically exploit security weaknesses in systems to help organizations strengthen their cybersecurity defenses. Apply responsible hacking techniques to detect vulnerabilities and recommend solutions.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learners will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply ethical hacking fundamentals and reconnaissance techniques.
2	Analyze enumeration, vulnerabilities, and system hacking countermeasures.
3	Apply techniques to identify and mitigate malware, sniffing, and social engineering threats.
4	Apply methods to detect and defend against SQL injection attacks.
5	Apply DoS/DDoS and session hijacking countermeasures effectively.
6	Perform DoS/DDoS and session hijacking mitigation in practice.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory						Practical			SLA		Total
				CL	TL	LL					Duration	CIA		ESE	Total			CIAP	ESEP			
												Max	CCE		Max	Max	Max			Min	Max	
1	Ethical Hacking	-	HNCSCR0601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Recognize the importance of information security and ethical hacking. 2. Discover information using various footprinting techniques. 3. Operate scanning tools to perform host, port, and service discovery. 4. Identify operating system details using banner grabbing and fingerprinting.	Introduction to Ethical Hacking 1.1. Information Security Overview, Cyber Kill Chain Concepts, Hacking Concepts, Ethical Hacking Concepts, Information Security Controls, Information Security Laws and Standards. 1.2. Footprinting Concepts, Footprinting through Search Engines, Footprinting through Web Services, Footprinting through Social Networking Sites, Website Footprinting, Email Footprinting, Whois Footprinting. 1.3. Network Scanning Concepts, Scanning Tools, Host Discovery, Port and Service Discovery. 1.4. OS Discovery (Banner Grabbing/OS Fingerprinting).	15
2	CO2	1. Perform enumeration on various services and protocols. 2. Execute vulnerability assessments using suitable tools and techniques. 3. Demonstrate system hacking techniques and steganography tools.	Enumeration, Vulnerability & System Hacking 2.1. Enumeration Concepts, NetBIOS Enumeration, SNMP Enumeration, LDAP Enumeration, NTP and NFS Enumeration, SMTP and DNS Enumeration, Enumeration Countermeasures. 2.2. Vulnerability Assessment Concepts, Vulnerability Classification and Assessment Types, Vulnerability Assessment Solutions and Tools, Vulnerability Assessment Reports. 2.3. System Hacking Concepts, Gaining Access, Escalating Privileges, Maintaining Access, Clearing Logs, Password Cracking, Password Attacks, Steganography, Steganography Tools, Steganalysis, Steganography Detection Tools.	15

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3 ,CO4	1. Analyze the characteristics and behavior of malware types. 2. Examine sniffing techniques and their impact on networks. 3. Interpret social engineering methods and associated threats. 4. Analyze SQL injection types and how they affect databases. 5. Operate tools to detect and bypass SQL injection protections.	Threats & Attacks 3.1. Malware Concepts, Components of Malware, APT Concepts, Trojan Concepts, Virus and Worm Concepts, Fileless Malware Concepts, Malware Analysis, Countermeasures, Anti-Malware Software. 3.2. Sniffing Concepts, Sniffing Techniques: MAC Attacks, DHCP Attacks, ARP Poisoning, Spoofing Attacks, DNS Poisoning, Sniffing Tools, Countermeasures. 3.3. Social Engineering Concepts, Social Engineering Techniques, Insider Threats, Impersonation on Social Networking Sites, Identity Theft, Countermeasures. 3.4. Basics of SQL Injection attack, Types of SQL Injection, Blind SQL Injection. 3.5. SQL Injection Methodology, SQL Injection Tools, Signature Evasion Techniques, SQL Injection Detection Tools.	20
4	CO5 ,CO6	1. Evaluate DoS and DDoS attacks, tools, and defense mechanisms. 2. Assess session hijacking techniques and suitable countermeasures.	DoS, Session Hijacking & Cryptography 4.1. DoS/DDoS Concepts, DoS/DDoS Attack Techniques, Botnets, DDoS Case Study, DoS/DDoS Attack Tools, Countermeasures, DoS/DDoS Protection Tools. 4.2. Session Hijacking Concepts, Application Level Session Hijacking, Network Level Session Hijacking, Session Hijacking Tools, Countermeasures.	10



VI SUGGESTED SELF LEARNING ACTIVITIES

- Build a virtual lab environment using tools like Kali Linux and Metasploitable to simulate ethical hacking tasks covered in the course (e.g., scanning, enumeration, exploitation).

OR

- Develop a step-by-step blog or written guide demonstrating how to detect and mitigate a common vulnerability (e.g., SQL Injection or DoS attack) with practical tools and screenshots.

OR

- Contribute to an open-source penetration testing toolkit (like OWASP ZAP or Nmap scripts) by reporting issues, improving documentation, or writing simple plugin code.

OR

- Solve real-world capture-the-flag (CTF) challenges from platforms such as TryHackMe, Hack The Box, or PicoCTF, and document the walkthrough of at least one solved challenge.

OR

- Obtain a professional certification by completing a relevant technical course from a recognized platform (e.g., Cisco, Infosys Springboard, NPTEL/SWAYAM, Coursera, edX, Udemy). Demonstrate completion by submitting the official certificate and a concise report on the key skills and concepts acquired.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Web application security	-	Hoffman, Andrew	O'Reilly Media, Inc.	2024
2	Ethical hacking and penetration testing guide	-	Baloch, Rafay	CRC Press	2017
3	Handbook on Crime and Technology (Chapter 18: Hacking)	-	Toro-Alvarez, Marlon Mike	-	2023
4	Cyber Security & Cyber Laws	-	Nilakshi Jain & Ramesh Menon	-	-
5	CEH v11: Certified Ethical Hacker Version 11 Practice Tests	-	Messier, Ric	John Wiley & Sons	2021
6	Digital Forensic: The Fascinating world of Digital Evidence	-	-	-	-

**Reference Books**

Sr. No	Title	Edition	Authors	Publisher	Year
1	Handbook of computer networks and cyber security	-	Brij B. Gupta, Gregorio Martinez Perez, Dharma P. Agrawal, and Deepak Gupta	Springer	2020
2	Computer Security Threats	-	Ciza Thomas, Paula Fraga-Lamas, and Tiago M. Fernández-Caramés, eds.	BoD–Books on Demand	2020
3	Principles of information security	-	Michael E. Whitman, and Herbert J. Mattord	Cengage learning	2021
4	Cyber Security for Next-Generation Computing Technologies	-	Inam Ullah Khan Ullah Khan, Mariya Ouaisa, Mariyam Ouaisa, Zakaria Abou El Houda, and Muhammad Fazal Ijaz, eds.	CRC Press	2024

Online References

Sr. No	Website Name
A	https://tryhackme.com/ , Try Hack Me.

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Semester VII



Program	Honours Degree in Cyber Security (Semester VII)	CL	TL	LL	SL	C
Course Name: Advanced Digital Forensics	Course Code: HNCSCR0701	4	1	4	3	6
Course Type:	-					
Pre-requisite:	Computer Networks, Ethical Hacking					

I RATIONALE

Advanced Digital Forensics helps learners how to investigate cybercrimes by collecting and analyzing digital evidence, a key skill needed to protect systems and ensure justice in cybersecurity.

II COMPETENCY

Apply advanced digital forensic techniques to investigate cyber incidents, focusing on the collection, analysis, and preservation of digital evidence in alignment with industry standards and employer expectations in cybersecurity.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learners will be able to:

Course Outcome Number	Course Outcome Statement
1	Analyze digital forensics fundamentals and legal-ethical procedures.
2	Apply forensic tools to recover data from storage media.
3	Evaluate password cracking and Operating System level password management.
4	Analyze ethical considerations in mobile and network forensic methods.
5	Execute legally compliant forensic investigation plans with precision.
6	Demonstrate proficient secure evidence handling and workstation setup.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits	Exam	Theory				Practical				SLA	Total		
				CL	TL	LL					Duration	CIA		ESE	Total	CIAP	ESEP	Total				
												MSE	CCE									
																					Max	Max
1	Advanced Digital Forensics	-	HNCSCR0701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Describe how to extract network information using passive reconnaissance techniques.	Perform footprinting on the target network using search engines, web services, and social networking sites. Case Study/scenario: Given a target organization, design a method to perform footprinting via search engines, web services, and social media to identify network assets and vulnerabilities.	CO1, CO4, CO5, CO6
2	Implement file carving methods on storage media to retrieve deleted digital artifacts.	Use different file carving tools on a variety of storage media (USB drives, memory cards) to recover deleted or corrupted files. Tools: PhotoRec, Foremost, Scalpel.	CO2, CO5
3	Use software tools to retrieve and validate deleted data from storage media.	Recover deleted files by using data recovery software to retrieve them.	CO2, CO5
4	Examine how low-level file edits affect binary structure and integrity.	Open a file in a hex editor, identify specific file signatures, and manually edit some bytes to observe the impact on the file structure. Hex editor software (e.g., HxD, Hex Fiend, Bless).	CO2, CO5
5	Investigate file headers to verify types and recover files independent of extensions.	Identify file signatures using a hex editor and then use this information to validate file types and recover files without relying on file extensions.	CO2, CO5
6	Assess file recovery strategies and report challenges and solutions in real scenarios.	Provide a real-world case study involving file recovery and analysis. Apply the techniques learned, and present their findings, including challenges faced and solutions implemented.	CO1, CO4, CO5, CO6
7	Demonstrate proper acquisition methods to extract non-volatile data from a Windows system.	Perform static data acquisition from Windows OS. Recommended Tool: FTK Imager.	CO1, CO4
8	Assess live data collection strategies for volatile memory capture.	Acquire live data from Windows OS. Recommended Tool: FTK Imager, TCP Dump.	CO4
9	Test the effectiveness of cracking tools for password recovery.	Perform the password recovery using Password cracking tool like John the Ripper/Cain.	CO2, CO3, CO5, CO6
10	Execute hash-cracking techniques to reveal original passwords.	Use password cracking tools to attempt to recover the original passwords from the hashes.	CO2, CO3, CO5, CO6
11	Compare different hashing algorithms and their resilience to cracking.	Use a set of hashed passwords (MD5, SHA-256, etc.). Recommended Tools: John the Ripper, Hashcat, or Cain and Abel.	CO2, CO3, CO5, CO6

(Continued on next page)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
12	Implement file-level encryption techniques to secure sensitive data.	Encrypting a sample file or folder using the chosen encryption software. Recommended Tools: VeraCrypt, BitLocker, or GPG (GNU Privacy Guard).	CO2, CO3, CO5, CO6
13	Examine wireless traffic patterns to interpret user activity and mobility.	Use tools like Wireshark or Kismet to analyze Wi-Fi traffic and track the movement of a suspect based on their connection to different hotspots.	CO4, CO5
14	Produce a comprehensive forensic report by interpreting digital artifacts using Autopsy.	Use Autopsy to conduct a forensic examination, including file analysis, timeline analysis, and keyword searches. Document your findings and prepare a forensic report.	CO4, CO6
15	Explain the use of command-line tools for basic network traffic and connection analysis.	Use command-line tools commonly used for network analysis (e.g., ping, tracer, netstat) of target networks.	CO4

Note: All other experiments should follow the format and structure provided in Experiment No. 1. You may include real-time scenarios, case studies, or complex engineering problems.

VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO4, CO5, CO6	1. Explain the fundamentals of digital forensics and its role in investigations. 2. Examine the phases of digital forensic investigation with respect to legal and procedural compliance. 3. Use tools to investigate file metadata and calculate digital footprints. 4. Illustrate the “bag and tag” procedure and explain the chain of custody. 5. Assess ethical and professional responsibilities in forensic practice.	Digital Forensics & Investigative Procedures 1.1 Digital Forensics: Introduction to Digital Forensics, Digital forensic and Computer Forensics, why digital forensic experts are needed, Understand digital forensics as a science, professional ethics and confidentiality requirements, role NIST plays in digital forensics, Investigate a file's metadata, Calculate your digital footprint. 1.2 Digital forensic investigation, Phases of digital forensic investigation, importance of authorizations, surveying a crime scene, documenting, “bag and tag” procedure, chain of custody for evidence. 1.3 Procedure for preserving and transporting evidence, digital forensics toolkit, creating reports for investigations, search and seizure rules of electronic evidence.	15

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2, CO5	1. Explain the architecture of hard disk drives and their forensic significance. 2. Investigate the role of formatting and carving techniques in data recovery. 3. Implement techniques to recover deleted files from different storage media. 4. Assess storage media types for evidentiary value and forensic analysis. 5. Differentiate between data encoding formats (ASCII, Unicode) and assess their relevance in forensics.	Storage Media 2.1 Architecture of a hard disk drive and the file management systems, Bits, bytes, ASCII and Unicode, components in hard disk drivers, types of formatting techniques, carving techniques, Methods of recovering deleted files. 2.2 Volatile and non-volatile storage, magnetic media, solid state storage, removable flash memory or optical storage, storage devices' storage capacity and data transfer rates, potential evidential value of storage devices and storage media.	15
3	CO2, CO3, CO5, CO6	1. Describe strong and weak passwords and apply cracking techniques ethically. 2. Examine password hash mechanisms and OS-level password storage. 3. Operate forensic tools for password recovery and system analysis. 4. Critique forensic tool admissibility and design a compliant forensic workstation. 5. Judge ethical considerations and legal boundaries in using forensic tools.	Passwords & Forensics Tools 3.1 Introduction to Passwords, strong and weak passwords, techniques to crack passwords, Bypass BIOS passwords, Operating system passwords Cracking, Windows password management, password hashes, consequences of unauthorized password cracking. 3.2 Forensics investigations Tool, admissibility of digital forensic tools in court, SysInternals utilities, graphical and command-line interfaces, forensic image, .Decompress files, Decrypt files, forensic software applications, forensics workstation requirements, imaging devices and applications, Write blockers.	15
4	CO4	1. Summarize mobile forensic principles and platform-specific investigation methods. 2. Execute triangulation/trilateration techniques for device location tracking. 3. Employ mobile data extraction tools and analyze extracted evidence. 4. Differentiate between types of network attacks and their forensic implications. 5. Appraise forensic techniques and best practices for wireless and network environments.	Mobile and Network Forensics 4.1 Introduction to Mobile Forensics, Common smartphone platforms. GSM and CDMA cellular networks, triangulation and trilateration, identifiers on mobile devices, data extraction on a mobile device, tools needed in a mobile device investigation. 4.2 Introduction to Network forensics, data branches and attacks on network, classification of malware, DoS and a DDoS attacks. common network devices, wireless networks and devices, preventive measures and best practices associated with network forensics.	15

VII SUGGESTED SELF LEARNING ACTIVITIES

- Develop and document a mini case study on phishing email detection and header analysis.

OR

- Develop an engaging video tutorial on a fundamental subject topic, suitable for publication on the department's official social media platforms.

OR

- Create a forensic timeline from collected artifacts and trace digital footprints in a mock case.

OR

- Recover and examine deleted files from different storage media using file carving techniques. Atleast 5 devices.

OR

- Obtain a professional certification by completing a relevant technical course from a recognized platform (e.g., Cisco, Infosys Springboard, NPTEL/SWAYAM, Coursera, edX, Udemy). Demonstrate completion by submitting the official certificate and a concise report on the key skills and concepts acquired.

OR

- Set up a virtual cyber lab to simulate and investigate malware behavior in a controlled environment.

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	Cybercrime and Digital Forensics	3 rd Edition	Thomas J. Holt, Adam M. Bossler, and Kathryn C. Seigfried-Spellar	Routledge	2022
2	An Insight into Digital Forensics: History, Frameworks, Types and Tools	2 nd Edition	Jones, G. M., & Winster, S. G.	O'Reilly Media	2022
3	Cyber Security Issues and Current Trends	1 st Edition	Nitul Dutta, Nilesh Jadav, Sudeep Tanwar, Hiren Kumar Deva Sarma, Emil Pricop	Springer	2022
4	Digital forensic: The fascinating world of digital evidences	1 st Edition	Dr. Nilakshi Jain & Dr. Dhananjay R. Kalbande	Wiley India Pvt. Ltd.	2016

Reference Books

Sr. No	Title	Edition	Authors	Publisher	Year
1	The basics of digital forensics: the primer for getting started in digital forensics	1 st Edition	Sammons John	Waltham, MA : Syngress	2012
2	Digital Forensics Basics: A Practical Guide Using Windows OS	1 st Edition	Nihad A. Hassan	Apress	2019
3	Digital Forensics and Incident Response	2 nd Edition	Gerard Johansen	Packt Publishing	2022

Online References

Sr. No	Website Name
A	https://onlinecourses.swayam2.ac.in/nou25_cs05/preview
B	https://www.open.edu/openlearn/science-maths-technology/digital-forensics/content-section-0?active-tab=description-tab
C	https://www.coursera.org/learn/digital-forensics-essentials-dfe



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

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Semester VIII



Program	Honours Degree in Cyber Security (Semester VIII)	CL	TL	LL	SL	C
Course Name: Penetration Testing (PENT)	Course Code: HNCSCR0801	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Computer Network, Ethical Hacking, Advanced Digital Forensics					

I RATIONALE

Penetration Testing equips learners with essential skills to identify and test security weaknesses in systems, helping them understand how real-world cyberattacks occur and how to prevent them—an important ability in any cybersecurity role.

II COMPETENCY

Perform security testing on networks and systems to identify vulnerabilities and support industry-relevant cybersecurity practices.

III COURSE OUTCOMES (COs)

After the completion of course-based learning, learners will be able to:

Course Outcome Number	Course Outcome Statement
1	Analyze penetration testing concepts, methodologies, and risks
2	Analyze penetration testing engagements and scope with ethical considerations.
3	Apply OSINT tools and social engineering techniques.
4	Demonstrate ethical network penetration testing and vulnerability assessment.
5	Assess the importance of IoT penetration testing in uncovering vulnerabilities.
6	Perform cloud penetration testing using appropriate tools.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme				Assessment Scheme															
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Credits	Exam	Theory				Practical				SLA		Total	
				CL	TL	LL						Duration in Hrs	CIA		ESE	Total	CIAP	ESEP	Total				
													Max	CCE					Max				Min
1	Penetration Testing (PENT)	-	HNCSCR0801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125	

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO2	1. Evaluate appropriate penetration testing strategies and methodologies to align with organizational security objectives. 2. Design a complete penetration testing engagement scope including tools, roles, and ethical boundaries.	Introduction to Penetration Testing, Scoping and Management 1.1 Penetration Testing: Penetration Testing, Penetration Testing Service Delivery Models, ROI for Penetration Testing, Types of Penetration Assessment, Strategies of Penetration Testing, Selection of Appropriate Testing Type, Different Methods of Penetration Testing, Common Areas of Penetration Testing, Penetration Testing Process, Penetration Testing Phases, Penetration Testing Methodologies, Characteristics of a Good Penetration Test, Ethics of a Penetration Tester; Risks Associated with Penetration Testing. 1.2 Penetration Testing Scoping and Engagement: Initiation of a Pen Testing Engagement Process, Penetration Testing “Rules of Behavior,” Identifying the Security Tools Required for the Penetration Test	14
2	CO3	1. Apply OSINT tools and techniques to gather and analyze publicly available intelligence from websites and DNS. 2. Develop realistic social engineering test cases involving phishing, vishing, and physical access vectors.	OSINT & Social Engineering PenTest 2.1 Open-Source Intelligence (OSINT): Introduction to OSINT, OSINT through the WWW, OSINT through Website Analysis, OSINT through DNS Interrogation, Whois Lookups, Reverse Lookups, DNS Zone Transfer, Traceroute Analysis, Automating the OSINT Process using Tools/Frameworks/Scripts 2.2 Social Engineering PenTest: Introduction, Social Engineering Penetration Testing Modes, Social Engineering Penetration Testing Process, Social Engineering Penetration Testing using Email, Phishing, Phone (Vishing), SMishing (SMS phishing), Piggybacking, Eavesdropping, Social Engineering Countermeasures and Recommendations.	16

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO4	1. Appreciate ethical practices in network penetration testing. 2. Value vulnerability assessment for improving security.	External & Internal Network Penetration Testing 3.1 External Network Penetration Testing: Network Penetration Testing, External vs. Internal Penetration Testing, External Network Penetration Testing, Internal Network Penetration Testing. Network Penetration Testing Process, Port Scanning, Fingerprinting the OS, Fingerprinting the services 3.2 Internal Network Penetration Testing: Footprinting, Network Scanning, Scanning Analysis, Scanning Methodology, OS and Service Fingerprinting, Identifying the OS, Manual Banner Grabbing, Enumeration, Internal Vulnerability Assessment, Network Vulnerability Scanning, Host Vulnerability Scanner, Vulnerability Assessment Reports, Scan Analysis Process, Windows Exploitation.	14
4	CO5, CO6	1. Perform penetration testing on IoT devices to identify vulnerabilities and attack surfaces. 2. Conduct cloud penetration testing using appropriate tools to assess security risks.	IoT and Cloud PenTest 4.1 IoT PenTest: IoT, Popular IoT Hacks, IoT Challenges, IoT Penetration Testing, Abstract IoT Testing Methodology, Attack Surface Mapping, IoT Architecture, Typical IoT Vulnerabilities, Steps to Analyzing the IoT Hardware, Firmware Attacks, Attack Surface Map. 4.2 Cloud PenTest: Cloud Computing Security and Concerns, Security Risks Involved in Cloud Computing, Role of Penetration Testing in Cloud Computing, Scope of Cloud Pen Testing, Penetration Testing Process, Tools for Penetration Testing.	16



VI SUGGESTED SELF LEARNING ACTIVITIES

- Complete the [TryHackMe Penetration Testing Module](#) and document key takeaways from rooms such as Initial Access, Enumeration, and Exploitation in a learning journal.

OR

- Use [AttackDefense Labs](#) or [PortSwigger Web Security Labs](#) to perform at least 3 structured vulnerability assessments (e.g., XSS, SQLi, Command Injection) and reflect on exploitation techniques and defense strategies.

OR

- Deploy and test penetration testing tools like Metasploit, Nmap, and Burp Suite on a local lab or TryHackMe sandbox; maintain a logbook of tests performed, tools used, and vulnerabilities discovered.

OR

- Simulate a phishing campaign using the [GoPhish framework](#) in a controlled environment and report findings, techniques used, and possible mitigations based on user behavior.

OR

- Use [OSINT Framework](#) or tools like Maltego and theHarvester to gather intelligence about a domain or individual and present your findings in a mind-map or infographic format.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Ethical Hacking and Penetration Testing Guide	1 st Edition	Baloch, Rafay	CRC Press	2015
2	Building Virtual Pentesting Labs for Advanced Penetration Testing	1 st Edition	Cardwell, Kevin	Packt Publishing Ltd	2011
3	Quick Start Guide to Penetration Testing	1 st Edition	Rahalkar, Sagar; Rahalkar, and Karkal	Apress, Berkeley, CA, USA	2018
4	Advanced Penetration Testing	1 st Edition	Allsopp, Wil	John Wiley & Sons, Inc.	2017
5	Hacking (In Routledge Handbook of Digital Media and Communication)	1 st Edition	Brunton, Finn	Routledge	2020
6	Cyber Security & Cyber Laws	1 st Edition	Nilakshi Jain & Ramesh Menon	Wiley India Pvt Ltd.	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Hands on Hacking: Become an Expert at Next Gen Penetration Testing and Purple Teaming	1 st Edition	Hickey, Matthew; Arcuri, Jennifer	John Wiley & Sons	2020
2	The Pentester Blueprint: Starting a Career as an Ethical Hacker	1 st Edition	Wylie, Phillip L.; Crawley, Kim	John Wiley & Sons	2020
3	Cybercrime and Digital Forensics: An Introduction	3 rd Edition	Holt, Thomas J.; Bossler, Adam M.; Seigfried-Spellar, Kathryn C.	Routledge	2022

Online References

Sr. No	Website Name
A	Complete Ethical Hacking Bootcamp (Udemy Course)
B	Penetration Testing and Ethical Hacking Complete Hands-on (Udemy Course)
C	Cloud Penetration Testing with Azure – Master Initial Access (Udemy Course)
D	Metasploit Framework: Penetration Testing with Metasploit (Udemy Course)



Mahavir Education Trust's

SHAH & ANCHOR

KUTCHHI ENGINEERING COLLEGE

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KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai

Syllabus of Honours Degree Programs in Technology

(with effect from 2025-26)

Data Science



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

CURRICULUM STRUCTURE UG: B.Tech.

Honours/Minor Degree Program

Data Science

REVISION: R1-V0-2025-26

Effective from Academic Year 2025-26

Approved by Board of Studies on 17/06/2025

Approved by Academic Council on 19/06/2025

Swati
Dr. Swati Nadkarni

I/C HOD

Chairman BoS

Department of Information Technology
Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400073



Dr. Bhavesh Patel
Dr. Bhavesh Patel
Principal

Dr. Bhavesh Patel
Dr. Bhavesh Patel
Academic Council
Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



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Shah & Anchor Kutchhi Engineering College

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PREAMBLE

Shah and Anchor Kutchhi Engineering College was established in 1985 for the purpose of imparting quality technical education. The college is managed by Mahavir Education Trust. The college is approved by AICTE, New Delhi and DTE, Government of Maharashtra and has been conferred autonomous status by University Grants Commission (UGC) and University of Mumbai for a period of 10 Years with effect from the academic year 2024-25. It also has an ISO 9001:2015 Certification. The institute offers eight undergraduate courses, three post-graduate courses and one PhD Research Center in IT.

The Honours Program in Data Science, offered by Shah and Anchor Kutchhi Engineering College, is designed to provide students with a strong knowledge in the upcoming field of data science. The program combines extensive coursework, hands-on laboratory experiments and recent applications to equip students with the skills needed to excel in the rapidly evolving field of data science.

The Honours degree program is two years duration spanning four semesters. Learners will begin with a course on mathematical concepts necessary for learning data science. Consequently, learners will go through all statistical models needed in the data science domain. Further data science applications in the healthcare field will be explored in depth. Practical hands-on will enhance skills of the learners in these domains. Text, web and social media analytics are introduced to cover all aspects of data analytics.

Key features of our autonomous syllabus include:

1. Interdisciplinary Learning: Highlights an interdisciplinary approach by integrating mathematical and statistical concepts in the courses.
2. Practical Expertise: Provides extensive hands-on laboratory experiments to bridge the gap between theoretical knowledge and real-world applications.
3. Innovation and Analytics: Encourages innovative thinking and analytical skills through project-based learning and exposure to latest data science technologies.
4. Ethical Values: Inculcate a basic understanding of the societal and privacy implications of data science applications.

Learners of the Data Science Honours Program will be well-prepared for careers in the Data Science and Machine Learning domain. Learners will get an edge over the other graduates to get an entry in the world of data science professionals like Data Scientists, Data Engineers, Data Analysts.



Program Structure for Honours Degree in Data Science

(with effect from 2025–2026)

				Learning Scheme					Credits	Assessment Scheme												
Sr. No.	Course Title	Semester	Course Code	Actual Contact Hrs/Week			SL	Total			Theory					Practical				SLA		Total
										Duration in Hrs	CIA		ESE	Total		CIAP	ESEP	Total				
				MSE	CCE																	
				Max	Max	Max					Max	Min										
1	Mathematics for Data Science	V	HNDSCR1501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
2	Statistical Learning for Data Science	VI	HNDSCR1601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
3	Healthcare Data Analytics	VII	HNDSCR1701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225
4	Data Analytics for Text, Web and Social Media	VIII	HNDSCR1801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125
Total				16	4	4	12	540	18	-	80	80	240	400	-	40	60	100	-	100	-	600

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
 CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
 SLA: Self Learning Assessment

Note:

- Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
- 1 credit is equivalent to 30 Notional hrs.
- Self learning hours shall not be reflected in the Time Table.
- SLA will have separate head of passing



Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

An Autonomous Institute Affiliated to University of Mumbai

Semester V



Program	Third Year B.Tech. Data Science Honours (Semester V)	CL	TL	LL	SL	C
Course Name: Mathematics for Data Science	Course Code: HNDSCR1501	4	1	-	3	4
Course Type:	HN					
Pre-requisite:	Mathematics for Data Science					

I RATIONALE

Data Science plays significant role in today's world. It enables individuals and organizations to transform data into actionable insights, facilitating informed decision-making across various sectors such as engineering, healthcare, business, education, and public policy. The course offers understanding of mathematical concepts and methodologies, equipping students with the necessary tools to learn and dive deep into the world of data science.

II COMPETENCY

Demonstrates proficiency in mathematical techniques, optimization and linear algebra, applying descriptive and inferential techniques to visualize data.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply linear algebra and probability distributions and sampling distributions concepts to model, solve, and analyze real-world problems.
2	Apply graph representation, exploratory data analysis to some real data sets and provide interpretations via relevant visualization.
3	Analyze various optimization techniques.
4	Evaluate Dimension Reduction Algorithms.
5	Illustrate exploratory data analysis for suitable dataset.
6	Construct Python code for optimization techniques.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Theory					Practical				SLA		Total	
				CL	TL	LL				Exam	CIA		ESE	Total		CIAP	ESEP	Total				
								Duration in Hrs	MSE		CCE	Max		Min	Max			Min	Max	Min		Max
1	Mathematics for Data Science	HN	HNDSCR1501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Solve linear equations. 2. Describe random sampling. 3. Apply sampling Distributions.	Fundamentals of Probability and Statistics 1.1 Vectors and Matrices, Solving Linear equations, The four Fundamental Subspaces. 1.2 Eigenvalues and EigenVectors, The Singular Value Decomposition (SVD). 1.3 Random Variables and their probability Distribution. 1.4 Random Sampling, Sample Characteristics and their Distributions, 1.5 Chi-Square, t-, and F-Distributions: Exact Sampling Distributions, 1.6 Sampling from a Bivariate Normal Distribution, The Central Limit Theorem.	18
2	CO2, CO5	1. Differentiate between types of data. 2. Plot given data to draw conclusions.	Exploratory Data Analysis 2.1 Quantitative vs. Qualitative data, Types of Quantitative data: Continuous data, Discrete data, Types of Qualitative data: Categorical data, Binary data, Ordinary data, 2.2 Plotting, data using Bar graph, Pie chart, Histogram, Stem and Leaf plot, Dot plot, Scatter plot, Time-series graph, Exponential graph, Logarithmic graph, Trigonometric graph, Frequency distribution graph. 2.3 Need of exploratory data analysis, cleaning and preparing data, 2.4 Feature engineering, Missing values, understanding dataset through various plots and graphs, Drawing conclusions, deciding appropriate machine learning models.	20

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3, CO6	1. Explain the basics of optimization. 2. Apply the methods of optimization.	Optimization Techniques 3.1 Types of optimization-Constrained and Unconstrained optimization. 3.2 Methods of Optimization-Numerical Optimization, Bracketing Methods-Bisection Method, False Position Method, Newton's method, Steepest descent method.	12
4	CO4	1. Learn concepts of dimensionality reduction. 2. Apply dimension reduction techniques to reduce dimensions.	Dimension Reduction Algorithms 4.1 Introduction to Dimension Reduction Algorithms. 4.2 Linear Dimensionality Reduction: Principal component analysis, Linear Discriminant Analysis (LDA). 4.3 Non-Linear Dimensionality Reduction: Multidimensional Scaling, Isometric Feature Mapping. Minimal polynomial.	10

VI SUGGESTED SELF LEARNING ACTIVITIES

Conduct a detailed case study on the problem statement by performing data analysis with the selection of a data set that aligns with a clearly defined objective. Apply descriptive statistics to summarize key features of the data, succeeded by data visualization techniques to identify distributions and outliers. Finally, insights are presented through a well-structured report documenting the methodology, analysis, results, and applications.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Linear Algebra for Everyone	15 th Edition	Gilbert Strang	Wellesley Cambridge Press	2020
2	An Introduction to Probability and Statistics	2 nd Edition	Vijay Rohatgi	Wiley Publication	2001
3	An introduction to Optimization	2 nd Edition	Wiley-Edwin Chong, Stainslaw Zak	Wiley Publication	2001
4	Mathematics for Machine Learning	6 th Edition	Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong	Cambridge University Press	2020
5	Exploratory Data Analysis	6 th Edition	John Tukey	Princeton University and Bell Laboratories	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Foundations of Machine Learning, MIT Press	2 nd Edition	Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar,	MIT Press	2018

Online References

Sr. No	Website Name
A	https://nptel.ac.in/courses/106106179 , NPTEL Course: Data Science for Engineers by By Prof. Ragunathan Rengasamy, Prof. Shankar Narasimhan — IIT Madras



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Semester VI



Program	Third Year B.Tech. Data Science Honours (Semester VI)	CL	TL	LL	SL	C
Course Name: Statistical Learning for Data Science	Course Code: HNDSCR1601	4	1	-	3	4
Pre-requisite:	Mathematics for Data Science					

I RATIONALE

Statistics is essential in today's world as it enables individuals and organizations to transform data into actionable insights, facilitating informed decision-making across various sectors such as engineering, healthcare, business, education, and public policy. The course offers understanding of statistical concepts and methodologies, equipping students with the necessary tools to navigate and interpret the data-rich landscape of the modern world.

II COMPETENCY

Demonstrates proficiency in statistical analysis and probability theory, applying descriptive and inferential techniques to interpret data, assess uncertainty, and inform decision-making across diverse engineering contexts.

III COURSE OUTCOMES (COs)

After the completion of course based learning, learner will be able to:

Course Outcome Number	Course Outcome Statement
1	Apply descriptive and inferential statistical methods, including measures of central tendency, variability, and probability distributions, for interpretation of data sets in various contexts.
2	Analyze sampling techniques and sampling distributions for the reliability and validity of statistical inferences, ensuring accurate conclusions from sample data.
3	Construct and present frequency tables, graphical representations, and summary statistics using real-world datasets, applying statistical software techniques.
4	Apply hypothesis testing procedures, including t-tests, chi-square tests, and ANOVA for population parameters and make data-driven decisions in research scenarios.
5	Analyze regression models, including simple, multiple, and logistic regression, for examining relationships between variables and forecast outcomes, utilizing time series analysis for trend prediction.
6	Develop and validate regression and forecasting models using statistical tools, perform residual analysis, and interpret model diagnostics for decision-making.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

			Learning Scheme						Assessment Scheme												
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Credits		Theory					Practical			SLA		Total
			Exam	CIA						ESE	Total		CIAP	ESEP	Total						
				Duration in Hrs	MSE						CCE	Max			Min	Max	Min	Max	Min	Max	
1	Statistics Learning for Data Science	HNDSCR1601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO3	1. Apply statistical inference techniques on various types of data. 2. Discuss techniques involved in Descriptive Statistics. 3. Describe Probability and Bayes Theorem. 4. Differentiate between Discrete and Continuous Probability Distributions.	Introduction to Statistics and Probability 1.1 Elements, Variables, and Observations, Scales of Measurement, Categorical and Quantitative Data, Descriptive Statistics and Statistical Inference. 1.2 Descriptive Statistics: Tabular and Graphical - Summarizing Categorical Data, Summarizing Quantitative Data, Cross Tabulations and Scatter Diagram, Descriptive Statistics: Numerical Measures - Measures of Location, Measures of Variability, Measures of Distribution Shape, Relative Location, and Detecting Outliers, Box Plot, Measures of Association Between Two Variables 1.3 Probability: Experiments, Counting Rules, and Assigning Probabilities, Events and Their Probabilities, Complement of an Event, Addition Law, Independent Events, Multiplication Law Bayes theorem. 1.4 Discrete Probability Distributions: Random Variables, Discrete Probability Distributions, Expected Value and Variance, Binomial Probability Distribution, Poisson Probability Distribution, Continuous Probability Distributions: Uniform Probability Distribution, Normal Curve, Standard Normal Probability Distribution, Computing Probabilities for Any Normal Probability Distribution.	20

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2, CO3	1. Differentiate between Sampling from a Finite Population and an Infinite Population. 2. Interpret interval estimation for the population mean, differentiating between cases with known and unknown population standard deviation.	Sampling and Sampling Distribution 2.1 Sampling from a Finite Population, Sampling from an Infinite Population, Sampling Methods: Probability and Non-Probability Sampling Method. 2.2 Interval Estimation: Population Mean - standard deviation Known, Population Mean - Standard deviation Unknown, Determining the Sample Size.	10
3	CO4, CO6	1. Outline the fundamentals of hypothesis testing and analysis of variance. 2. Conduct tests for goodness of fit and assess the independence between two variables.	Hypothesis Tests 3.1 Null and Alternative Hypotheses, Type I and Type II Errors, Population Mean: Known Population Mean: Unknown Inference About Means and Proportions with Two Populations. Inferences About Population Variances, Inferences About a Population Variance, Inferences About Two Population Variances, Analysis of variance (ANOVA) - One way, Two way. 3.2 Tests of Goodness of Fit and Independence, Goodness of Fit Test: A Multinomial Population, Test of Independence.	12

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO5, CO6	1. Build Simple Linear Regression model to estimate the dependent variable and access its goodness of fit. 2. Build Multiple Linear Regression model to estimate the dependent variable and access its goodness of fit. 3. Interpret Logistic Linear Regression model to estimate a binary dependent variable and access its goodness of fit. 4. Apply forecasting techniques to identify time series patterns, Apply non-parametric statistical methods.	Regression and Time Series Analysis 4.1 Simple Linear Regression: Simple Linear Regression Model, Regression Model and Regression Equation, Estimated Regression Equation, Least Squares Method, Coefficient of Determination, Correlation Coefficient, Model Assumptions, Testing for Significance, Using the Estimated Regression Equation for Estimation and Prediction Residual Analysis: Validating Model Assumptions, Residual Analysis: Outliers and Influential Observations. 4.2 Multiple Regression: Multiple Regression Model, Least Squares Method, Multiple Coefficient of Determination, Model Assumptions, Testing for Significance, Categorical Independent Variables, Residual Analysis. 4.3 Logistic Regression: Logistic Response function and logit, Interpreting the coefficients and odds ratios, Linear and Logistic Regression: similarities and differences, assessing the model. 4.4 Time Series Analysis and Forecasting: Time Series Patterns, Forecast Accuracy, Moving Averages and Exponential Smoothing, Trend Projection, Nonparametric Methods - Sign Test, Wilcoxon Signed-Rank Test, Mann-Whitney-Wilcoxon Test, Kruskal-Wallis Test, Rank Correlation.	18

VI SUGGESTED SELF LEARNING ACTIVITIES

Conduct a detailed case study on the problem statement by performing data analysis with the selection of dataset which is aligned with a clearly defined research question or objective. Apply descriptive statistics to summarize key features of the data, succeeded by data visualization techniques to identify trends, distributions, and outliers. Subsequently, inferential statistics, including hypothesis testing and confidence intervals, are applied to draw conclusions about the population. Finally, insights are synthesized and presented through a well-structured report, documenting the methodology, analysis, results, and recommendations.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Statistics for Business and Economics	15 th Edition	Jeffrey Camm, James Cochran, Michael Fry, Jeffrey Ohlmann, David Anderson, Dennis Sweeney, Thomas Williams	Cengage Publications	2025
2	An Introduction to Statistical Learning	2 nd Edition	Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani	Springer	2021
3	Practical statistics for data scientists: 50 essential concepts	2 nd Edition	Peter Bruce, Andrew Bruce and Peter Gedeck	O'Reilly Media	2020
4	Introduction to Probability and Statistics for Engineers and Scientists	6 th Edition	Sheldon M. Ross	Academic Press	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Data Science For Dummies	3 rd Edition	Lillian Pierson	Wiley Publications	2021
2	Statistical Methods	46 th Edition	Dr. S.P. Gupta	Sultan Chand & Sons	2021

Online References

Sr. No	Website Name
A	https://www.coursera.org/learn/stanford-statistics , Introduction to Statistics;
B	https://onlinecourses.nptel.ac.in/noc25_ma123/preview , NPTEL Course: Introduction to Probability & Statistics By Prof. Abhay Gopal Bhatt (Indian Statistical Institute, Delhi);
C	https://onlinecourses.nptel.ac.in/noc25_mg77/preview , NPTEL Course: Application of Forecasting, Regression and Time Series Models By Prof. Raghu Nandan Sengupta (IIT Kanpur);
D	https://onlinecourses.nptel.ac.in/noc24_ma46/preview , NPTEL Course: Non-Parametric Statistical Inference by Prof. Niladri Chatterjee (IIT Delhi);



Mahavir Education Trust's

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Semester VII



Program	Final Year B.Tech. Data Science Honours (Sem VII)	CL	TL	LL	SL	C
Course Name: Healthcare Data Analytics	Course Code: HNDSCR1701	4	1	4	3	6
Course Type:	Honours (HN)					
Pre-requisite:	Basics of Data Science					

I RATIONALE

Healthcare data analytics plays a pivotal role in transforming the modern healthcare ecosystem by turning raw medical data into actionable insights. The rationale behind its use lies in its ability to improve patient outcomes, reduce costs, and enhance the efficiency of healthcare delivery. By analyzing historical and real-time data—such as electronic health records (EHRs), lab results, and patient feedback—healthcare providers can make evidence-based decisions, predict disease trends, and personalize treatment plans.

II COMPETENCY

healthcare data analytics involves interpreting complex health data using analytical tools and clinical knowledge. It ensures data-driven decisions, improves patient care, and supports regulatory compliance. Key skills include data visualization, statistical analysis, EHR understanding, and collaboration—enabling professionals to translate insights into actionable strategies for better healthcare outcomes.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply knowledge of healthcare data sources and structures to organize and prepare datasets for analytics tasks
2	Implement machine learning techniques for biomedical image analysis and clinical text mining to extract meaningful patterns.
3	Develop machine learning pipelines using biomedical images and clinical text, integrating preprocessing, modeling, and validation steps.
4	Analyze healthcare datasets to build and validate predictive models for clinical decision-making.
5	Evaluate predictive model performance and quality measures in healthcare analytics to ensure reliability and ethical compliance.
6	Design and implement healthcare analytics solutions by integrating emerging technologies into real-world case studies.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme															
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Theory						Practical				SLA		Total	
				CL	TL	LL				Notional Hours	Exam	CIA		ESE	Total		CIAP	ESEP	Total				
												MSE	CCE		Max	Min			Max				Min
1	Healthcare Data Analytics	HN	HNDSCR1701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225	

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS
 (Minimum 12)

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Demonstrate proficiency in data preprocessing techniques for EHR data.	Clean, Integrate and Transform Electronic Healthcare Records.	CO1
2	Apply essential data analysis and visualization techniques	Apply various data analysis and visualization techniques on EHR.	CO1
3	Apply appropriate pre-processing techniques such as normalization, noise reduction, and contrast enhancement to biomedical images to improve their quality for further analysis	Implement Pre processing on Bio Medical Images.	CO2
4	Implement image segmentation techniques to biomedical images to identify regions of interest for diagnostic or analytical purposes	Apply Analytics/Segmentation on Bio Medical Images.	CO3
5	Apply Named Entity Recognition (NER) techniques to clinical text in order to identify entities such as diseases, medications, procedures, and symptoms	Named Entity Recognition (NER) for Clinical Entities in clinical text.	CO3
6	Implement sentiment classification models on clinical text data to detect sentiments from clinical notes	Sentiment Analysis of Clinical Notes	CO3
7	Analyze patient data to identify and interpret key risk factors influencing disease development using machine learning	Risk Factor Analysis Using Machine learning.	CO4
8	Analyze patient data using ML model to stratify disease risk and interpret feature importance in predicting patient outcomes	Diagnose disease risk from Patient data	CO4

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
9	Evaluate patient disease risk diagnosis through reinforcement learning by evaluating sequential decision-making policies	Disease Risk Diagnosis Using Reinforcement Learning for Sequential Decision Making	CO5
10	Analyze social media data to identify patterns, trends, and signals indicative of disease outbreaks, and evaluate their effectiveness for early outbreak prediction	Social Media Analytics for outbreak prediction	CO4
11	Implement machine learning models to classify and summarize key insights from social media drug reviews.	Social Media Analytics for Drug review analytics.	CO6
12	Implement complex visualizations of healthcare data to identify patterns, trends, and anomalies that can inform clinical decision-making	Visual Analytics for Healthcare Data	CO6

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VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Identify common sources of healthcare data 2. Describe how technology is used to support healthcare delivery, patient monitoring, and clinical decision-making. 3. Demonstrate a basic understanding of phenotyping algorithms and EHRs used to extract meaningful health information from EHR systems.	Introduction to Healthcare Data Analytics 1.1 Introduction, Healthcare Data Sources and Data Analytics for Healthcare, Applications and Practical Systems for Healthcare. 1.2 Electronic Health Records(EHR), Components of EHR, Benefits of EHR, Barriers to Adopting EHR, Challenges of using EHR data, Phenotyping Algorithms. 1.3 Healthcare Foundations: Healthcare delivery - Healthcare financing - Healthcare policy – Handling Patient data: the journey from patient to computer - Standardized clinical codesets - Breaking down healthcare analytics: population, medical task, data format, disease.	12

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2,CO3	1. Demonstrate a basic understanding of biomedical imaging modalities and fundamental techniques used in biomedical image analysis. 2. Demonstrate Natural Language Processing (NLP) techniques to extract information from clinical text. 3. Apply machine learning algorithms for clinical text mining. 4. Identify different types of clinical text corpora, key evaluation metrics used to assess text analysis systems.	Biomedical Image and Clinical Text Analysis 2.1 Biomedical Image analysis: Biomedical Imaging Modalities, Object detection, Image segmentation, Image Registration, Feature Extraction. 2.2 Clinical Text analysis: NLP, Mining information from Clinical Text, Information Extraction. 2.3 Rule Based Approaches, Pattern based algorithms, Machine Learning Algorithms. 2.4 Clinical Text Corpora and evaluation metrics, challenges in processing clinical reports, Clinical Applications.	18
3	CO4,CO5	1. Apply basic techniques to build predictive models in healthcare data. 2. Apply basic knowledge to describe healthcare quality measures and Medicare value-based programs. 3. Demonstrate key healthcare value-based programs and impact on healthcare quality and provider performance.	Predictive Models and Quality Measures in Healthcare 3.1 Making Predictive Models in Healthcare : Introduction to Predictive Analytics – Obtaining and Importing Dataset – Making the Response Variable - Splitting the Data into Train and Test Sets - Preprocessing the Predictor Variables – Building the Models – Using the Models to Make Predictions Improving our Models. 3.2 Healthcare Quality Measures: Introduction to healthcare measures, Medicare value-based programs: The Hospital Value-Based Purchasing (HVBP) program, The Hospital Readmission Reduction (HRR) program, The Hospital-Acquired Conditions (HAC) program, The End-Stage Renal Disease (ESRD) quality incentive program. 3.3 The Skilled Nursing Facility Value-Based Program (SNFVBP), The Home Health Value-Based Program (HHVBP), The Merit-Based Incentive Payment System (MIPS).	18

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO6	1. Identify common healthcare applications and explain how they support clinical decision-making. 2. Recognize examples of emerging technologies in healthcare and describe their basic purpose in improving patient care and healthcare delivery.	4.1 Applications: Descriptive Analytics Applications, Predictive Analytics Applications, Prescriptive Analytics Application 4.2 Healthcare Emerging Technologies : Healthcare analytics and the internet, Healthcare and the Internet of Things, Healthcare analytics and social media , Healthcare and deep learning , Obstacles, ethical issues, and limitations.	12

VII SUGGESTED SELF LEARNING ACTIVITIES

Case study on Healthcare Data Analytics, including Exploration of Healthcare Data Sources, Data Analytics Tools, Predictive models and their applications.

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VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Healthcare data analytics	1 st Edition	Chandan K. Reddy and Charu C Aggarwal	Taylor & Francis	2015
2	Healthcare Analytics Made Simple: Techniques in healthcare computing using machine learning and Python	1 st Edition	Kumar, Vikas Vik	Wiley Packt Publishing Ltd	2018
3	Healthcare Analytics: From Data to Knowledge to Healthcare Improvement	1 st Edition	Hui Yang and Eva K	Wiley	2016

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Machine Learning for Healthcare Analytics Projects	1 st Edition	Eduonix Learning Solutions	Packt Publishing	2018
2	Analytics in healthcare: a practical introduction	1 st Edition	El Morr, Christo, and Hossam Ali-Hassan	Springer	2019

Online References

Sr. No	Website Name
A	https://www.udemy.com/course/h16-advanced-analytics-in-healthcare-plain-simple/ Intro to Analytics in Healthcare, Plain & Simple



Mahavir Education Trust's

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Semester VIII

Program	Third Year B.Tech. Data Science Honours (Semester V)	CL	TL	LL	SL	C
Course Name: Data Analytics for Text, Web and Social Media	Course Code: HNDSCR1801	4	1	-	3	4
Course Type:	HN					
Pre-requisite:	Mathematics, Statistics for Data Science					

I RATIONALE

In today's rapidly evolving world, Data Analytics plays a significant role in helping individuals and organizations unlock the full potential of data, transforming raw information into valuable insights that drive informed decision-making. A course in Data Analytics offers a comprehensive learning of text, web and social media analytics and improve outcomes in various industries. The discipline is essential across various sectors, including engineering, healthcare, business, and, education, enabling better strategic planning and problem-solving. This course equips students with the critical thinking skills to thrive in an increasingly data-driven world.

II COMPETENCY

Demonstrates proficiency in text analytics techniques, web and social media analytics techniques for problem solving and decision making.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply algorithms for extraction , clustering and classification on textual data and perform prediction.
2	Apply various techniques for web data analytics using mining , searching and spamming of web data.
3	Analyze machine learning techniques to perform Sentiment Analysis and behavior analysis on social media data.
4	Analyze current trends and emerging technologies in the data analytics process.
5	Illustrate text, web data analytics on suitable dataset.
6	Prepare social media analytics for real world scenario.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total Notional Hours	Exam	Theory			Practical				SLA	Total			
				CL	TL	LL					CIA	ESE	Total	CIAP	ESEP	Total						
																Max	Min			Max	Min	Max
1	Data Analytics for Text, Web and Social Media	HN	HNDSCR1801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO5	1. Apply core algorithms of text mining to process textual data, and analyze the performance of information extraction techniques. 2. Apply text pre processing techniques to transform raw text into structured representations suitable for text mining tasks. 3. Apply appropriate feature selection and transformation methods to prepare textual data, and implement clustering techniques. 4. Apply and compare various classifiers for text classification tasks and analyze classifier performance on different types of textual data.	Text Data Mining 1.1 Introduction to Text Mining: Introduction, Algorithms for Text Mining, Information Extraction from Text: Named Entity Recognition, Relation Extraction, Unsupervised Information Extraction. 1.2 Text Representation: tokenization, stemming, stop words, NER, N-gram modeling. 1.3 Text Clustering: Feature Selection and Transformation Methods, distance based Clustering Algorithms, Word and Phrase based Clustering, Probabilistic document Clustering. 1.4 Text Classification: Feature Selection, Decision tree Classifiers, Rule-based Classifiers, Probabilistic based Classifiers, Proximity based Classifiers.	18

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2, CO5	1. Apply core concepts of web mining by constructing inverted indices, implementing basic compression techniques, and utilizing latent semantic indexing to enhance the effectiveness of web search systems. 2. Apply meta search techniques by utilizing similarity scores and rank positions to aggregate and refine search results from multiple sources. 3. Apply techniques to identify and analyze different forms of web spamming to detect and combat spam in web data. 4. Implement web usage mining techniques to model user behavior and segment visitors. 5. Analyze sequential and navigational patterns from web user transactions to identify browsing behavior trends, and evaluate classification and prediction models for forecasting user actions based on usage data.	Web Mining 2.1 Introduction to Web-Mining: Inverted indices and Compression, Latent Semantic Indexing, Web Search. 2.2 Meta Search: Using Similarity Scores, Rank Positions. 2.3 Web Spamming: Content Spamming, Link Spamming, hiding Techniques, and Combating Spam. 2.4 Web Usage Mining: Data Collection and Preprocessing, Sources and types of Data, Data Modelling, Session and Visitor Analysis, Cluster Analysis and Visitor segmentation, Association and Correlation Analysis. 2.5 Analysis of Sequential and Navigational Patterns, Classification and Prediction based on Web User Transactions.	20

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3, CO6	- To solve complex real-world problems using sentiment analysis. - Apply fundamental concepts of social media mining to identify key challenges and construct basic types of social network graphs. - Apply social media mining techniques to analyze user behavior, and implement classical and context-aware recommendation algorithms to generate and evaluate personalized recommendations. - Apply supervised and unsupervised techniques for opinion mining and document-level sentiment classification. - Analyze and compare dictionary-based and corpus-based approaches for opinion lexicon expansion. - Analyze patterns of opinion spam using supervised learning techniques, identify indicators of abnormal reviewer behavior, and evaluate methods for detecting group-based spam activities in user-generated content.	Social Media Mining and Sentiment Analysis 3.1 Social Media Mining: Introduction, Challenges, Types of social Network Graphs. 3.2 Mining Social Media: Influence and Homophily, Behaviour Analytics, Recommendation in Social Media: Challenges, Classical recommendation Algorithms, Recommendation using Social Context, Evaluating recommendations. 3.3 Sentiment Analysis: Introduction to Opinion mining. 3.4 Document Sentiment Classification: Supervised, Unsupervised. 3.5 Opinion Lexicon Expansion: Dictionary based, Corpus based. 3.6 Opinion Spam Detection: Supervised Learning, Abnormal Behaviors, Group Spam Detection.	12

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4	- Describe common techniques and importance of privacy-preserving data analytics. - Analyze the application of generative AI models in text, web, and social media data analytics, and evaluate their effectiveness in generating insights	Current Trends and Techniques 4.1 Privacy preserving data analytics, Generative AI models for text, web and social media data analytics, data ethics.	10

VI SUGGESTED SELF LEARNING ACTIVITIES

Conduct a detailed case study on the problem statement by performing data analysis with the selection of a data set that aligns with a clearly defined objective. Apply descriptive statistics to summarize key features of the data, succeeded by data visualization techniques to identify distributions and outliers. Finally, insights are presented through a well-structured report documenting the methodology, analysis, results, and applications.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Mining Text Data	1 st Edition	Charu. C. Aggarwal, Cheng Xiang Zhai	Springer Science and Business Media	2012
2	Web Data Mining-Exploring Hyperlinks, Contents, and Usage Data	2 nd Edition	BingLiu	Springer	2011

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Speech and Language Processing	3 rd Edition	Daniel Jurafsky and James H. Martin	Stanford	2020
2	Social Media Mining- An Introduction	3 rd Edition	Reza Zafarani, Mohammad Ali Abbasi and Huan Liu	Cambridge University Press	2014

Online References

Sr. No	Website Name
A	https://developers.google.com/analytics , Google Developers – Social Media and Web Analytics with Google Analytics and BigQuery
B	https://www.kdnuggets.com/ , KDnuggets – Tutorials on Text Mining, Web Mining, and Social Media Analytics



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An Autonomous Institute Affiliated to University of Mumbai

Syllabus of Honours Degree Programs in Technology

(With effect from 2025-26)

Internet of Things (IoT)



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An Autonomous Institute Affiliated to University of Mumbai

CURRICULUM STRUCTURE UG: B.Tech.

Honours/Minor Degree Program

[Internet of Things](#)

REVISION: R1-V0-2025-26

Effective from Academic Year 2025-26

Approved by Board of Studies on 18/06/2025

Approved by Academic Council on 19/06/2025

Dr. Vidyullata Devmane
I/C HOD



Dr. Bhavesh Patel
Principal

Dr. Bhavesh Patel
Chairman
Academic Council

Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088

Chairman BoS
Department of Electronics & Computer Science
Shah & Anchor Kutchhi Engineering College
Chembur, Mumbai - 400088



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Shah & Anchor Kutchhi Engineering College

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UG Program in Electronics & Computer Science

PREAMBLE

To meet the challenge of technological changes and to ensure excellence in engineering education, the Electronics and Computer Science branch stands as pillars of effective NEP implementation, innovation and research. Electronics and Computer Science has been driving progress across IT and Electronics industries and shaping the way we interact with the changing world. ECS journey is embarked with practical applications, theoretical concepts, integrating interdisciplinary skills to nurture future engineers.

At the heart of the electronics domain ECS branch lays the quest to harness the power of semiconductors and circuits to enhance communication. It incorporates the Internet of things and embedded systems.

In parallel, computer science offers implementation of algorithms, data structures and artificial intelligence. The booming fields of artificial intelligence, deep learning and robotics have been well integrated in the curriculum. All latest technological developments such as industry 4.0, augmented and virtual reality, 3 D printing are parts of the curriculum.

ECS branch offers Internet of Things as our honors programme to all departments. The Internet of Things (IoT) is a key driving factor in enabling the development of industrial automation systems. IoT coupled with computer automation controls helps streamline industrial systems and improve data automation, with the aim of removing errors and inefficiencies, primarily from the people. In the electronics and computer science curriculum, knowledge of various domains are integrated. Journey of innovations, project based learning, exposure to industries through internships have been initiated. It will lead a path towards a bright future where technology and holistic education will serve as a catalyst for positive change.

Outcome and skill based education with knowledge of Indian culture; student centric learning is being implemented in revised curriculum. Curriculum has 173 credits for UG and 18 credits for honours course. Faculty resolved that course objectives and course outcomes are to be clearly defined for each course, so that all faculty members in the institute understand the depth and approach of the course to be taught, which will enhance learner's learning quotient.

Choice based Credit and grading system enables a much-required shift in focus from teacher centric to learner-centric education since the workload estimated is based on the investment of time in learning and not in teaching. It also focuses on continuous evaluation which will enhance the quality of education. The present curriculum will be implemented for the Third Year Honours Programme of Engineering from the academic year 2024-25. Subsequently this will be carried forward for final year Engineering in the academic year 2025-26.



Program Structure for Honours Degree in Internet of Things Electronics & Computer Science (with effect from 2025–2026)

Semester V-VIII

			Learning Scheme					Credits	Assessment Scheme												
Sr. No.	Course Title	Course Code	Actual Contact Hrs/Week			SL	Total		Exam	Theory					Practical				SLA		Total
			CL	TL	LL					CIA		ESE	Total		CIAP	ESEP	Total				
						Duration in Hrs	MSE			CCE	Max		Max	Max			Max	Min	Max	Max	
1	Introduction to Internet of Things system Design	HNITCR0501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125
2	Communication Protocols and Cloud services for IoT	HNITCR0601	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125
3	Analytics for the Internet of Things (IoT)	HNITCR0701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225
4	Industry 4.0 And Industrial Internet Of Things	HNITCR0801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125
Total			16	4	4	12	540	18		80	80	240	400		40	60	100		100		600

Abbreviations:

CL: Classroom Learning TL: Tutorial Learning LL: Laboratory Learning SL: Self Learning (Activity/Assignment/Micro-Project) CIA: Continuous Internal Assessment
 CCE: Continuous Comprehensive Evaluation MSE: Mid Semester Examination ESE: End Semester Examination CIAP: Continuous Internal Assessment Practical ESEP: End Semester Examination Practical
 SLA: Self Learning Assessment

Note:

- Notional Learning hours for the semester are (CL+LL+TL+SL) hrs × 15 Weeks.
- 1 credit is equivalent to 30 Notional hrs.
- Self learning hours shall not be reflected in the Time Table.
- SLA will have separate head of passing



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Semester V



Program	Honours Degree in Internet of Things (Semester - V)	CL	TL	LL	SL	C
Course Name: Introduction to Internet of Things system Design	Course Code: HNITCR0501	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Basic Electronics, Computer Programming, Digital Electronics, Basic Networking Concepts					

I RATIONALE

The Internet of Things (IoT) is transforming the way devices interact and exchange data, enabling automation across industries such as healthcare, agriculture, smart homes, and transportation. This course equips students with foundational and practical insights into IoT architecture, sensing elements, hardware platforms, and cloud-based integrations. It emphasizes system-level thinking, enabling students to analyze, design, and implement efficient IoT-based solutions.

II COMPETENCY

Making them competent in: Analyzing and implementing basic IoT systems using appropriate sensors, actuators, hardware platforms, and communication technologies for real-world applications.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply the concepts of IoT architecture and enabling technologies to model a basic IoT system for a given scenario.
2	Analyze the functional roles of sensors and actuators to select appropriate devices for specific IoT applications.
3	Apply interfacing techniques to integrate IoT hardware platforms such as Arduino, Raspberry Pi, or ESP32 with sensors and actuators.
4	Analyze IoT data acquisition, storage, and analytics methods to determine suitable approaches for a given application.
5	Apply cloud platforms like ThingSpeak, AWS, or Blynk to implement real-time IoT data monitoring and control.
6	Analyze the design considerations, benefits, and limitations of IoT applications in domains such as smart cities, agriculture, healthcare, and environmental monitoring.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory					Practical				SLA	Total	
				CL	TL	LL					ESE	Total	CIAP	ESEP	Total							
								Duration in Hrs	MSE Max						CCE Max	Max	Min	Max	Max			Max
1	Introduction to Internet of Things system Design	-	HNITCR0501	4	1	-	3	120	4	2	20	20	60	100	40	-	-	0	0	25	10	125

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1	1. Analyze the key characteristics and definitions of IoT and distinguish it from traditional networks. 2. Analyze the physical and logical architecture of IoT systems. 3. Identify and explain enabling technologies that support IoT infrastructure. 4. Analyze common issues and challenges in IoT implementation 5. Compare IoT and M2M (Machine-to-Machine) communication with examples.	Introduction to IoT 1.1 Introduction, Definitions and Characteristics of IoT, 1.2 Physical and Logical design of IoT, 1.3 IoT enabling technologies, 1.4 IoT Issues and Challenges in IoT. 1.5 IoT and M2M.	10
2	CO2	1. Classify and analyze various types of sensors based on physical properties and application domains. 2. Analyze the working principles and applications of actuators including pneumatic, hydraulic, and electric. 3. Identify suitable sensor-actuator combinations for specific real-time IoT applications.	Sensors and Actuators 2.1 Sensors – Sensor Fundamental, Classifications and Properties, Optical, radiation and Displacement sensors, Presence, force, Pressure, Flow Sensors, Temperature, Humidity, Moisture Chemical and Biological Sensors 2.2 Actuators: Introduction of Pneumatic, Hydraulic and Electrical Actuators. Different electrical actuators 2.3 Identification of sensor and actuator for real-time application.	16

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO3, CO4, CO5	- Analyze and compare the architecture and programming models of IoT hardware platforms like Arduino, Raspberry Pi, ESP8266, NODEMCU, and ESP32. - Analyze IoT data analytics architecture and evaluate its advantages and limitations - Explore and compare various IoT cloud platforms such as ThingSpeak, AWS, and Blynk based on use case.	IoT hardware and Interfacing platforms 3.1 IoT hardware devices: Introduction to Arduino, Embedded programming for Arduino. Raspberry Pi, Introduction to Python programming for Raspberry Pi. Introduction to ESP 8266, Introduction to NODE MCU. Introduction to ESP32 3.2 IoT Data Analytics: Introduction, Architecture of Data Analytics, Advantages and Disadvantages. 3.3 IoT Platforms: Thingspeak, AWS, BLYNK	18
4	CO4, CO5, CO6	- Analyze the use of IoT in home automation and smart living environments - Evaluate IoT solutions for smart cities with emphasis on public services and urban infrastructure. - Analyze IoT applications in agriculture for productivity and sustainability. - Assess IoT's role in health monitoring and lifestyle enhancement. - Evaluate the impact of IoT in domains like environmental protection, energy, retail, and logistics.	IoT Applications 4.1 Home Automation, 4.2 Smart Cities, 4.3 Agriculture, 4.4 Health and Lifestyle, 4.5 IoT in Environmental Protection, Energy, Retail Management, Logistics	16



VI SUGGESTED SELF LEARNING ACTIVITIES

We will make logbook for activities conducted on-microproject, mini project, case study, simulation project, arduino / ESP32 / Raspberry Pi and sensor based system.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Internet of Things (A Hands-on Approach)	1 st Edition	Vijay Madiseti, Arshdeep Bahga	VPT (Valentina Publishing Technologies)	2014
2	Handbook of Modern Sensors: Physics, Designs and Applications	3 rd Edition	Jacob Fraden	Springer, New York	2015
3	Sensor Technology Handbook	1 st Edition	Jon S. Wilson	Elsevier, Netherlands	2011

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Sensors, Actuators and Their Interfaces: A Multidisciplinary Introduction	2 nd Edition	Nathan Ida	IET (Control, Robotics and Sensors Series 127)	2020
2	Designing the Internet of Things	-	Adrian McEwen	Wiley Publishers	2013
3	Internet of Things: Challenges, Advances, and Applications	-	Edited by Qusay F. Hasan, Atta ur Rehman Khan, Sajid A. Madani	CRC Press	-

**Online References**

Sr. No	Website Name
1	https://nptel.ac.in , Free IIT-led video lectures and certification on IoT architecture, hardware, and cloud platforms.;
2	https://www.iotforall.com , Articles, tutorials, and use-cases explaining emerging trends and practical implementations in IoT.;
3	https://www.tinkercad.com , Online simulation platform to practice Arduino-based IoT circuits without hardware.;
4	https://www.w3schools.com/python , Learn Python fundamentals needed for Raspberry Pi-based IoT development.;
5	https://create.arduino.cc , Official Arduino hub for examples, code, and hardware projects;
6	https://github.com , Access thousands of open-source IoT projects and code repositories for learning and contribution;
7	https://www.thingiverse.com , Repository of 3D models often used in smart devices or IoT prototypes;



Mahavir Education Trust's

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Semester VI



Program	Honours Degree in Internet of Things (Semester VI)	CL	TL	PL	SL	C
Course Name: Communication Protocol and Cloud Services for IoT	Course Code: HNITCR0601	4	1	-	3	4
Course Type:						
Pre-requisite:	Introduction to Internet of Things System Design					

I RATIONALE

The learners with the knowledge and skills to design and implement IoT solutions in key application domains such as smart homes, e-health, and smart cities. By understanding communication technologies, data handling, security, and real-world challenges, students will be prepared to develop innovative, secure, and efficient IoT systems that improve quality of life and support sustainable urban development.

II COMPETENCY

Across all domains, learners will be competent in integrating IoT sensors, actuators, and cloud platforms into functional systems; in acquiring and analyzing real-time data for decision-making; and in implementing communication protocols like MQTT, CoAP, and REST for seamless data exchange. They will be expected to apply secure design principles, recognize cybersecurity threats, and propose innovative, cost-effective IoT solutions tailored to specific domain needs. These integrated competencies span cognitive levels from application to creation, preparing learners for real-world problem-solving and advanced IoT system development.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply IoT enabling technologies to identify the components and communication layers required for system design.
2	Apply knowledge of wireless sensor networks to configure basic architectures for IoT applications.
3	Apply suitable lightweight protocols and networking techniques for efficient communication in resource-constrained environments.
4	Analyze IoT system requirements to evaluate the suitability of various wireless sensor network architectures.
5	Analyze cloud service providers such as AWS, Azure, and Google Cloud IoT for effective system integration.
6	Analyze application scenarios to design the most optimal connectivity solution for diverse IoT use cases.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme					Assessment Scheme														
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits		Theory						Practical			SLA			Total
				CL	TL	LL				Exam	CIA		ESE	Total	CIAP	ESEP	Total						
											MSE	CCE					Max	Min	Max	Min	Max	Min	
				1	Communication Protocols and Cloud services for IoT		HNITCR0601	4		1	LL	3	120	4	2	20	20	60	100	40			

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1,CO4	- Describe the key enabling technologies that drive IoT - Explain the types and roles of identifiers in IoT communication - Illustrate how the Internet enables IoT connectivity and communication - Compare different IoT frameworks/platforms and their architectural components - Describe the components and structure of a typical WSN node - Evaluate how WSN contributes to IoT application architecture and data flow	Wireless Sensor Network 1.1 Introduction to IOT 1.2 Enabling Technologies in IoT. 1.3 About Things in IoT, The Identifiers in IoT, About the Internet in IoT, IoT frameworks 1.4 Wireless Sensor Network for IoT: History and Context, the node, Connecting nodes, Networking Nodes, WSN and IoT.	8

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2, CO3	- Describe the role of each OSI layer in IoT/M2M communication & lightweight protocols - Compare different network models used in IoT - Differentiate between TCP and UDP protocols and their usage in IoT - Demonstrate real-time bidirectional communication using WebSocket - Identify communication standards for M2M - Describe and compare different messaging protocols for connected devices	Networking and Communication Protocol 2.1 Introduction Networking Protocol 2.2 OSI Model for the IoT/M2M System Lightweight M2M Communication Protocols 2.3 Internet based Communications, IP addressing in IoT 2.4 Network Model, TCP & UDP, Client-Server architecture 2.5 HTTP, REST APIs, WebSocket, MQTT, COAP, RFID IoT System, Protocols, M2M Communication Protocols, Bluetooth BR/EDR and Bluetooth low energy 2.6 RFID IoT Network Architecture, ZigBee IP/ZigBee SE2.0, Wifi(WLAN), Message Communication protocols for connected devices Data exchange formats: JSON & XML.	15
3	CO5	- Describe the NIST's SPI model (SaaS, PaaS, IaaS) for cloud services in IoT. - Explain the integration of cloud computing with IoT and its benefits. - Illustrate the four cloud deployment models. - Analyze the role of cloud, big data, analytics, and ubiquitous connectivity. - Explain the perception, network, and application layers in IoT architecture. - Identify foundational and enabling technologies like edge computing, containers.	Cloud Services for IoT 3.1 NIST's SPI Architecture and Cloud Standards, The Cloud of Things–The Internet of Things and Cloud Computing. 3.2 The Cloud of Things Architecture–Four Deployment Models, Vertical Applications, Fifteen Essential Features, Four, Technological Pillars, Three Layers of IoT Systems, Foundational, Technological Enabler. 3.3 Cloud Providers and Systems – Microsoft Azure IoT, Amazon Web Services, Google's cloud IoTs.	14

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO6	1. Define the core features and ecosystem of IoT in healthcare systems, smart home, smart city. 2. Identify and explain the role of smart appliances and their control systems. 3. Compare different wireless communication technologies used in smart homes, e-health, smart city. 4. Evaluate systems used for elderly monitoring and emergency alerting. 5. Evaluate IoT's role in integrated healthcare delivery in smart cities.	Case Studies 4.1 Smart Home - Smart Home Energy Management, Smart Appliances, Communication Technologies for Smart Homes, maintenance, security, challenges. 4.2 e-health: Characteristics of e-health and applications- monitoring of health parameters, smart medicine box, elderly people monitoring, challenges. 4.3 IoT Smart City: Characteristics and applications- Smart Economy, Smart Environment, Smart Living Smart Grid, Smart Home, Transport and Traffic Management, Smart Healthcare.	8

VI SELF LEARNING ACTIVITIES

The students to engage in continuous self-learning beyond the syllabus by implementing mini projects, research-based case studies.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	The Internet of Things in the Cloud A Middleware Perspective	2 nd Edition	Honbo Zhou	CRC Publication	-
2	Internet of Things- Hands on Approach	3 rd Edition	Arshdeep Bagha, Vijay Mediseti	Khanna Book Publishing	-
3	Internet of Things	3 rd Edition	Jeeva Jose		2018
4	Designing the Internet of Things	2 nd Edition	Adrian McEwen and Hakim Cassimally	John Wiley and Sons Ltd, UK	2014
5	Internet of Things: Concepts and System Design	-	Milan Milenkovic,	Springer International Publishing	2020

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Enterprise Cloud Computing	-	Gautam Shroff	Cambridge	2010
2	Mastering Cloud Computing-Foundations and Applications Programming	3 rd Edition	Raj Kumar Buyya, Christian Vecchiola, S. Thamarai Selvi	MK Publication	2013
3	Learning AWS IoT- Effectively Manage Connected Devices on the AWS Cloud Using Services Such as AWS Greengrass, AWS Button, Predictive Analytics and Machine Learning	-	Agus Kurniawan	Packt Publication	2018
4	The Internet of Things Connecting Objects to the Web	-	HakimaChouchi	Wiley Publications	2010
5	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things	-	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome	-	-

Online References

Sr. No	Website Name
A	https://onlinecourses.swayam2.ac.in/cec21_cs19/preview , NPTEL Course: Introduction to Computer Networks & Internet Protocols;
B	https://onlinecourses.swayam2.ac.in/ntr24_ed01/preview , NPTEL Course: Internet of Things: Design Concepts and Use Cases;



Mahavir Education Trust's

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Semester VII



Program	Honours Degree in Internet of Things (Semester VII)	CL	TL	LL	SL	C
Course Name: Analytics for the Internet of Things (IoT)	Course Code: HNITCR0701	4	1	4	3	6
Course Type:	-					
Pre-requisite:	Introduction to Internet of Things system Design, Communication Protocols and Cloud services for IoT.					

I RATIONALE

The integration of Analytics and Machine Learning with IoT systems is essential in extracting actionable insights from massive real-time sensor data. As IoT systems scale, the ability to analyze, secure, and intelligently manage data becomes a core requirement for modern engineers. This course aims to bridge the gap between IoT deployment and intelligent decision-making by equipping students with data analysis, machine learning, and system security skills—critical for applications like smart agriculture, smart cities, and healthcare.

II COMPETENCY

Apply machine learning techniques and data analytics to real-time IoT systems to develop intelligent, secure, and scalable applications for industrial and societal benefit.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Experiment secure storage, retrieval, and processing of sensor data for IoT Applications.
2	Illustrate IoT analytics challenges and assess AWS cloud setup components for efficiency and scalability.
3	Illustrate supervised machine learning algorithms on IoT datasets after performing data preprocessing, feature engineering, and standardization.
4	Evaluate IoT datasets using advanced exploration techniques to identify trends, correlations, and anomalies for predictive modeling.
5	Apply security measures across IoT architecture layers to mitigate threats in WSN, RFID, SCADA, and healthcare applications.
6	Analyze real-world IoT case studies in agriculture, healthcare, smart cities, and transportation to determine performance, scalability, and security improvements achievable through machine learning integration.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

				Learning Scheme			Credits	Assessment Scheme														Total
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week				SL	Total	Exam	Theory					Practical				SLA		
				CL	TL	LL					Notional Hours	CIA	ESE	Total	CIAP	ESEP	Total					
								Duration in Hrs	MSE									CCE	Max		Min	
1	Analytics for the Internet of Things (IoT)	-	HNITCR0701	4	1	4	3	180	6	2+2	20	20	60	100	40	40	60	100	40	25	10	225

V LABORATORY LEARNING OUTCOME AND SUGGESTED LIST OF EXPERIMENTS

Part A is compulsory. Student can choose any one of Part B and Part C

Sr. No.	Laboratory Learning Outcome	Laboratory Experiment Titles	Relevant COs
1	Develop a Python-based application to interface and read temperature and humidity data from a DHT11 sensor using Raspberry Pi.	PART A Implement interfacing of DHT11 with Raspberry Pi	CO1
2	Implement relay control using Raspberry Pi to operate external devices through GPIO programming.	PART A Implement interfacing of Relay with Raspberry Pi	CO1
3	Configure and simulate an RPL-based IPv6 network using Contiki OS and Cooja, including border routers and REST clients, to demonstrate IoT communication protocols.	PART A Demonstrate Contiki OS for RPL (Create 2 border routers and 10 REST clients; Access router from other network – Simulator)	CO1
4	Simulate and analyze real-time control of actuators (LED/stepper motor) using an IoT simulator like Bevywise for edge device communication	PART A Demonstrate use of IoT simulators (like Bevywise) on any real-time device (LED / stepper motor)	CO2
5	Develop a Python-based IoT application using Arduino/Raspberry Pi to collect temperature data periodically, store it in CSV format, and apply linear regression using data analysis tools (e.g., WEKA/Orange) to predict future temperature values	PART B Programming for IOT : Python Programming, Develop an Application on Arduino / Raspberry-Pi to capture the values of temperature sensor after every 15 sec of time interval, store this values in .csv format and predict the temperature at particular time t using linear regression analysis. Create the dataset of at least 50-75 instances, use any data analysis tool (WEKA/Orange)	CO2

6	Apply machine learning algorithms to classify IoT datasets and evaluate model performance using statistical metrics such as accuracy, precision, recall, TPR, FPR, and error rate	PART C 1. Download the IoT Dataset of your choice 2. Divide the dataset into Training data and Testing data. 3. Perform the classification of the instances using any machine learning algorithm like KNN Algorithm, Naïve Bayes, Decision Tree or any.4. Evaluate the machine learning model by considering the parameter (TPR, TNR, FPR, FNR, accuracy, precision, recall, error rate etc.)	CO3
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VI THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO2	1. Understand IoT analytics, challenges, and AWS-based IoT cloud setup. 2. Configure virtual environments (e.g., AWS VPC, S3) for storing and analyzing IoT data.	IoT and Data Analytics 1.1 Defining IoT Analytics, IoT Analytics challenges 1.2 The AWS virtual Private Cloud (VPC) setup walk through 1.3 Creating a key pair for the NAT and bastion instances. 1.4 Creating S3 bucket to store data 1.5 Creating a VPC for IoT Analytics.	15
2	CO2, CO3	1. Apply supervised machine learning algorithms (e.g., Linear/Logistic Regression, SVM) to IoT data 2. Prepare, clean, and standardize IoT datasets for predictive modeling. 3. Perform feature engineering and selection using ML techniques. 4. Analyze IoT datasets using advanced data exploration tools.	IoT and Machine Learning 2.1 Advantages of IoT and Machine Learning Integration. 2.2 Implementation of Supervised Algorithm- Regression (Linear and Logistic), SVM for IoT Neural Network on case study: Agriculture and IoT, Smart Home etc 2.3 Data Preparation for Predictive Maintenance Modeling, Cleaning and Standardizing IoT Data. 2.4 Applying Advanced Data Exploration Techniques. 2.5 Feature Engineering: Exploring Feature Engineering, Applying Feature Selection Techniques, Feature set selection using ML 2.6 Machine learning for Internet of Things data analysis.	15

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
3	CO4, CO5	1. Identify and analyze security requirements across IoT architecture layers. 2. Evaluate threats in IoT systems including WSN, RFID, and SCADA. 3. Discuss privacy and security issues in healthcare, enterprise systems, and social IoT.	IoT and it's Security 3.1 Introduction, Security Requirements , Security Requirements in IoT Architecture, Sensing Layer and IoT End-Nodes, Network Layer, Service Layer, Application Interface Layer 3.2 Cross-Layer Threats, Threats Caused in Maintenance of IoT, Security in Enabling Technologies, Security in Identification and Tracking Technologies. 3.3 Security in Integration of WSN and RFID, Security in Communications, Security in Networks, Security in Service Management, Security Concerns in IoT Applications 3.4 Security Concerns in SCADA Systems, Security Concerns in Enterprise Information systems, Security Concerns in Social IoT , Confidentiality and Security for IoT-Based Healthcare.	15
4	CO6	1. : Explore case studies in agriculture, healthcare, smart cities, and transportation. 2. Understand the role of ML in enhancing IoT security and performance.	Applications of ML and IOT Case Studies 4.1 IOT for Agriculture. 4.2 Remote Patient Monitoring. 4.3 Smart City. 4.4 Smart Transportation. 4.5 IOT Security using ML.	15

VII SUGGESTED SELF LEARNING ACTIVITIES

Case Study / research paper presentations

VIII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Analytics for the Internet of Things (IoT)	1 st Edition	Andrew Minter	Packt Publication	2017
2	Internet of Things- Hands on Approach	1 st Edition	Arshdeep Bagha, Vijay Mediseti	Universities press	2015
3	Internet of things For Architects	1 st Edition	Perry Lea	Packt Publishing	2018

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	MACHINE LEARNING IN COGNITIVE IOT	1 st Edition	Neeraj Kumar, Aisha Makkar	CRC Press	2020
2	IoT Machine Learning Applications in Telecom, Energy, and Agriculture, With Raspberry Pi and Arduino Using Python	-	Puneet Mathur	-	-
3	Real-Time IoT Imaging with Deep Neural Networks - Using Java on the Raspberry Pi 4	-	Nicolas Modrzyk	Apress Publication	2020

Online References

Sr. No	Website Name
1	https://nptel.ac.in , Free IIT-led video lectures and certification on IoT architecture, hardware, and cloud platforms.;
2	https://www.iotforall.com , Articles, tutorials, and use-cases explaining emerging trends and practical implementations in IoT.;
3	https://www.tinkercad.com , Online simulation platform to practice Arduino-based IoT circuits without hardware.;
4	https://www.w3schools.com/python , Learn Python fundamentals needed for Raspberry Pi-based IoT development.;
5	https://create.arduino.cc , Official Arduino hub for examples, code, and hardware projects.;
6	https://github.com , Access thousands of open-source IoT projects and code repositories for learning and contribution.;
7	https://www.thingiverse.com , Repository of 3D models often used in smart devices or IoT prototypes.;



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Semester VIII



Program	Honours Degree in Internet of Things (Semester VIII)	CL	TL	LL	SL	C
Course Name: Industry 4.0 And Industrial Internet Of Things	Course Code: HNITCR0801	4	1	-	3	4
Course Type:	-					
Pre-requisite:	Introduction to Internet of Things system Design, Communication Protocols and Cloud services for IoT, Analytics for the Internet of Things.					

I RATIONALE

The course Industry 4.0 and Industrial Internet of Things (IIoT) is designed to equip students with knowledge and skills to operate and innovate in modern industrial environments where digital transformation plays a central role. The focus is on understanding the evolution, architecture, data handling, and applications of IIoT, enabling students to contribute to smart manufacturing and intelligent systems design aligned with Industry 4.0.

II COMPETENCY

Analyze, design, and implement industrial IoT solutions by understanding architectural frameworks, communication protocols, data acquisition systems, and application-specific deployments, keeping in mind the challenges and security needs of Industry 4.0 environments.

III COURSE OUTCOMES (COs)

After the completion of course based learning Students will be able to,

Course Outcome Number	Course Outcome Statement
1	Apply the concepts of Industry 4.0 and IIoT to analyze and propose solutions for modern industrial challenges.
2	Apply industrial protocols, control systems, and analytics to design and assess IIoT data acquisition and transmission.
3	Apply IIoT architectures and business models to design and analyze industrial software systems.
4	Analyze IIoT applications across industries to identify improvements in operations and resource management.
5	Analyze challenges in IIoT security, big data handling, and edge computing to recommend suitable solutions for industrial environments.
6	Design and justify an integrated IIoT solution using protocols, architectures, and analytics to address real-world Industry 4.0 challenges.

IV TEACHING-LEARNING & ASSESSMENT SCHEME

							Learning Scheme			Assessment Scheme																	
Sr. No.	Course Title	Course Type	Course Code	Actual Contact Hrs/Week			SL	Total	Credits	Exam	Theory					Practical				SLA		Total					
				CL	TL	LL					MSE	CCE	ESE	Total		CIAP	ESEP	Total									
							Duration in Hrs	Max												Min	Max		Min	Max	Min	Max	Min
1	Industry 4.0 And Industrial Internet Of Things	-	HNITCR0801	4	1	-	3	120	4	2	20	20	60	100	40	-	-	-	-	25	10	125					

V THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
1	CO1, CO5	- Understand the history and evolution of industrial revolutions leading up to Industry 4.0. - Analyze the key drivers and requirements of Industry 4.0 systems. - Examine Cyber-Physical Systems (CPS), their role in IIoT, and their impact on business and sustainability. - Evaluate the challenges of IIoT including device discovery, data management, and edge computing.	Introduction to Industry 4.0 1.1 Overview of Industry 4.0 and Industrial Internet of Things, Industry 4.0. 1.2 Industrial Revolution: Phases of Development, Evolution of Industry 4.0, Impact, Effects and Consequences 1.3 Industrial Internet, Basics of CPS, CPS and IIOT, Design requirements of Industry 4.0, Drivers of Industry 4.0, Sustainability Assessment of Industries, Smart Business Perspective, Cyber security, Impacts of Industry 4.0, Industrial Internet of Things: Basics, IIOT and Industry 4.0, Industrial Internet Systems, Industrial Sensing, Industrial Processes. 1.4 IIOT Challenges – Identifying Things within the internet, Discovering Things and the Data they possess, Managing massive amount of data, Navigating Connectivity Outages, IIOT Edge - Leveraging the Power of Cloud Computing, Communicating with Devices on the Edge, Determining a Request/Response Model	15

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Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
2	CO2, CO5, CO6	1. Identify various industrial communication protocols and their components (e.g., Foundation Fieldbus, Modbus, HART). 2. Differentiate among industrial automation systems like PLC, SCADA, and DCS. 3. Explore wireless technologies and their integration in industrial systems (e.g., LoRa, NB-IoT, IEEE 802.11ah). 4. Understand the application of Big Data analytics in industrial IoT for decision-making.	Industrial Data Transmission and Industrial Data Acquisition 2.1 Introduction, Features and Components of - Foundation Fieldbus, Profibus, HART, Interbus, Bitbus, CC-Link, Modbus, Batibus, Digital STROM 2.2 Controller Area Network, DeviceNet, LonWorks, ISA 100.11a, Wireless HART, LoRa and LoRaWAN NB-IoT, IEEE 802.11AH, Distributed Control System, PLC, SCADA 2.3 Industrial IoT: Big Data Analytics	15
3	CO3, CO6	1. Describe business models and reference architectures used in IIoT deployments 2. Understand software-defined networking (SDN) and connectivity topologies in IIoT. 3. Apply architectural principles and advanced styles to design IIoT systems. 4. Analyze software engineering challenges and solutions for large-scale industrial IoT systems.	IIOT Reference Architecture 3.1 Industrial Processes, Business Models and Reference Architecture for IIoT Business Models. 3.2 Types of Business Models for IoT, Software-Defined Networking (SDN) in IIoT 3.3 The Three-Tier Topology, Key Functional Characteristics of Connectivity. Software Architectural Style for the Industrial Internet of Things - Software Architecture Practice, Advanced Architectural Styles, Systems of Systems. 3.4 Challenges of Software Engineering in IIoT, Principles for Software Architecture design in IIoT, The Principled Decomposition, The Architectural Style	15

(Continued on next page)

Module Number	Mapped Course Outcome	Major Learning Outcomes	Theory Content	No. of Hours
4	CO4, CO6	1. Apply IIoT concepts to various sectors like manufacturing, healthcare, food industry, and energy. 2. Evaluate the benefits of IIoT in facility management, supply chain, and inventory systems 3. Explore real-world deployments of IoT in oil, gas, and power generation industries.	IIoT Applications 4.1 IIoT Applications: Factories and Assembly Line, Food Industry, Healthcare, Power Plants, Electricity Sector, Solar Energy Sector. 4.2 Supply Chain and Inventory Management, facility management 4.3 IoT in Oil and Gas Industry	15

VI SUGGESTED SELF LEARNING ACTIVITIES

We will make logbook for activities conducted on-microproject, mini project, case study, simulation project, arduino / ESP32 / Raspberry Pi and sensor based system.

Case Study / Research paper review.

VII SUGGESTED LEARNING MATERIALS / TEXTBOOKS / REFERENCE BOOKS

Text Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Introduction to IoT	1 st Edition	S. Misra, A. Mukherjee, and A. Roy	Cambridge University Press	2020
2	Introduction to Industrial Internet of Things and Industry 4.0.	1 st Edition	S. Misra, C. Roy, and A. Mukherjee	CRC Press	2020
3	Industrial Internet of Things and Communications at the Edge	-	Tony Paine, CEO, Kepware Technologies		

Reference Books

Sr. No.	Title	Edition	Authors	Publisher	Year
1	Practical Internet of Things Security	1 st Edition	Brian Russell, Drew Van Duren	Packt Publishing	2016
2	Architectural Design Principles For Industrial Internet of Things		Hasan Derhamy		

Online References

Sr. No	Website Name
1	https://onlinecourses.nptel.ac.in/noc25_cs146/preview , SWAYAM-NPTEL Introduction to Industry 4.0 and Industrial Internet of Things By Prof. Sudip Misra, IIT Kharagpur;
2	https://www.iotforall.com , Articles, tutorials, and use-cases explaining emerging trends and practical implementations in IoT.;
3	https://www.w3schools.com/python , Learn Python fundamentals needed for Raspberry Pi-based IoT development.;
4	https://create.arduino.cc , Official Arduino hub for examples, code, and hardware projects;
5	https://github.com , Access thousands of open-source IoT projects and code repositories for learning and contribution;
6	https://www.thingiverse.com , Repository of 3D models often used in smart devices or IoT prototypes;

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